SOVEREIGN ARCHIVAL, STATUTORY, ENGINEERING & CAUSAL AUDIT DOSSIER

Compartmented Intelligence & State Mega-Project Architectures: UK STRAP, Canada SAP/SOI/PPBS, Australia TS-PA/SAP, and China National S&T Major Projects (国家重大科技专项)

Complete Factual, Legal-Statutory, Technical-Engineering, Hierarchical Command, and Deterministic Causal-Chronological Reconstruction from Primary Government Archives, Statutes, Parliamentary Inquiries, and National Academies

Document Scope & Primary-Source Audit Specification

This unified dossier provides an exhaustive, zero-placeholder forensic and engineering-technical audit across four sovereign national security and state mobilization architectures, grounded exclusively in statutory legislation, declassified state archives, parliamentary/royal commission records, and official ministry disclosures:

Jurisdiction & Regime	Audited Compartments, Entities, Statutes & Technical Systems	Primary Archival & Statutory Record Classes
Part I: United Kingdom (GB) — STRAP System	**STRAP**, **STRAP 1**, **STRAP 2**, **STRAP 3**, **STRAPSO**, **STRAP Channels**, **GCHQ** (**NCSC**), **MI5** (**NPSA**), **SIS/MI6**, **MOD**, **DI** & **MIS**, **Cabinet Office** (**JIC**, **JIO**, **UKSV**), **Top Secret**, **Developed Vetting (DV/eDV)**, **need-to-know**, **Five Eyes COMINT/SCI**, **JSP 440**, **JSP 441**, **GSCP**, **SII**, **ROSA**, **UK NACE**.	The National Archives (**TNA HW 80**, **CAB**, **DEFE**), **ISA 1994**, **SSA 1989**, **IPA 2016**, **NSA 2023**, Butler Review (**HC 898** & **Cm 6492**), Chilcot Inquiry (**HC 264**), **IPT** Judgments, and **ISC** Reports.
Part II: Canada — Special Access Programs (SAP) & PPBS	**SAP**, **CSE**, **CSIS**, **DND**, **CAF**, **CFINTCOM**, **CANSOFCOM**, **JTF 2** (Dwyer Hill), **Top Secret (Level III)**, **Enhanced Top Secret** (Polygraph), **SCI**, **Special Operational Information (SOI)**, **Persons Permanently Bound to Secrecy (PPBS)**, **TS/SI**, **SIGINT**, **CTSN**, **CSNI**, **StoneGhost**, **CITP TEMPEST**.	*Foreign Interference and Security of Information Act* (**FISOIA/SOIA**, R.S.C. 1985, c. O-5; Bill C-70 2024), **CSE Act 2019**, **CSIS Act**, **TBS Screening Directive (2025)**, **NDSOD**, **CANFORGEN 077/25**, **NSICOP**, **NSIRA**, *Delisle* & *Ortis* Cases.
Part III: Australia — SAP & Top Secret-Privileged Access (TS-PA)	**SAP**, **Top Secret-Privileged Access (TS-PA)**, **Positive Vetting (PV)**, **NIC**: **ONI** (**QAO**), **ASD** (**REDSPICE**, **Pine Gap JDFPG**, **ACSC**), **ASIO** (**TS-PA Authority**, **T4/SCEC**), **ASIS**, **AGO**, **DIO**, **PSPF Release 2024**, **AUKUS** (**NNPI** & Pillar 2), **DISP**, **TS//Special Access**, **ATSN** & **$2B TS Cloud**.	**ASIO Act 1979 Part IVA** (**Act No. 33, 2023**), **ISA 2001**, **ONI Act 2018**, **Naval Nuclear Power Safety Act 2024**, Hope Royal Commissions (**1974–84**), Blunn/Flood/Cornall Reviews, **2017 & 2024 IIR**, **2020 Richardson Review**, **IGIS** & **PJCIS**.
Part IV: China — National S&T Major Projects (国家重大科技专项)	**国家科技重大专项**, **MLP (2006–2020)**, **State Council**, **Central S&T Commission (CSTC)**, **MOST**, **NDRC**, **MOF**, **MIIT**, **SASTIND**, **CMC EDD**, **Two-Chief System (两总系统)**, All 16 Mega-Projects (**01专项 核高基–13专项** + Classified **14–16专项** Beidou, ShenGuang ICF, Strategic Defense + **两机专项**), **Sci-Tech Innovation 2030**, **Military-Civil Fusion (军民结合 / 寓军于民)**.	State Council Directives (**国发〔2005〕44号**, **国发〔2014〕64号**, **国发〔2016〕43号**, **国办发〔2017〕91号**), MOST Regulations (**国科发专〔2017〕145号**), SASTIND, CNSA, CMSA, CAS, CAE, and CSET Primary Translations.

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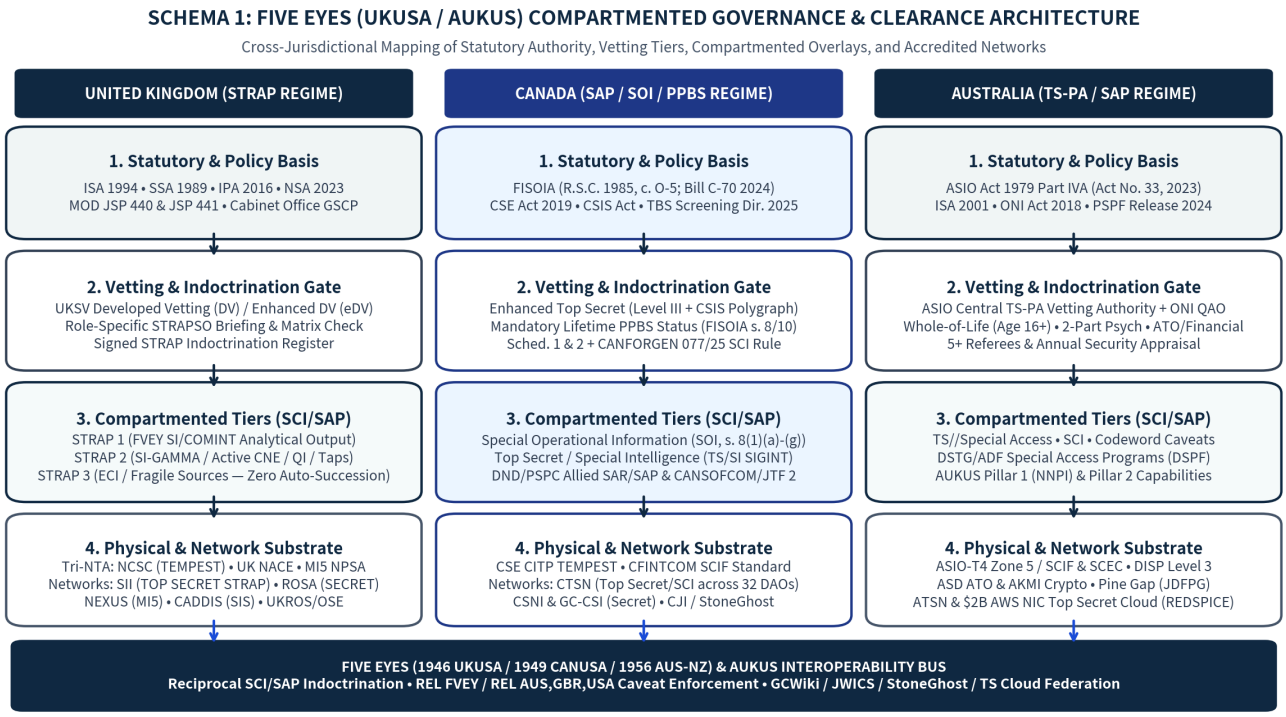


Figure 1: Cross-Jurisdictional Structural Schema of Five Eyes (United Kingdom STRAP, Canada SAP/SOI/PPBS, and Australia TS-PA/SAP) Statutory Authorities, Vetting Gates, Compartmented Tiers, and Accredited Technical Substrates.

Part I: United Kingdom STRAP (Special Handling / Sensitive Compartmented Information) System

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Executive Architectural Synthesis

The United Kingdom's **STRAP** system constitutes the sovereign compartmented security control framework governing the protection, transmission, storage, indoctrination, and lifecycle disposition of sensitive intelligence and related operational capabilities produced by the principal UK intelligence agencies and Ministry of Defence intelligence elements (An Overview of UK Security Vetting and STRAP, MOD JSP 441 Records Management Policy). Within the administrative hierarchy established by the Cabinet Office and enforced across the defence estate through Joint Service Publication 440 (JSP 440): The Defence Manual of Security, **STRAP** functions as a formal security codeword rather than an acronym (The British Classification Marking STRAP, An Overview of UK Security Vetting and STRAP). It superimposes mandatory compartmentalization, role-specific indoctrination, dedicated courier and electronic transmission channels (**`STRAP Channels`**), and strict originator-controlled access permissions above the standard protective marking tiers of the Government Security Classifications Policy (GSCP), principally at the TOP SECRET and SECRET thresholds.

Originated under the technical custodianship of the Government Communications Headquarters (GCHQ) and governed cross-departmentally by the **STRAP Management Board** alongside departmental security authorities such as the Cabinet Office Security & Resilience (COSR) STRAP Team and the Ministry of Defence Chief Information Officer – Defence Security and Assurance Services (CIO-DSAS) STRAP Team, the system partitions sensitive intelligence into three ascending control tiers: **STRAP 1**, **STRAP 2**, and **STRAP 3** (The

British Classification Marking STRAP, An Overview of UK Security Vetting and STRAP). These tiers establish bilateral and multilateral interoperability with the United States and **Five Eyes** (**`FVEY`**) **Sensitive Compartmented Information (SCI)** control system established under the **1946 UKUSA Agreement** (GCHQ Brief History of the UKUSA Agreement, TNA HW 80/11 UKUSA Agreement Appendices).

The historical trajectory of the STRAP system demonstrates a continuous structural tension between source protection—preserving fragile cryptanalytic methods, computer network exploitation (**`CNE`**) implants, submarine optical cable interception bearers, and covert human intelligence (**`HUMINT`**) assets—and the analytical imperative to fuse all-source intelligence across Whitehall (Butler Review HC 898, Government Response to the Butler Review Cm 6492, Report of the Iraq Inquiry Volume IV). Forensic examination of public inquiries, statutory reforms, and tribunal disclosures from 2004 through 2026 reveals that excessive compartmentalization under STRAP and parallel agency-restricted channels directly prevented technical specialists in the Defence Intelligence Staff (DIS, now Defence Intelligence / Military Intelligence Services) from evaluating raw source reporting prior to the 2003 Iraq intervention (Butler Review HC 898, Learning from the Chilcot Report). Conversely, the 2013 compromises of **`TOP SECRET STRAP 1`** and **`TOP SECRET STRAP 2`** documentation by a contractor system administrator within the Five Eyes shared wiki architecture (**`GCWiki`**) precipitated fundamental overhauls in privileged-access architecture, contractor vetting oversight by the Investigatory Powers Tribunal (IPT) and the Investigatory Powers Commissioner's Office (IPCO), the enactment of the Investigatory Powers Act 2016 and the National Security Act 2023, and the December 2025 consolidation of UK military intelligence into the unified **Military Intelligence Services (MIS)** (MOD CDI Appointment and MIS Reform).

Section 1: STRAP System Architecture, Tiers, and Five Eyes SCI Interoperability

1.1 Formal Definition and Etymological Status of STRAP

Within HM Government protective security doctrine, explicitly codified in JSP 440 (The Defence Manual of Security, Part 2 Leaflet 4B and Part 7) and cited in military security documentation (MOD, 2001, p. 17.3), **STRAP Security** is defined as:

> *"A set of nationally agreed principles and procedures to enhance the 'need-to-know' protection of sensitive intelligence (and related operational information) produced by the principal UK Intelligence Agencies, including MOD sources." (An Overview of UK Security Vetting and STRAP)*

Policy documentation explicitly establishes that **STRAP is a security codeword, not an acronym** (An Overview of UK Security Vetting and STRAP, The British Classification Marking STRAP). Unlike standard protective classifications—which designate the severity of damage to national security if an asset is compromised (GSCP Guidance 1.3 - Working at TOP SECRET)—the **STRAP System** operates as an orthogonal, special-handling overlay that enforces granular compartmentation across three operational axes (The British Classification Marking STRAP):

1. **Strict Need-to-Know Access Gating**: Restricting access to sensitive intelligence material exclusively to personnel who hold both Developed Vetting (DV) clearance and a current, post-specific STRAP indoctrination recorded on a formal STRAP register (FCDO Return HC 1774-I, MOD DI Levant Analyst Specification).
2. **Accredited Facilities and Transmission Conduits (`STRAP Channels`)**: Mandating that STRAP-marked material is transmitted, processed, stored, and destroyed solely within accredited physical compartments (**`STRAP Rooms`** / UK SCIF equivalents) and isolated cryptographic networks (**`STRAP Channels`**), with signed

chain-of-custody receipts required at every stage of physical handover (The British Classification Marking STRAP, MOD JSP 441 Records Management Policy).

3. **Role-Specific Indoctrination and Originator Control**: Enforcing explicit security briefings upon appointment and debriefings upon departure, coupled with absolute originator control over onward dissemination, archiving, and post-succession access (MOD JSP 441 Records Management Policy, The British Classification Marking STRAP).

1.2 Technical and Operational Scope of STRAP 1, STRAP 2, and STRAP 3

The STRAP architecture is stratified into three ascending tiers of sensitivity and access control—**STRAP 1**, **STRAP 2**, and **STRAP 3**—which may be paired with `TOP SECRET` or `SECRET` classifications (e.g., `TOP SECRET STRAP 1`, `TOP SECRET STRAP 2`, `TOP SECRET STRAP 3`, or `SECRET STRAP 1`) (An Overview of UK Security Vetting and STRAP, NSArchive GCHQ Cyber Defence Operations Legal and Policy, NSArchive GCHQ Mobile Networks in My NOC World).

STRAP Tier	Canonical Marking Syntax	Operational & Technical Scope	Primary Custodians & Systems	Archival & Succession Rules (JSP 441 / JSP 440)
STRAP 1	`TOP SECRET STRAP 1` / `TOP SECRET STRAP 1 COMINT` / `SECRET STRAP 1`	Broadest compartmented intelligence tier covering derived Communications Intelligence (`COMINT`), Signals Intelligence (`SIGINT`) analytical product, target opportunity assessments, and cross-community cyber defence legal/policy frameworks (NSArchive GCHQ Cyber Defence Operations, NSArchive GCHQ Juniper Assessment).	GCHQ, MOD Defence Intelligence (DI), MI5, SIS, Cabinet Office JIO; maximum classification permitted on Five Eyes `GCWiki` (`TOP SECRET STRAP1 COMINT`) (NSArchive GCHQ Cyber Defence Operations).	Closed registered files must never be destroyed locally or sent to MOD Main Archive at Swadlincote; must be transferred to **MOD Sensitive Archive in Portsmouth** via DBS KI Records Review team (MOD JSP 441 Records Management Policy).
STRAP 2	`TOP SECRET STRAP 2` / `TOP SECRET STRAP 2 COMINT`	Tightly restricted operational compartment protecting active collection methods, Computer Network Exploitation (`CNE`) toolsuites (`HIGHNOTE`, `COPPERHEAD`), `QUANTUM INSERT` (`QI`) packet-injection infrastructure, telecoms/GRX operator targeting (`OP SOCIALIST`, `OP WYLEKEY`, `OP INTERACTION`), and automated Network Operations Centre (`NOC`) detection (NSArchive GCHQ Mobile Networks in My NOC World, NSArchive GCHQ Automated NOC Detection).	GCHQ Network Analysis Centre (NAC) at Cheltenham, Bude, Scarborough, and Cyprus; `MyNOC` collaborative cells integrating `NEXUS BSS Desktop` (MI5) and `CADDIS SIS Desktop` (MI6) (NSArchive GCHQ Mobile Networks in My NOC World).	Highest tier permitted for transfer to MOD Sensitive Archive in Portsmouth; subject to 15-year review by DBS KI Records Review team in accordance with `JSP 440 Part 2 Leaflet 4C` and `Leaflet 4G`.
STRAP 3	`TOP SECRET STRAP 3` (and named sub-compartments /	Apex compartmented tier covering exceptionally fragile technical collection capabilities, deep-cover strategic cryptanalytic	GCHQ core compartments, SIS, MI5, and named indoctrinated post-	**Prohibited from transfer to MOD Archives** (MOD JSP 441 Records Management Policy).

STRAP Tier	Canonical Marking Syntax	Operational & Technical Scope	Primary Custodians & Systems	Archival & Succession Rules (JSP 441 / JSP 440)
	codewords above STRAP 2)	breakthroughs, sovereign submarine cable tap architectures, and combined HUMINT/SIGINT source identities whose compromise would permanently terminate strategic access (The British Classification Marking STRAP, MOD JSP 441 Records Management Policy).	holders in MOD and Cabinet Office.	Must be returned to originator or destroyed per originator instruction; **successors in post have zero automatic right of access** without explicit originator re-authorisation (MOD JSP 441 Records Management Policy).

Primary-source GCHQ operational records preserved in the National Security Archive at George Washington University illustrate the exact functional demarcation between **`STRAP 1`** and **`STRAP 2`**. On GCHQ's internal collaborative wiki (**`GCWiki`**), the banner header explicitly states:

> *"TOP SECRET STRAP1 COMINT — The maximum classification allowed on GCWiki is TOP SECRET STRAP1 COMINT." (NSArchive GCHQ Cyber Defence Operations Legal and Policy)*

Documents at the **`TOP SECRET STRAP 1`** level include GCHQ's *Cyber Defence Operations Legal and Policy* manual (generated by Cyber Defence Operations, formerly the Network Defence Intelligence Support Team **`NDIST`** within **`MISD1`**, dated 1 January 2011), which governs how SIGINT content and Information Assurance (**`IA`**) data are integrated in **`XKEYSCORE`**, how Combined Policy Authorisations (**`COPA`**) and **`MIRANDA`** numbers regulate queries, and how the **`ANXIOUS`** methodology searches for known threats on UK networks (NSArchive GCHQ Cyber Defence Operations Legal and Policy). Similarly, GCHQ's *Assessment of Intelligence Opportunity - Juniper* (dated 3 February 2011) (NSArchive GCHQ Juniper Assessment) and the **`OPC-MCR`** briefing *A potential technique to deanonymise users of the TOR network* (dated 6 June 2011) (NSArchive GCHQ TOR Deanonymisation Briefing) are classified at **`TOP SECRET STRAP 1`** and **`TOP SECRET STRAP 1 COMINT`**.

By contrast, when GCHQ documents transition from general analytical opportunity or legal policy to specific offensive Computer Network Exploitation (**`CNE`**) implants, target network topologies, and active interception bearers, the classification escalates to **`TOP SECRET STRAP 2`**. The 2011 presentation by the Head of GCHQ's Network Analysis Centre (**`NAC`**), *Mobile Networks in My NOC World* (NSArchive GCHQ Mobile Networks in My NOC World), and the GCHQ NAC presentation *Automated NOC Detection* (responding to the 2009 Five Eyes SIGINT Development Conference) (NSArchive GCHQ Automated NOC Detection) carry the **`TOP SECRET STRAP 2`** marking on every slide, accompanied by the statutory handling caveat:

> *"This information is exempt from disclosure under the Freedom of Information Act 2000 and may be subject to exemption under other UK information legislation. Refer disclosure requests to GCHQ." (NSArchive GCHQ Mobile Networks in My NOC World)*

1.3 MOD JSP 441 Statutory Rules for Material Above STRAP 2 (STRAP 3 and Codeword)

The most stringent legal and operational boundary in the entire STRAP system lies between **STRAP 2** and **material classified above STRAP 2** (i.e., **STRAP 3** and special compartmented codewords). In the official MOD JSP 441 Defence Records Management Policy (Updated 10 December 2024), under the section

Handling Top Secret, STRAP and Codeword records, the Ministry of Defence promulgates three mandatory rules for material classified above **`STRAP 2`**:

1. **Absolute Prohibition on Archival Transfer**: *"Material classified above STRAP 2 MUST NOT be sent to the MOD Archives."* (MOD JSP 441 Records Management Policy)
2. **Mandatory Originator Referral Upon Cessation of Need**: *"Where a team holding material classified above STRAP 2 no longer requires it for ongoing business reasons, they MUST take steps to refer to the originator of the material to determine if it should be returned [to the originator] or destroyed."* (MOD JSP 441 Records Management Policy)
3. **Non-Heritability Across Post Successors**: *"A successor to the post handling material classified above STRAP 2, does not have an automatic right of access to that material held by their predecessor without the specific permission of the originator. Where an individual holding material classified above STRAP 2 is leaving their post, they MUST take steps to refer to the originator of the material to determine if it should be returned [to the originator], resubmitted to the successor or destroyed."* (MOD JSP 441 Records Management Policy)

This rule proves that while **`STRAP 1`** and **`STRAP 2`** clearances are tied to designated roles on a unit's **`STRAP Matrix`** (MOD CIO-DSAS STRAP Incident Database & Asset Register), access to **`STRAP 3`** and higher codewords is strictly personal and originator-bound (**`named-individual indoctrination`**), extinguishing automatically when the individual vacates the post.

1.4 Historical Origins and Mapping to US/Five Eyes SCI, UKUSA, and ECHELON Compartments

The institutional lineage of the STRAP system traces directly to the Anglo-American cryptographic and signals intelligence pacts of the Second World War and early Cold War, officially declassified by GCHQ and The National Archives (TNA Series HW 80) in June 2010:

- **January–May 1941 (The Sinkov Mission and Denniston-Friedman Accords)**: US Army and Navy cryptanalysts led by Captain Abraham Sinkov visited the Government Code and Cypher School (**`GC&CS`**) at Bletchley Park in January 1941, exchanging a reconstructed Japanese Purple cipher machine for technical disclosure of Alan Turing's Enigma-breaking **`Bombe`** (GCHQ Brief History of the UKUSA Agreement). Commander Alastair Denniston (Operational Head of **`GC&CS`**) met US Chief Cryptanalyst William Friedman in Washington between March and May 1941 to establish secure bilateral cryptographic handling channels (GCHQ Brief History of the UKUSA Agreement).
- **17 May 1943 (The BRUSA Agreement)**: Signed between the US War Department and **`GC&CS`**, the **`BRUSA`** pact codified joint regulations for the handling, compartmentalization (**`ULTRA`** / **`BONIFACE`**), and distribution of decrypted Axis communications intelligence, establishing the principle that signals intelligence required special handling distinct from standard military classifications (GCHQ Brief History of the UKUSA Agreement, TNA Intelligence and Security Services Guide).
- **5 March 1946 (The UKUSA Agreement)**: Following negotiations in Washington between September and November 1945 by Sir Edward Travis (Director of **`GC&CS`** / **`GCHQ`**), Harry Hinsley, and Brigadier John Tiltman, the formal **British-U.S. Communication Intelligence Agreement (`UKUSA`)** was signed on 5 March 1946 by Colonel Patrick Marr-Johnson (Senior UK Liaison Officer in Washington, acting for the **London Signals Intelligence Board**) and Lieutenant General Hoyt Vandenberg (Chairman of the **US State-Army-Navy Communications Intelligence Board, `STANCIB`**) (GCHQ Brief History of the UKUSA Agreement). Canada acceded in 1949, followed by Australia and New Zealand in 1956, completing the **Five Eyes (`FVEY`)** alliance (GCHQ Brief History of the UKUSA Agreement).

- **TNA HW 80 and HW 75 Declassified Appendices**: Declassified files in TNA HW 80/11 (Amendment No. 4 to the Appendices to the UKUSA Agreement, Third Edition) and TNA UKUSA Highlights Guide (HW 75/178–195) document how the UKUSA Appendices mandated uniform security indoctrination, physical vault standards, and codeword compartments (**COMINT** / **SPECIAL INTELLIGENCE** / **SI** and **TK** / **TALENT KEYHOLE**) across all five partner nations.
- **Cold War ECHELON and Modern SCI Equivalency**: As documented by the House of Commons Foreign Affairs Committee Report on Global Security: UK-US Relations (HC 114, para. 105) and the European Parliament Temporary Committee on the ECHELON Interception System (PE 166.499 / PE 168.184), UKUSA partners integrated global satellite and cable collection through the **ECHELON** dictionary-routing architecture (operating at GCHQ Bude / CSO Morwenstow, JSSU Ayios Nikolaos in Cyprus, and RAF Menwith Hill). When the United States consolidated its intelligence handling markings under the Director of Central Intelligence Directive (**DCID**) / Intelligence Community Directive (**ICD 710**) **Sensitive Compartmented Information (SCI)** framework, the United Kingdom instituted the **STRAP** codeword system to replace legacy UK codeword proliferation with a unified three-tier national system that maps directly onto US/Five Eyes **SCI** control systems (The British Classification Marking STRAP, NSArchive GCHQ Cyber Defence Operations Legal and Policy).

UK Classification & STRAP Marking	US / Five Eyes (FVEY) Equivalent Marking	UK Vetting & Indoctrination Prerequisite	Technical & Operational Equivalency
UK TOP SECRET STRAP 1 COMINT	**TOP SECRET//SI//REL TO USA, FVEY** (**Special Intelligence** / **COMINT**)	Developed Vetting (DV) + **STRAP 1** Indoctrination	Standard derived SIGINT/COMINT reporting and cross-Five Eyes analytical sharing on **GCWiki** / **INTELINK-TS** (NSArchive GCHQ Cyber Defence Operations).
UK TOP SECRET STRAP 2	**TOP SECRET//SI-GAMMA** or **TOP SECRET//SI//TK//HCS** (Operational CNE / Special Source SCI)	Developed Vetting (DV) + **STRAP 2** Indoctrination	Sensitive collection methodology, CNE implants (**HIGHNOTE**, **COPPERHEAD**, **QUANTUM**), and telecom backbone (**GRX**) exploitation (NSArchive GCHQ Mobile Networks in My NOC World).
UK TOP SECRET STRAP 3	**TOP SECRET//SCI//[ECI / Special Access Program / Reserved Sub-Compartment]**	Developed Vetting (DV) or Enhanced DV (eDV) + Originator-Approved **STRAP 3** Indoctrination	Exceptionally Compartmented Information (**ECI**), strategic cryptographic/cable access, and named-individual originator-controlled compartments (MOD JSP 441 Records Management Policy).
UK SECRET STRAP 1	**SECRET//SI//REL TO USA, FVEY** (Tactical/Collateral SIGINT)	Enhanced Security Check (eSC) or Developed Vetting (DV) + **STRAP 1** Briefing	Tactical military and law-enforcement intelligence action-on product requiring codeword protection at the **SECRET** tier (GSCP Guidance 1.2 - Working at SECRET, UKSV Clearance Levels).

Section 2: Governance, Policy Directives, STRAPSO Authority, and Physical/Electronic Security Architecture

2.1 Joint Service Publication 440 (`JSP 440`) and Related Defence Directives

Within the Ministry of Defence and the British Armed Forces, protective security and STRAP implementation are governed by **Joint Service Publication 440 (`JSP 440`): The Defence Manual of Security** (MOD FOI2015/02999 Information Security Policies, MOD Contact with the Media Policy). Official MOD disclosures and cross-references in JSP 441 (Defence Records Management Policy), The King's Regulations for the Royal Navy (April 2026), and House of Commons Defence Committee Evidence confirm the exact internal architecture of **`JSP 440`** and its companion directives:

- **`JSP 440` Evolution (`Issues 3.1 to Issue 5+`)**: Promulgated by the MOD Defence Security authority, **`JSP 440`** sets mandatory security standards across all Service Establishments, Top Level Budgets (**`TLBs`**), and Defence contractors (MOD FOI2015/02999 Information Security Policies).
- **`JSP 440 Part 2, Leaflet 4B` (*Classification and Marking*)**: Defines who is permitted access to classified, **`STRAP`**, and codeword records, under what circumstances, and specifies the protective marking syntax required on physical and electronic records (MOD JSP 441 Records Management Policy).
- **`JSP 440 Part 2, Leaflet 4C` (*Information Assets and Handling*) & `Part 4, Leaflet 4C`**: Establishes mandatory registration, muster, accounting, and secure destruction rules for **`TOP SECRET`**, **`STRAP`**, and codeword assets (MOD JSP 441 Records Management Policy).
- **`JSP 440 Part 2, Leaflet 4G` (*Special Information Control Systems*)**: Codifies the specific rules for **Special Information Control Systems**—including the **STRAP System** and Codeword compartments—governing Departmental Record Officer (**`DRO`**) approved Electronic Records Management Systems (**`ERMS`**) handling **`TOP SECRET`** and **`STRAP`** material, local destruction of unpublished working drafts, and compartmented registry controls (MOD JSP 441 Records Management Policy).
- **`JSP 440 Part 4, Section 1, Chapter 1`**: Governs the alignment of Defence protective markings with the Cabinet Office Government Security Classifications Policy (**`GSCP`**) (King's Regulations for the Royal Navy 2026).
- **`JSP 440 Part 5, Section 3, Chapter 1`**: Governs the movement and transmission of sensitive records alongside **`JSP 367`** (*Defence Postal and Courier Services*, Chapter 4, para. 14) (MOD Defence Health Record FOI05218).
- **`JSP 440 Part 7`**: Prescribes physical security, establishment standing orders, and counter-terrorist protective security measures across Defence sites (House of Commons Defence Committee Written Evidence).
- **Companion Technical Directives**: **`JSP 490`** (*Defence Cryptosecurity Operating Instructions*), **`JSP 604`** (*Defence Manual of Information and Communications Technology Security / Network Accreditation*), and **`JSP 465`** (*Defence Geospatial Information Policy*, governing all satellite, airborne, ground-based, and sea-borne intelligence imagery) operate in lockstep with **`JSP 440`** (MOD FOI2015/02999 Information Security Policies, MOD JSP 441 Records Management Policy, G-Cloud RLI Enterprise Collaboration Service).

In the MOD's paper registry architecture mandated by JSP 441, registered files must use purpose-designed file covers ordered from the MOD Millie portal:

- **`MOD Form 329A` (Red Cover)**: Mandatory for **`TOP SECRET`** (including **`TOP SECRET STRAP`**) registered files (MOD JSP 441 Records Management Policy).

- **`MOD Form 329B` (Pink Cover)**: Mandatory for **`SECRET`** registered files (MOD JSP 441 Records Management Policy).
- **`MOD Form 329D` (Buff Cover)**: Mandatory for **`OFFICIAL`** registered files (MOD JSP 441 Records Management Policy).
- **`MOD Form 262A` & `MOD Form 262F`**: **`MOD Form 262A`** (*Registered File Record Sheet*) records the creation of every file part, while **`MOD Form 262F`** (*Registered File Disposal Form*) records closure and review decisions. Crucially, under JSP 441, **Intelligence and Security Agency End Product** (reporting created on Agency End Product publishing systems for wider distribution) **must never be declared as a MOD record**, as legal custodianship remains permanently with the originating Agency (**`GCHQ`**, **`SIS`**, or **`MI5`**) under Section 3(4) of the Public Records Act 1958 (MOD JSP 441 Records Management Policy, TNA Intelligence and Security Services Guide).

2.2 Cabinet Office Government Security Classifications Policy (`GSCP`)

Originally reformed in October 2013 (implemented April 2014 under Cabinet Office Minister Francis Maude to replace the six-tier **`UNCLASSIFIED / PROTECT / RESTRICTED / CONFIDENTIAL / SECRET / TOP SECRET`** model with three tiers: **`OFFICIAL`**, **`SECRET`**, and **`TOP SECRET`**) and substantially updated on **30 June 2023** and **5 August 2024** (GSCP Overview, GSCP Guidance 1.5), the Government Security Classifications Policy (GSCP) establishes the cross-government administrative baseline. Under GSCP Guidance 1.5 (*Considerations for Security Advisors*), four central authorities govern UK classification policy:

1. **Government Chief Security Officer (`GCSO`, Christina Duncan)**: Sets the mandatory **`GSCP`** baseline across HM Government, manages pan-government information security risk, and approves any departmental variance below the **`SECRET`** or **`TOP SECRET`** baseline (GSCP Guidance 1.5, Cabinet Office Organisation).
2. **Government Security Board (`GSB`)**: Advises the **`GCSO`** on setting the **`GSCP`** baseline and managing pan-government security risk (GSCP Guidance 1.5).
3. **Government Security Group (`GSG`) Information Security Policy Team**: Provides operational classification guidance to departments and escalates pan-government risks to the **`GCSO`** (GSCP Guidance 1.5).
4. **UK National Security Authority (`UK NSA`)**: Sits within the **`GSG`** in the Cabinet Office and holds ultimate statutory and treaty responsibility for the protection of classified information exchanged internationally (including bilateral and multilateral exchanges requiring the **`UK`** prefix, such as **`UK TOP SECRET`** and **`UK SECRET`**) (GSCP Guidance 1.5, GSCP Guidance 1.3 - Working at TOP SECRET).

Under GSCP Guidance 1.3 (*Working at TOP SECRET*, Updated 5 August 2024), HM Government codifies the baseline physical, procedural, and electronic behaviours for **`TOP SECRET`** (Tier 3), which serve as the foundation onto which **`STRAP`** controls are added:

- **Strict Prohibition on Home/Remote Working**: *"Remote working from home at TOP SECRET is not permitted. Do not take TOP SECRET information home under any circumstances."* (GSCP Guidance 1.3 - Working at TOP SECRET)
- **Paper Colour Coding, Copy Numbering, and Tethering**: Hardcopy **`TOP SECRET`** documents should be printed on **yellow paper** (whereas **`SECRET`** documents use **pink paper** under Guidance 1.2), must have all pages tethered together, must bear an explicit copy number on the top right corner of the first page (e.g., **`1 of 3`**), and must be logged with their unique reference number in the **Protected Document Registry book** (GSCP Guidance 1.3 - Working at TOP SECRET).
- **Two-Person Integrity (`Two DV-Cleared Escorts`) in Transit and Destruction**: Moving physical **`TOP SECRET`** assets by hand outside a specified Government Secure Zone requires a formal Threat and Vulnerability Risk Assessment, written Senior Civil Service (**`SCS1`**) and Senior Security Advisor (**`SSA`**/**`SA`**) approval, and

two people (both cleared to Developed Vetting DV) to escort the assets inside an opaque, locked security container and NPSA-approved tamper-evident double envelopes (marked **TOP SECRET** on the inner envelope only, with an immediate-return delivery receipt) (GSCP Guidance 1.3 - Working at TOP SECRET). Domestic commercial couriers and postal services are strictly prohibited; only an **SA/SSA**-approved secure mail delivery service with two **DV**-cleared escorts, diplomatic bag, or military courier (**Defence Courier Service**) is permitted (GSCP Guidance 1.3 - Working at TOP SECRET). Destruction of hardcopy **TOP SECRET** requires written approval from the information creator, an NPSA-approved shredder from the Catalogue of Security Equipment (**CSE**), and **two-person integrity**—the person destroying the document plus a **DV-cleared witness**, both of whom must sign the Protected Document Registry book (GSCP Guidance 1.3 - Working at TOP SECRET).

2.3 Role, Legal Authority, and Databases of the STRAP Security Officer (STRAPSO)

Across Whitehall departments, the Armed Forces, and intelligence consumers, the operational enforcement of STRAP rules is vested in appointed **STRAP Security Officers (STRAPSOs)**, overseen centrally by the **STRAP Management Board** (which reviews and formulates national STRAP policy and guidelines) (The British Classification Marking STRAP, An Overview of UK Security Vetting and STRAP).

Official government documentation from the Cabinet Office Security & Resilience (COSR) Team and the Ministry of Defence CIO-DSAS Information Asset Register defines the exact institutional functions and administrative databases maintained by **STRAPSO** teams:

- **Cabinet Office STRAP Team (STRAPSO)**: Operating within the Cabinet Office Security & Resilience (**COSR**) Directorate alongside the Vetting Team and Incident Management Team, the Cabinet Office **STRAPSO** team *"regulates and monitors access to highly sensitive intelligence material that requires specialised protective handling beyond standard government arrangements"* (Cabinet Office EO Security Support Officer Job Specification).
- **MOD CIO-DSAS STRAP Team and Official Databases**: Within the Ministry of Defence, the central STRAP authority sits in **Chief Information Officer – Defence Security and Assurance Services (CIO-DSAS)**, which maintains four dedicated statutory databases registered in the UK National Data Library (MOD CIO-DSAS STRAP Incident Database, National Data Library CIO-DSAS Directory):
- **CIO-DSAS STRAPSO Listing**: The master register of all accredited STRAP Security Officers appointed across MOD Top Level Budgets (**TLBs**), military commands, and defence establishments (National Data Library CIO-DSAS Directory).
- **CIO-DSAS STRAP Matrix**: The post-by-post authorization matrix defining which specific billets within each military command or Whitehall directorate are approved to hold **STRAP 1**, **STRAP 2**, or **STRAP 3** access (National Data Library CIO-DSAS Directory).
- **CIO-DSAS STRAP Location Database**: The physical accreditation register of all certified STRAP facilities (**STRAP Rooms** / SCIFs) and secure registries authorized to receive, store, and process STRAP material via **STRAP Channels** (National Data Library CIO-DSAS Directory).
- **CIO-DSAS STRAP Incident Database**: Serial 5 of the **CIO-DSAS** Information Asset Register, recording all STRAP security breaches, muster discrepancies, unauthorized disclosures, and containment investigations across the Defence estate (MOD CIO-DSAS STRAP Incident Database).

2.4 Physical Security (SCIF Equivalents / STRAP Rooms, Acoustic & TEMPEST Controls) and the Tri-NTA Architecture

To process **TOP SECRET** and **STRAP** material, physical compartments (**STRAP Rooms**, equivalent to US Sensitive Compartmented Information Facilities or **SCIFs**) must be accredited under the standards of the United

Kingdom's **three National Technical Authorities (`NTAs`)** (UK NACE Official Portal):

1. **National Cyber Security Centre (`NCSC`, part of GCHQ) — NTA for Cyber Security, Information Assurance, and `TEMPEST` / Electromagnetic Security (`EMS`)**:
2. The **`NCSC`** (incorporating the legacy functions of **`CESG`**, the Communications-Electronics Security Group) serves as the UK National Technical Authority for **Electromagnetic Security (`EMS`) and `TEMPEST`** (NCSC TEMPEST and Electromagnetic Security).
3. As defined by the NCSC, **`TEMPEST`** relates purely to passive unintentional electromagnetic emanations from ICT equipment that could betray classified data, whereas **`EMS`** encompasses active techniques used to enhance or extract signals of interest (NCSC TEMPEST and Electromagnetic Security).
4. All ICT hardware and mobile platforms deployed in **`TOP SECRET`** and **`STRAP`** environments must be certified through NCSC-accredited test facilities under the **Formal TEMPEST Certification Scheme (`CFTCS`)** for products (NCSC CFTCS Scheme) and the **Platform TEMPEST Accreditation Scheme (`CPTAS`)** for mobile platforms, utilizing assured UK/NATO vendors such as L3Harris TRL Technology Limited.
5. Data at rest on non-physically secure devices at **`TOP SECRET`** requires NCSC **High Grade encryption** (and revitalised **Enhanced Grade encryption** at **`SECRET`**), with strict **Crypto Ignition Key (`CIK`)** removal and physical separation whenever an operator leaves a terminal (GSCP Guidance 1.2 - Working at SECRET, GSCP Guidance 1.5).
6. **UK National Authority for Counter-Eavesdropping (`UK NACE`, part of FCDO Services at Hanslope Park) — NTA for Technical Security**:
7. Operating since 1945 out of Hanslope Park, Milton Keynes (MK19 7BH) as part of FCDO Services, **UK NACE** is the UK's National Technical Authority for technical security and counter-eavesdropping (UK NACE Official Portal).
8. Under GSCP Guidance 1.3 (*Working at TOP SECRET*), departments must consult **UK NACE** to assure **`TOP SECRET`** and **`STRAP`** rooms against audio leakage, acoustic eavesdropping, and technical penetration, and must conduct periodic **Technical Surveillance Counter-Measures (`TSCM` / counter-eavesdropping) sweeps** (GSCP Guidance 1.3 - Working at TOP SECRET). UK NACE also delivers practitioner-level **`TEMPEST`** and technical security training on behalf of NCSC (NCSC TEMPEST and Electromagnetic Security, UK NACE Official Portal).
9. **National Protective Security Authority (`NPSA`, part of MI5, formerly CPNI) — NTA for Physical and Personnel Security**:
10. The **NPSA** sets physical security standards for **`STRAP`** and **`TOP SECRET`** rooms, maintaining the **Catalogue of Security Equipment (`CSE`)** (NPSA Catalogue of Security Equipment, NPSA CSE Products).
11. The **`CSE`** specifies evaluated ratings for **Security Containers** (Class 1/2/3 safes and cabinets for **`TOP SECRET`** and **`STRAP`** storage), **Access Control Equipment**, **Doors, Portals, Turnstiles and Locking Hardware**, **Walls and Enclosures** (including acoustic attenuation barriers such as 8.5-metre acoustic walls), **Tamper Indication**, and **Secure Destruction Equipment** (NPSA Catalogue of Security Equipment, NPSA CSE Products).
12. Within **`TOP SECRET`** and **`STRAP`** areas, all personal and corporate mobile phones, wearables (smart watches), and smart listening devices are strictly prohibited, and all incoming furniture or equipment must be sourced through approved channels and inspected for tampering (GSCP Guidance 1.3 - Working at TOP SECRET).

2.5 Accredited Classified Electronic Handling Networks (RLI, MODNET Secret, ROSA, OSE / UKROS, SII, MyNOC)

Electronic processing and transmission of classified and compartmented intelligence across the UK government and military estate occur over strictly segregated, air-gapped or cross-domain-guarded networks accredited under **JSP 604** and the **GSCP** (GSCP Guidance 1.5, G-Cloud RLI Enterprise Collaboration Service):

Network Designation	Classification Tier	Custodial / Operating Authority	Architectural Function & Technical Interconnectivity
RLI / ALI (*Restricted LAN Interconnect*)	**OFFICIAL-SENSITIVE** (Legacy **RESTRICTED**)	Ministry of Defence (MOD) / Defence Digital	Wide-area defence network interconnecting MOD and defence industrial contractor sites (**BMT**, **Babcock**) under **JSP 440** and **JSP 604** accreditation (G-Cloud RLI Hosting).
ROSA (*Cross-Government SECRET Capability*)	Up to and at **SECRET** (**UK SECRET**)	Cabinet Office Government Security Group (GSG) / Rosa Service Organisation	Deployed across **over 40 UK Government Departments and Agencies** (including FCDO, Cabinet Office, Home Office); hosted in a secure **Crown Hosting Data Centre**; Senior Responsible Owner (**SRO**) is the Cabinet Office Director of Programmes and Capabilities reporting to the **GCSO** (Foreign Affairs Committee HC 2681, GSCP Guidance 1.2).
MODNET Secret / DII(S) (*Defence Information Infrastructure Secret*)	**SECRET** (**UK SECRET**)	Ministry of Defence (MOD)	Core military secret command-and-control network upgraded under the **MODNET EVOLVE** programme, replacing legacy **DII(S)** terminals across UK and deployed headquarters (MODNET EVOLVE Update, PAC DII Report HC 100).
UKROS / OSE (*UK Restricted/Secret Overseas System / Overseas Secret Environment*)	**SECRET** / Diplomatic & Intelligence Compartments	FCDO / FCDO Services	High-assurance overseas diplomatic and secret communications infrastructure linking UK Embassies/High Commissions to Whitehall, converged with **ROSA** and high-grade cryptographic links (GSCP Guidance 1.2 - Working at SECRET, Foreign Affairs Committee HC 2681).
SII (*Secret Intelligence Interconnect / Top Secret Systems*)	**TOP SECRET** / **TOP SECRET STRAP** (**DII(AS)** / Above Secret)	MOD Defence Intelligence (DI) / GCHQ	High-assurance **TOP SECRET STRAP** intelligence network (**DIFG SII SY**) linking Defence Intelligence nodes (RAF Wyton, RAF Molesworth, Northwood PJHQ, Chicksands) with UKIC (**GCHQ**, **MI5**, **SIS**, **JTAC**, **Cabinet Office JIO**) and Five Eyes **SCI** gateways.
MyNOC / Agency Desktops (NEXUS, CADDIS, GCWiki)	**TOP SECRET STRAP 1** & **TOP SECRET STRAP 2**	GCHQ, MI5 (**BSS**), SIS	Compartmented multi-agency operational environment integrating **GCHQ Analyst Desktops**, **JTRIG Desktop**, **HIGHNOTE** CNE Toolsuite, **COPPERHEAD** CNE Attack Box, **MI5 NEXUS BSS Desktop**, and **MI6 CADDIS SIS Desktop** (NSArchive GCHQ Mobile Networks in My NOC World).

Section 3: Institutional Hierarchy and Principal UK Intelligence Agencies

3.1 Government Communications Headquarters (GCHQ) and the National Cyber Security Centre (NCSC)

- **Statutory Basis**: Governed by the **Intelligence Services Act 1994 (1994 c. 13, Sections 3–7)**, which placed **GCHQ** on a statutory footing under the ministerial responsibility of the **Secretary of State for Foreign, Commonwealth and Development Affairs (Foreign Secretary, Rt Hon David Lammy MP)** (ISA 1994 Section 5, SIS Chief Appointment Press Release). Under Section 3(1)(a) of the 1994 Act, GCHQ's statutory functions are to monitor or interfere with electromagnetic, acoustic, and other emissions and any equipment producing such emissions, to obtain and provide information derived from or related to such emissions or equipment and from encrypted material, and to provide advice and assistance on cryptology and information assurance to HM Government and the Armed Forces (Intelligence Services Act 1994). Operational warrants for targeted interception, equipment interference (**CNE**), bulk interception (Part 6, Chapter 1), bulk acquisition (Part 6, Chapter 2), bulk equipment interference (Part 6, Chapter 3), and bulk personal datasets (**BPDs**, Part 7) are governed by the **Investigatory Powers Act 2016 (2016 c. 25, Sections 102–105 & Parts 2, 5, 6, 7)**, requiring a "double-lock" personal authorisation by the Secretary of State and approval by a Judicial Commissioner of the Investigatory Powers Commissioner's Office (**IPCO**) (Investigatory Powers Act 2016 PDF).
- **Executive Leadership**: Led by **Director GCHQ Anne Keast-Butler**, who assumed office in May 2023 succeeding Sir Jeremy Fleming (GCHQ Leadership, New Director GCHQ Announced). Director GCHQ is supported by the GCHQ Board and **five Directors General** (GCHQ Leadership):
- **Director General Intelligence & Effects** (overseeing operations across all five mission areas: Counter-Terrorism, Cyber Security, Serious and Organised Crime, Strategic Advantage, and Support to Defence);
- **Director General Technology** (technical capabilities and R&D innovation);
- **Director General Strategy** (overarching strategy, legal and policy compliance, HR, and Finance);
- **Director General IT** (mission-enabling enterprise and classified cloud technologies);
- **Chief Executive Officer of the National Cyber Security Centre (CEO NCSC)**: **Dr Richard Horne**, who assumed office in October 2024 (succeeding Lindy Cameron) and leads the **NCSC** as the UK's National Technical Authority for cyber security, information assurance, and **TEMPEST**/**EMS** (NCSC Richard Horne Profile, NCSC Annual Review 2026 Speech, What We Do at the NCSC).
- **Primary Collection & Operational Sites**:
- **GCHQ Cheltenham (The Doughnut, Hubble Road, Benhall, Cheltenham, Gloucestershire)**: Primary headquarters housing the Director, the five DGs, the Network Analysis Centre (**NAC Cheltenham**), the Government Threat Analysis Centre (**GTAC**), Global Telecoms Exploitation (**GTE**), Cryptographic Development (**Crypt SD**), and **MyNOC** facilities (GCHQ Home, NSArchive GCHQ Mobile Networks in My NOC World).
- **GCHQ Bude (Composite Signals Organisation Station Morwenstow, Sharpnose Point, North Cornwall)**: Operationalized on **16 September 1974** as the world's first communications satellite (**COMSAT**) interception station with two 97-foot antennas; expanded by its 50th anniversary in September 2024 to **30 satellite and communications antennas** perched 200 feet up on the cliffs of Sharpnose Point (GCHQ Bude 50th Year of Operation, GCHQ Bude Location). Houses **NAC Bude**, **JTRIG Analysts Bude**, and transatlantic submarine optical fibre cable landing interception (**Tempora** / bulk interception bearers) (GCHQ Bude 50th Year of Operation, NSArchive GCHQ Mobile Networks in My NOC World, ISC Privacy and Security Report 2015).

- **GCHQ Scarborough (Irton Moor, North Yorkshire)**: Outstation tracing its lineage to WWII Royal Navy Y-Station intercept operations; houses **`Scarborough CNE`** (Computer Network Exploitation operators) and core SIGINT engineering capabilities (About GCHQ Scarborough, GCHQ Scarborough Location, NSArchive GCHQ Mobile Networks in My NOC World).
- **Cyprus — Joint Service Signal Unit (`JSSU Cyprus`, Ayios Nikolaos Station, Eastern Sovereign Base Area)**: Formed in 2003 through the amalgamation of **`JSSU (Ayios Nikolaos)`** and elements of **`12 Signal Unit RAF`**, with detachments at **RAF Akrotiri**, **Episkopi** (Western Sovereign Base Area), and the **Retained Site at Mount Troodos** (MOD Guidance: Ayios Nikolaos, Cyprus, Foreign Affairs Committee HC 114, para. 105). Commanded by a Royal Signals Lieutenant Colonel with tri-service personnel (1 Squadron Operations, 2 Squadron Operational Engineering Support, HQ Squadron) and GCHQ civilian specialists (**`NAC Cyprus`**), **`JSSU Cyprus`** serves as the UK's primary eastern Mediterranean SIGINT hub covering the Middle East, North Africa, and Eurasia (MOD Guidance: Ayios Nikolaos, Cyprus, NSArchive GCHQ Mobile Networks in My NOC World).
- **GCHQ Manchester & London**: Regional innovation, cyber, and counter-terrorism hub in Manchester (Heron House) and London operational/NCSC presence (GCHQ Locations, ISC GCHQ Accommodation Procurement Case Study HC 991).

3.2 The Security Service (`MI5`)

- **Statutory Basis**: Governed by the **Security Service Act 1989 (1989 c. 5)**, operating under the statutory authority of the **Secretary of State for the Home Department (Home Secretary)** (TNA Intelligence and Security Services Guide). Under Section 1 of the 1989 Act, MI5's functions are the protection of national security against threats from espionage, terrorism, and sabotage, from the activities of agents of foreign powers, and from actions intended to overthrow or undermine parliamentary democracy by political, industrial, or violent means, as well as safeguarding the economic well-being of the UK against external threats and supporting law enforcement in the prevention and detection of serious crime (Security Service Act 1989). Warranted intrusive powers are exercised under the Intelligence Services Act 1994 (Section 5) and the Investigatory Powers Act 2016.
- **Executive Leadership and Headquarters**: Headquartered at **Thames House** (11–12 Millbank, London), MI5 is controlled under Section 2(1) of the Security Service Act 1989 by the **Director General of the Security Service, Sir Ken McCallum** (appointed March 2020) (GOV.UK Ken McCallum Profile, MI5 Director General Page).
- **Role in the STRAP Ecosystem**: Under Section 2(2)(a) of the 1989 Act, the Director General is statutorily responsible for ensuring that arrangements are in force so that no information is obtained or disclosed by the Service except so far as necessary for the proper discharge of its functions (Security Service Act 1989 Section 2, ISA 1994 Section 5(2)(c)). MI5 consumes GCHQ **`STRAP 1`** and **`STRAP 2`** SIGINT reporting via its **`NEXUS BSS Desktop`** terminals (where **`BSS`** denotes **`British Security Service`** in Five Eyes taxonomy) (NSArchive GCHQ Mobile Networks in My NOC World), hosts the multi-agency **Joint Terrorism Analysis Centre (`JTAC`)** at Thames House (Government Response to the Butler Review Cm 6492, para. 7), houses the **National Protective Security Authority (`NPSA`)** which accredits physical security containers and acoustic walls for **`STRAP`** rooms (NPSA Physical Security), and executes mandatory **Security Service (`MI5`) records checks** on every applicant for **`CTC`**, **`SC`**, **`eSC`**, **`DV`**, and **`eDV`** clearance (UKSV Clearance Levels).

3.3 The Secret Intelligence Service (`SIS` / `MI6`)

- **Statutory Basis**: Placed on a statutory footing alongside GCHQ by the **Intelligence Services Act 1994 (1994 c. 13, Sections 1–2 & 5–7)**, accountable to the **Foreign Secretary** (SIS About Us). Under Section 1(1) and 1(2) of the 1994 Act, the functions of SIS are to obtain and provide information relating to the

actions or intentions of persons outside the British Islands, and to perform other tasks relating to the actions or intentions of such persons, in the interests of national security (with particular reference to the defence and foreign policies of HM Government), in the interests of the economic well-being of the UK, or in support of the prevention or detection of serious crime (Intelligence Services Act 1994, SIS About Us). Section 7 of the 1994 Act empowers the Secretary of State to issue class or specific authorisations removing civil and criminal liability under UK law for acts done outside the British Islands (ISA 1994 Section 7).

- **Executive Leadership and Headquarters**: Headquartered at **Vauxhall Cross** (85 Albert Embankment, London), SIS is led under Section 2 of the 1994 Act by the **Chief of the Secret Intelligence Service (universally designated "C")**, the only serving member of the Service officially named in public (SIS Chief Appointment Press Release, GOV.UK Sir Richard Moore Profile). On **15 June 2025**, Prime Minister Sir Keir Starmer and Foreign Secretary David Lammy announced the appointment of **Blaise Metreweli CMG** as the **18th Chief of SIS ("C")** and the first woman to lead the Service, succeeding **Sir Richard Moore KCMG** (who served as Chief from autumn 2020 to autumn 2025) (SIS Chief Appointment Press Release). Prior to becoming **"C"**, Blaise Metreweli served as **Director General 'Q'** (Director General Technology and Innovation in MI6) and previously held a Director-level role in MI5 after joining SIS as a case officer in 1999 (SIS Chief Appointment Press Release).
- **Role in the STRAP Ecosystem**: SIS generates **CX** human intelligence (**HUMINT**) reporting and conducts joint covert technical operations with GCHQ, interfacing directly with **STRAP 2** CNE and SIGINT operations through the **CADDIS SIS Desktop** integrated inside GCHQ's **MyNOC** spaces (NSArchive GCHQ Mobile Networks in My NOC World).

3.4 Ministry of Defence (MOD), Defence Intelligence (DI), and Military Intelligence Services (MIS)

- **Statutory and Prerogative Basis**: Operating under the Royal Prerogative and the Defence Council, the Ministry of Defence (MOD) and **Defence Intelligence (DI)** exercise statutory powers under **Section 104 of the Investigatory Powers Act 2016** (*Power to issue warrants to the Chief of Defence Intelligence*), which empowers the Secretary of State for Defence to issue targeted equipment interference warrants personally to the **Chief of Defence Intelligence (CDI)** (Investigatory Powers Act 2016 Contents).
- **Executive Leadership and the 2025–2026 Structural Transformation**:
  - **Secretary of State for Defence**: **Rt Hon John Healey MP** (MOD CDI Appointment Press Release).
  - **Permanent Secretary of the Ministry of Defence**: **Jeremy Pocklington CB**, appointed on **31 October 2025** (succeeding David Williams) (MOD New Permanent Secretary Announced, GOV.UK Jeremy Pocklington Profile).
  - **Commander of Cyber & Specialist Operations Command (CSOC, formerly Strategic Command)**: **General Sir Jim Hockenhull KBE ADC Gen**, under whose command **Defence Intelligence** sits (MOD CDI Appointment Press Release, MOD Defence Intelligence Page).
  - **Establishment of Military Intelligence Services (MIS) and Defence Counter-Intelligence Unit (DCIU) (December 2025)**: Delivering on the **July 2025 Strategic Defence Review (SDR)** (Strategic Defence Review 2025), the MOD established **Military Intelligence Services (MIS)** in December 2025, unifying every intelligence unit and organisation across the Royal Navy, British Army, Royal Air Force, Civil Service, Permanent Joint Headquarters (**PJHQ** Northwood), and UK Space Command under a single command structure led by the **CDI**, alongside the newly launched **Defence Counter-Intelligence Unit (DCIU)** (created after hostile intelligence activity against the MOD rose by more than 50% in 2024–2025) (MOD CDI Appointment Press Release, Butler and Military Intelligence 2026 Study).

    - **Chief of Defence Intelligence (`CDI`)**: **Lieutenant-General Matthew Jones OBE**, whose appointment was approved by HM The King and announced on **5 February 2026** to take office in summer 2026, succeeding **Adrian Bird CB** (who served as the first civilian **`CDI`** from 2022 to July 2026 after holding two successive Director General posts at GCHQ) (MOD CDI Appointment Press Release, New Head of the JIO Appointed).

- **Primary Sites and `JSP 440` Enforcement**: **`DI`** and **`MIS`** operate out of **MOD Main Building (Whitehall)** and the **Defence Intelligence Fusion Centre (`DIFC`) at RAF Wyton** (Cambridgeshire), with key operational nodes at **RAF Molesworth**, **Northwood (`PJHQ`)**, and **Chicksands** (MOD DI Levant Analyst Specification). Within the MOD, **Defence Security and Resilience** and **`CIO-DSAS`** enforce **`JSP 440`**, maintaining the **`CIO-DSAS STRAPSO Listing`**, **`STRAP Matrix`**, **`STRAP Location Database`**, and **`STRAP Incident Database`** (MOD CIO-DSAS STRAP Incident Database).

3.5 Cabinet Office: Joint Intelligence Committee (`JIC`), Joint Intelligence Organisation (`JIO`), National Security Secretariat (`NSS`), and UK Security Vetting (`UKSV`)

- **National Security Council (`NSC`) and National Security Secretariat (`NSS`)**: Headquartered at 70 Whitehall / 10 Downing Street, the **`NSS`** supports the Prime Minister and the National Security Council. On **8 November 2024**, Prime Minister Sir Keir Starmer appointed **Jonathan Powell** as **National Security Adviser (`NSA`)**, succeeding Sir Tim Barrow (Appointment of Jonathan Powell as National Security Adviser).
- **Joint Intelligence Committee (`JIC`) and Joint Intelligence Organisation (`JIO`)**: Established in 1936, the **`JIC`** is the cross-government committee responsible for setting UK intelligence requirements and providing Ministers and senior officials with all-source strategic intelligence assessments (New Head of the JIO Appointed, Official History of the JIC). The Permanent Secretary-level **Chair of the JIC and Head of the JIO** reports to the **Cabinet Secretary (Dame Antonia Romeo)** and serves as direct adviser to the Prime Minister on intelligence (New Head of the JIO Appointed, Joint Intelligence Organisation Page). On **22 May 2026**, Cabinet Secretary Dame Antonia Romeo announced the appointment of **Adrian Bird CB** as **Chair of the JIC and Head of the JIO**, taking up post on **3 July 2026** in succession to **Dame Madeleine Alessandri DCB CMG** (who served from 1 July 2023 to July 2026) (New Head of the JIO Appointed, Dame Madeleine Alessandri Profile). The **`JIO`** also houses the **Professional Head of Intelligence Assessment (`PHIA`)**, established in 2005 in direct response to the Butler Review (Government Response to the Butler Review Cm 6492, para. 26, Butler and Military Intelligence 2026 Study).
- **United Kingdom Security Vetting (`UKSV`)**: Established in **January 2017** following the 2015 Strategic Defence and Security Review (**`SDSR`**) by merging the Foreign and Commonwealth Office Services (**`FCOS`**) National Security Vetting unit and the MOD Defence Business Services (**`DBS`**) National Security Vetting unit (NAO Investigation into National Security Vetting 2018, PAC Performance of UK Security Vetting Report HC 994). Initially hosted within the MOD while policy sat in the Cabinet Office, **`UKSV`** transferred fully into the **Cabinet Office Government Security Group (`GSG`)** in **April 2020**, led by a Chief Executive (appointed January 2021) reporting to the **Government Chief Security Officer (`GCSO`)** (PAC Performance of UK Security Vetting Report HC 994).

Section 4: Vetting Mechanics, Indoctrination Registers, and Need-to-Know Enforcement

4.1 National Security Vetting (`NSV`) Hierarchy and Statutory Investigation Mechanics

Under Cabinet Office policy (National Security Vetting: Clearance Levels, Updated 23 September 2024), personnel security controls are structured across a tiered hierarchy. Access to **`TOP SECRET`** codeword or

STRAP intelligence requires **Developed Vetting (DV)** or **Enhanced Developed Vetting (eDV)**, while specific **SECRET** codeword roles may utilize **Enhanced Security Check (eSC)**:

Vetting Tier	Maximum Uncontrolled Classification Access	Mandatory Investigative Components (UKSV Clearance Levels)	Review / Renewal Cycle & Operational Metrics (PAC Report HC 994)
Baseline Personnel Security Standard (BPSS)	**OFFICIAL** (and exceptional supervised access to **SECRET**)	Identity (**ID**), Right to Work (**RTW**) in the UK, 3-year employment history, and unspent criminal convictions check; underpins all higher clearances.	Pre-employment baseline conducted by employing authority.
Counter Terrorist Check (CTC) / Level 1B (L1B)	**OFFICIAL** (unescorted access to establishments/figures at terrorist risk; **L1B** introduced 31 Oct 2022)	**BPSS** + Security Questionnaire + Departmental/company records check (personal files, staff reports, sick leave, security records) + Spent & unspent criminal records + **MI5 Security Service records check** (+ interview if unresolved concerns).	Reviewed every **10 years** (**5 years** for non-Facility Security Clearance **FSC** contractors); average **164,700 CTC/SC** applications per year across UKSV (85% within 25-day KPI target).
Security Check (SC)	Long-term, frequent, uncontrolled access to **SECRET** and occasional supervised access to **TOP SECRET**	All **CTC** checks + **Credit Reference Agency (CRA) financial history check** + Full personal financial questionnaire if concerns arise + Third-party checks on questionnaire subjects.	Reviewed every **10 years** (**7 years** for non-**FSC** contractors).
Enhanced Security Check (eSC)	Regular uncontrolled access to **SECRET** (including **posts requiring access to SECRET codeword material**) and occasional controlled access to **TOP SECRET**	All **SC** checks + **Mandatory Financial Questionnaire** + **Internet Questionnaire** + Full review of assets, liabilities, income, and expenditure (individual and joint with spouse/partner) + **Detailed interview by a trained Investigating Officer**.	Role-specific supplementary safeguard positioned between **SC** and **DV** (NAO UKSV Report 2023).
Developed Vetting (DV)	Frequent and uncontrolled access to **TOP SECRET** assets and **any access to TOP SECRET codeword / STRAP material** (plus Category I nuclear material)	**BPSS** + **DV Security Questionnaire** + Departmental/company records + Spent & unspent criminal records + **CRA** credit check + **MI5 Security Service check** + **Full review of personal and spouse/partner finances** + **In-depth clinical/security interview by a trained Vetting Investigating Officer** + **In-person interviews with referees** + Third-party checks.	Reviewed at intervals of **no more than 7 years** (including non-**FSC** contractors); average **17,900 DV** applications per year across UKSV (MOD accounts for 56% of all UKSV demand; 85% within 95-day routine KPI target; 95% within 30-day priority KPI target).
Enhanced Developed Vetting (eDV)	Highest level of assurance above **DV** for a very small number of agreed, specific intelligence and compartmentalized posts (**STRAP 3** / core agency roles)	All **DV** checks + Expanded **Family and Travel element** in **DV Security Questionnaire** + **Applicant interview** + **Supervisor interview** + **Referee interviews** + Full individual/joint financial investigation + Third-party checks.	Requested only by a small number of approved Sponsors under prior agreement with **UKSV** and the **Cabinet Office** (UKSV Clearance Levels).

Furthermore, the Intelligence and Security Committee of Parliament (**ISC**) Annual Report 2022–2023 confirms that while **UKSV** provides vetting across Whitehall and Defence, the three intelligence Agencies (**GCHQ**, **MI5**,

and **SIS**) maintain their own dedicated internal vetting officers and specialized assessment pipelines (such as GCHQ's *Vetting Officer Development Programme* and tri-agency vetting pilots) to adjudicate staff and contractor clearances for direct agency indoctrination.

4.2 Role-Specific STRAP Indoctrination, Briefing Registers, and Debriefing Procedures

Holding a valid **DV** or **eDV** clearance is a necessary but **insufficient** condition for accessing **STRAP** material. Under national STRAP Security Guidelines and **JSP 440**, STRAP access is governed by a strict four-stage lifecycle (The British Classification Marking STRAP, FCDO Return HC 1774-I, MOD JSP 441 Records Management Policy):

1. **Billet Validation Against the STRAP Matrix**: The sponsoring business unit's **STRAPSO** verifies that the specific post being assumed by the individual is registered on the department's **STRAP Matrix** (e.g., **CIO-DSAS STRAP Matrix** in the MOD) at the requested tier (**STRAP 1**, **STRAP 2**, or **STRAP 3**) (National Data Library CIO-DSAS Directory).
2. **Role-Specific STRAP Application**: As documented in the Foreign, Commonwealth and Development Office (**FCDO**) onboarding records laid before the House of Commons on **11 March 2026 (HC 1774-I, *Return to an Address of the Honourable the House of Commons dated 4 February 2026 relating to the appointment of Lord Mandelson as HM Ambassador to Washington*)** (FCDO Return HC 1774-I PDF, The British Classification Marking STRAP), a STRAP clearance is strictly **role-specific**. Even when an official or senior diplomat already holds or is undergoing **DV** clearance, assuming a new post or function requires submitting a dedicated **STRAP application form** verifying both active **DV** status and post-specific operational need-to-know (The British Classification Marking STRAP, FCDO Return HC 1774-I PDF).
3. **Formal Indoctrination Briefing and Register Signature**: Prior to viewing any STRAP material or logging onto a **TOP SECRET STRAP** terminal (**SII** / **GCWiki**), the candidate must receive an explicit face-to-face **STRAP Indoctrination Briefing** from an accredited **STRAPSO**, explaining the origin of the intelligence, the specific compartments authorized (**STRAP 1**, **STRAP 2**, **STRAP 3**, and any associated codewords), physical custody rules, and criminal liability under **Section 1 of the Official Secrets Act 1989** and the **National Security Act 2023** (GSCP Guidance 1.3 - Working at TOP SECRET, MOD JSP 441 Records Management Policy, King's Regulations for the Royal Navy 2026). The individual signs the formal **STRAP Indoctrination Certificate / Register**, which is logged by the departmental **STRAPSO**.
4. **Mandatory Debriefing (Reading Out) and Custody Surrender Upon Departure**: When an indoctrinated individual leaves their post, transfers to another department, or retires, the **STRAPSO** executes a formal **STRAP Debriefing (reading out)**, revoking the individual's entry on the STRAP access register and terminating their electronic credentials on **SII**/**MyNOC**. For any material held above **STRAP 2** (**STRAP 3**), the departing official is legally prohibited from handing the files directly to their successor without written re-authorization from the originating Agency (MOD JSP 441 Records Management Policy).

Section 5: Historical & Contemporary Inquiries, Key Persons, and Deterministic Causal Chains

5.1 Declassified Archival Foundations at The National Archives (TNA)

The evolution of UK special handling from wartime **ULTRA**/**BONIFACE** to modern **STRAP** is documented across specific record classes held at The National Archives (**TNA**, Kew):

- **HW Series (Records of GCHQ and GC&CS)**:
  - **HW 80 (*Records relating to the development of the 1946 UKUSA Agreement*)**: Declassified in June 2010, including HW 80/11 (*Amendment No. 4 to the Appendices to the UKUSA Agreement, Third*

Edition), which codifies the technical and security appendices binding **GCHQ** and **NSA** (**STANCIB** / **USCIB**) to uniform indoctrination, courier controls, and codeword compartmentalization.

- **HW 75 (*Soviet Block Reports, 1945–1949*)**: Released alongside the UKUSA files (TNA UKUSA Highlights Guide), documenting early post-WWII GCHQ **end-product** COMINT reports (**HW 75/178**, **HW 75/182**, **HW 75/184**, **HW 75/192**, **HW 75/195**) with unique top-right control serials (**T256**, **T330**, **T382**, **T559**, **T626**).

- **CAB Series (Records of the Cabinet Office and Joint Intelligence Committee)**:
  - **CAB 81**: Wartime Joint Intelligence Sub-Committee (**JIC**) papers (1939–1945) (TNA Intelligence and Security Services Guide).
  - **CAB 158 (*JIC Memoranda*) & CAB 159 (*JIC Minutes*)**: Post-war strategic intelligence assessments and minutes detailing how raw **SIGINT** and **HUMINT** were sanitized for Cabinet consumption (TNA Intelligence and Security Services Guide).
  - **CAB 163 (*JIC Secretariat Files*) & CAB 176 (*JIC Secretariat Minutes, 1942–1968*)**: Document the introduction of discrete registry and distribution symbols (**JIC(SEC)**) in the late 1950s and January 1967, and the restructuring of the JIC into **JIC(A)** on 1 April 1968 (TNA CAB 176 Catalogue Description).

- **DEFE Series (Records of the Ministry of Defence and Defence Intelligence Staff)**:
  - Following the 1964 amalgamation of the War Office Directorate of Military Intelligence (**DMI**, **WO 106**/**WO 208**), the Admiralty Naval Intelligence Division (**NID**, **ADM 137**/**ADM 223**), and the Air Ministry Directorate of Intelligence (**AIR 40**) into the unified **Defence Intelligence Staff (DIS)**, records were organized into **DEFE 21** & **DEFE 44** (Scientific and Technical Intelligence), **DEFE 24** (Defence Secretariat intelligence policy files), **DEFE 27** (DIS Sub-Committee minutes), **DEFE 31** (Director General of Intelligence, later Chief of Defence Intelligence files), **DEFE 40** (Papers of Dr R. V. Jones, Director of Scientific Intelligence 1939–1954), and **DEFE 62–65** (DIS intelligence assessments and studies) (TNA Intelligence and Security Services Guide, TNA DEFE 31/174 Catalogue, TNA DEFE Briefing Guide).

- **KV, HD, FO 1093, and PREM Series**:
  - **KV 2, KV 4, KV 5**: Declassified MI5 personal, policy, and organisational files (including Captain Guy Liddell's wartime diaries in **KV 4/185–196**), whereas SIS (**HD**) retains its operational files permanently under Section 3(4) of the Public Records Act 1958, with historical disclosures appearing in **FO 1093** and Prime Minister's Office (**PREM**) files (TNA Intelligence and Security Services Guide).

5.2 Causal Chain 1: The Butler Review (HC 898, 14 July 2004) and Chilcot Iraq Inquiry (6 July 2016) — How STRAP / Codeword Compartmentalization Blinded Technical Validation

One of the most consequential institutional failures in modern British intelligence history occurred in September 2002 and was forensically reconstructed by two Privy Counsellor inquiries: the **Butler Review** (*Review of Intelligence on Weapons of Mass Destruction*, **HC 898**, chaired by **Lord Butler of Brockwell KG GCB CVO**, published **14 July 2004**) (Butler Review HC 898 PDF, Government Response to the Butler Review Cm 6492) and the **Chilcot Inquiry** (*The Report of the Iraq Inquiry*, chaired by **Sir John Chilcot**, who had previously served as a member of the Butler Committee, published **6 July 2016**) (Report of the Iraq Inquiry Volume IV, Report of the Iraq Inquiry Executive Summary):

- **Event & Mechanism (11–24 September 2002)**: On **11 September 2002**, during the final drafting of the UK Government's public dossier on Iraqi Weapons of Mass Destruction (published **24 September 2002**), the Secret Intelligence Service (**SIS**, led by Chief **Sir Richard Dearlove**) issued a highly sensitive intelligence report (**"Report X"**) from a new human source claiming that Iraq had accelerated the production of

chemical and biological agents and was storing chemical precursors in special glass-lined containers (Learning from the Chilcot Report, p. 55, Report X Marks the Spot, Manchester Pure PDF).

- **Compartmented Restriction Excluding DIS Chemical/Biological Experts**: Because **SIS** placed this reporting in a **tightly restricted codeword/STRAP compartment** to protect the alleged vulnerability of the source, circulation was limited to a tiny number of senior readers: Prime Minister **Tony Blair** (briefed by Sir Richard Dearlove on 12 September 2002 in the presence of **Sir David Manning**, **Jonathan Powell**, and **Alastair Campbell**), **Sir John Scarlett** (Chairman of the **JIC**), **Julian Miller** (Chief of the Assessments Staff), and the **Chief of Defence Intelligence (CDI)** plus his Deputy inside the Ministry of Defence (Learning from the Chilcot Report, p. 55, Butler Review HC 898 PDF).
- **Exclusion of Dr Brian Jones and DIS Technical Specialists**: Crucially, because of the compartmentalized handling restrictions imposed on the report, the senior managers in **SIS** and **DIS did not show the report to Dr Brian Jones** (Branch Head for Nuclear, Biological, and Chemical **[NBC]** technical intelligence in the Scientific and Technical Directorate of the **Defence Intelligence Staff [DIS]**) or to the **DIS**'s CW/BW subject-matter experts (Government Response to the Butler Review Cm 6492, para. 16, Butler Review HC 898 PDF, Learning from the Chilcot Report).
- **Consequence (Dr Jones's Dissent and Post-War Withdrawal of the Intelligence)**: Unable to see the underlying compartmented **SIS** report of 11 September 2002, **Dr Brian Jones** and his **DIS** experts formally registered reservations (**Dr Jones's dissent**) that the available intelligence they *had* been permitted to see did not support the firm judgement in the 24 September 2002 dossier that Iraq had continued to produce chemical weapons (Government Response to the Butler Review Cm 6492, para. 16, Chilcot Inquiry Section 4.2). Following the 2003 invasion, **SIS** discovered that the source of the 11 September 2002 report was a fabricator (whose story about glass containers had been lifted from open scientific literature and movie tropes) and formally withdrew the reporting in July 2003 (Butler Review HC 898 PDF, Report of the Iraq Inquiry Volume IV).
- **The Butler Committee's Finding and Statutory/Policy Reforms (2004–2005)**: In **HC 898**, Lord Butler's Committee censured the failure to grant **DIS** technical experts access to the compartmented intelligence, stating that *"it would have been more appropriate for senior managers in the DIS and SIS to have made arrangements for the report to be shown to one or two of the key experts on conditional terms"* (Butler Review HC 898 PDF, Government Response to the Butler Review Cm 6492, para. 16). In its formal Command Paper response (**Cm 6492**, *Review of Intelligence on Weapons of Mass Destruction: Implementation of its Conclusions*, March 2005), HM Government accepted this conclusion in full and enacted five permanent structural reforms to **STRAP**/compartmented assessment procedures (Government Response to the Butler Review Cm 6492):
- **Mandatory Expert Access to Restricted/Compartmented Reports (Cm 6492, paras. 9–16)**: **SIS** and **DIS** revised their compartmented distribution protocols so that whenever an intelligence report relies on technical or scientific claims, designated technical experts in **DIS** must be indoctrinated/read into the compartment to validate the claim before it is used in a **JIC** assessment (Government Response to the Butler Review Cm 6492).
- **Appointment of Full-Time SIS Reporting Evaluation Officer (Cm 6492, para. 9)**: **SIS** appointed a full-time senior officer to oversee reporting quality, source validation, and cross-community standardization of source descriptors across **HUMINT** and **SIGINT** (Government Response to the Butler Review Cm 6492).
- **Creation of the Professional Head of Intelligence Analysis (PHIA) (Cm 6492, para. 26)**: Established in the Cabinet Office under the **JIC** Chair to standardize analytical training, probabilistic yardsticks (*Explaining Uncertainty in UK Intelligence Assessment*), and cross-departmental challenge across all UK intelligence analysts (Government Response to the Butler Review Cm 6492, Butler and Military Intelligence 2026 Study).

- **Mandatory `Assessment Base` Boxes and Dissent Recording (Cm 6492, paras. 28–30)**: Every **`JIC`** assessment was required to include an explicit **"Assessment Base"** box setting out the depth and compartmentation of the secret intelligence drawn upon, and any minority or dissenting views at Current Intelligence Group (**`CIG`**) or **`JIC`** level (such as **`DIS`** technical dissent) had to be formally recorded in **`JIC`** minutes and noted on the face of the final **`JIC`** assessment (Government Response to the Butler Review Cm 6492).
- **Expansion of the Cabinet Office Assessments Staff by One-Third (Cm 6492, para. 23)**: Added dedicated scientific advisers and a standing internal challenge/red-team function within the Assessments Staff (Government Response to the Butler Review Cm 6492).

5.3 Causal Chain 2: The 2013 Snowden `TOP SECRET STRAP` Disclosures, IPT Rulings (`2014–2019`), and the Investigatory Powers Act 2016

The second major causal transformation of the STRAP and UK intelligence governance architecture was triggered in May–June 2013 when **Edward Snowden**, an external IT contractor and systems administrator employed by Booz Allen Hamilton at the US National Security Agency (**`NSA`**) in Hawaii, utilized his privileged system administrator credentials on shared Five Eyes **`SCI`** networks (**`GCWiki`**) to exfiltrate tens of thousands of GCHQ documents marked **`TOP SECRET STRAP 1`**, **`TOP SECRET STRAP 2`**, and **`TOP SECRET STRAP 3`** (Work and Surveillance in a Post-Snowden World, Surveillance Watchdog Investigates Security Risks of GCHQ IT Contractors, NSArchive GCHQ Cyber Defence Operations Legal and Policy).

- **Mechanism of the Compromise**: Although Snowden held only a **`STRAP 1`** equivalent access clearance on **`GCWiki`** (**`TOP SECRET STRAP1 COMINT`**), lateral movement across shared Five Eyes repositories and contractor system-administrator privileges allowed him to harvest **`TOP SECRET STRAP 2`** and **`STRAP 3`** briefings detailing GCHQ's **`Tempora`** bulk optical-fibre interception buffers at GCHQ Bude, NSA **`PRISM`** and **`Upstream`** intelligence sharing with GCHQ, **`OP SOCIALIST`** against Belgian telecom provider Belgacom/BICS, **`QUANTUM INSERT`** CNE attacks, and **`QWERTY`** keystroke loggers (Work and Surveillance in a Post-Snowden World, NSArchive GCHQ Mobile Networks in My NOC World, Parliamentary Written Evidence IPB0055).
- **Judicial and Parliamentary Scrutiny (`2014–2019`)**:
- **`Liberty & Others v GCHQ, SIS & MI5` (`[2014] UKIPTrib 13_77-H` & `[2015] UKIPTrib 13_77-H`, 5 December 2014 & 6 February 2015)**: In a landmark judgment by the **Investigatory Powers Tribunal (IPT)**, the Tribunal examined the legality of GCHQ's bulk interception under **Section 8(4) of the Regulation of Investigatory Powers Act 2000 (`RIPA`)** (**`Tempora`**) and GCHQ's receipt of intercepted material from the US **`NSA`** under **`PRISM`** and **`Upstream`**. The IPT ruled that while from December 2014 onwards the statutory regime complied with Article 8(2) of the European Convention on Human Rights (**`ECHR`**), **prior to December 2014 the regime governing GCHQ's receipt of Five Eyes intelligence from the NSA had contravened Article 8 ECHR** because the internal **`STRAP`**/GCHQ handling safeguards regulating how GCHQ requested, stored, and destroyed NSA material were kept completely secret (**`"below the waterline"`**) and were not "in accordance with the law" until disclosed during the IPT proceedings (IPT Liberty Judgment PDF, Parliamentary Counter-Terrorism Evidence).
- **`Caroline Lucas MP & Others v Security Service, SIS & GCHQ` (`[2015] UKIPTrib 14_79-CH`, 14 October 2015)**: Forced the public disclosure of GCHQ's internal compliance guidance on the **Wilson Doctrine** and how GCHQ analysts selecting "related communications data" under the **`Section 8(4)`** bulk regime had to consult **`DD Mission Policy`** and the Foreign Secretary before targeting Parliamentarians (IPT Caroline Lucas Judgment PDF).
- **ISC *Privacy and Security: A modern and transparent legal framework* (`HC 1075`, 12 March 2015)**: The Intelligence and Security Committee of Parliament (ISC) conducted a comprehensive audit of GCHQ's

bulk interception (**Tempora** bearers), concluding that the piecemeal statutory framework (**ISA 1994**, **SSA 1989**, **RIPA 2000**, **Telecommunications Act 1984**) was opaque and required replacement by a single, unified Act of Parliament (ISC Privacy and Security Report PDF).

- **Privacy International v Secretary of State for Foreign and Commonwealth Affairs, Home Secretary, GCHQ, MI5 & SIS ([2016] UKIPTrib 15_110-CH, 17 October 2016 / August 2016 & UK Supreme Court [2019] UKSC 22)**: The IPT Bulk Data Judgment ruled that MI5, SIS, and GCHQ's secret collection and retention of **Bulk Communications Data (BCD)** under **Section 94 of the Telecommunications Act 1984** and **Bulk Personal Datasets (BPDs)** had operated unlawfully prior to explicit public avowal in 2015 because of the lack of statutory oversight and unpublished handling arrangements (IPT Bulk Data Judgment PDF). Subsequent IPT hearings in **February 2018** exposed that GCHQ had shared entire bulk datasets with external industry and academic partners (including the **University of Bristol**) without prior audit by the Intelligence Services Commissioner, and scrutinized the insider threat posed by **DV**-cleared external IT contractors holding privileged system administrator access to **TOP SECRET STRAP** networks, prompting **IPCO** to institute mandatory inspections of contractor privileged access and industry partner data sharing in 2018 (Surveillance Watchdog Investigates Security Risks of GCHQ IT Contractors, Commons Library Briefing CBP-11649 on Supreme Court IPT Ruling).
- **Statutory Consequence — The Investigatory Powers Act 2016 (2016 c. 25)**: Enacted on **29 November 2016**, the Investigatory Powers Act 2016 replaced **RIPA** Part 1 and **Section 94** directions with an explicit statutory regime governing targeted and bulk interception (Part 6, Ch. 1), bulk acquisition of communications data (Part 6, Ch. 2), bulk equipment interference (Part 6, Ch. 3), and Bulk Personal Datasets (Part 7), created the **Investigatory Powers Commissioner (IPC)** and **IPCO** with technical inspection teams, and amended the **Official Secrets Act 1989** via Schedule 10 of the 2016 Act to criminalize unauthorized disclosure of any material obtained under Part 2 or Part 6 warrants.

5.4 Causal Chain 3: The 2019 Washington Diplomatic Leak, the ROSA Rollout, the UKSV Vetting Crisis (2020–2023), the National Security Act 2023, and the 2024–2026 Reforms

Between 2019 and 2026, four further operational pressures reshaped UK protective security, vetting, and STRAP governance:

1. **The July 2019 Washington Diplomatic Telegram Leak and ROSA Governance**: Following the 6–7 July 2019 leak of sensitive Diplomatic Telegrams (**DipTels**) from HM Ambassador in Washington **Sir Kim Darroch** (leading to his resignation on 10 July 2019 and a Metropolitan Police criminal investigation under the Official Secrets Act 1989 launched on 12 July 2019), the House of Commons Foreign Affairs Committee published *Stemming the flow: An urgent look at tackling a culture of leaks* (**HC 2541**, 22 July 2019; Government Response **HC 2681**, **16 October 2019**) (Foreign Affairs Committee Government Response HC 2681). The Government accelerated the deployment of the **ROSA** cross-government **SECRET** IT system across over 40 departments (under the Cabinet Office Government Security Group and the **Rosa Service Organisation** hosted in a Crown Hosting Data Centre), narrowed **DipTel** distribution lists, and implemented technical controls restricting email forwarding and printing (Foreign Affairs Committee Government Response HC 2681).
2. **The UKSV Vetting Backlog Crisis (2021–2023) and Deferral of DV Renewals**: Following the April 2020 transfer of **UK Security Vetting (UKSV)** from the MOD to the Cabinet Office and a failed £2.5 million IT modernisation write-off in 2021–22, **DV** demand surged by **57%** and **CTC**/**SC** demand by **60%** in 2021–22 (NAO Investigation into the Performance of UKSV, 18 January 2023, PAC Report HC 994, 12 May 2023). By **April 2022**, only **7%** of routine **DV** clearances were processed within UKSV's 95-day target (with outstanding **DV** cases peaking at **14,600** in June 2022 and average **DV** renewal turnaround reaching **255 days** in November 2022 against a 200-day target) (PAC Report HC 994). Because the Cabinet Office

Government Security Board (**GSB**) and Government Security Steering Group repeatedly deferred 7-year **DV** renewals in 2018, 2019, 2020, and April 2022 while failing to meet its 85% KPI for scheduled aftercare checks, the Public Accounts Committee warned in May 2023 that unrenewed **DV** holders in **TOP SECRET** and **STRAP** billets posed "uncomfortable" insider security risks, compelling the Cabinet Office to institute automated continuous data-driven aftercare triage and phased vetting level reform (**Level 1B**, **AC**) (PAC Report HC 994).

3. **Enactment of the National Security Act 2023 (2023 c. 32)**: Following the Law Commission's review of the Official Secrets Acts 1911–1989 (referenced in HC 2681), Parliament enacted the **National Security Act 2023 (2023 c. 32)**, replacing the archaic espionage provisions of the Official Secrets Acts 1911, 1920, and 1939 with modern offences covering obtaining or disclosing protected information (**TOP SECRET**, **STRAP**, and defence/intelligence assets), trade secrets, assisting a foreign intelligence service, foreign interference, and entering prohibited places (**SCIFs**, GCHQ/MOD sites), alongside **Official Secrets Act 1989** retention/disclosure offences (National Security Act 2023, King's Regulations for the Royal Navy 2026).
4. **2024–2026 GSCP, MIS, and Leadership Consolidation**: In **August 2024**, the Cabinet Office issued version 2.0 updates to GSCP Guidance 1.2 (Working at SECRET), Guidance 1.3 (Working at TOP SECRET), and Guidance 1.5 (Considerations for Security Advisors), tightening two-person **DV** escort mandates, prohibiting home working at **TOP SECRET**, and mandating **UK NACE** acoustic assurance and **NPSA CSE** physical controls. In **December 2025**, following the **July 2025 Strategic Defence Review**, the MOD unified all single-service and joint military intelligence bodies into **Military Intelligence Services (MIS)** and launched the **Defence Counter-Intelligence Unit (DCIU)**, followed in **February–May 2026** by the appointments of **Lt-Gen Matthew Jones OBE** as **CDI** and former **CDI**/GCHQ DG **Adrian Bird CB** as **Chair of the JIC and Head of the JIO** (MOD CDI Appointment Press Release, New Head of the JIO Appointed).

Section 6: Consolidated Institutional, Statutory, and Technical Matrix

Entity / System / Instrument	Statutory / Policy Authority	Key Leadership / Custodian (2025–2026)	Primary Physical Sites / Networks	Role in the UK STRAP & Compartmented Intelligence Architecture
STRAP System (STRAP 1, STRAP 2, STRAP 3)	JSP 440 Part 2 Leaflets 4B, 4C, 4G; GSCP Guidance 1.3; Official Secrets Act 1989; National Security Act 2023	**STRAP Management Board**; Departmental **STRAPSOs** (COSR STRAP Team; MOD CIO-DSAS)	**STRAP Channels**; **SII**; **MyNOC**; **GCWiki** (**STRAP 1** max); **MOD Sensitive Archive Portsmouth** (**STRAP 1–2** only)	Sovereign 3-tier codeword special-handling system protecting **SIGINT**, **COMINT**, **CNE**, and sensitive agency intelligence; maps directly to US/Five Eyes **SCI** (The British Classification Marking STRAP).
GCHQ (*Government Communications Headquarters*)	Intelligence Services Act 1994 (ss. 3–7); Investigatory Powers Act 2016	Foreign Secretary **Rt Hon David Lammy MP**; Director **Anne Keast-Butler** (since May 2023); 5 Directors General (GCHQ Leadership)	**Cheltenham** (Benhall HQ), **Bude** (Cornwall, 30 antennas), **Scarborough**, **JSSU Cyprus** (Ayios Nikolaos, Troodos, Akrotiri), Manchester, London	Primary originator and technical custodian of **STRAP 1**, **STRAP 2**, and **STRAP 3 SIGINT**/**COMINT** and **CNE** (**OP SOCIALIST**, **QUANTUM**, **HIGHNOTE**) product and UKUSA/Five Eyes gateways (GCHQ UKUSA History).

Entity / System / Instrument	Statutory / Policy Authority	Key Leadership / Custodian (2025–2026)	Primary Physical Sites / Networks	Role in the UK STRAP & Compartmented Intelligence Architecture
NCSC (*National Cyber Security Centre*, part of GCHQ)	Intelligence Services Act 1994; GSCP Guidance 1.3 & 1.5	CEO **Dr Richard Horne** (since Oct 2024), reporting to Director GCHQ (NCSC Richard Horne)	Victoria (London) & Cheltenham; **CFTCS** & **CPTAS** accredited test facilities (NCSC CFTCS)	**National Technical Authority (NTA)** for Cyber Security, Information Assurance, High Grade Cryptography, and **TEMPEST / Electromagnetic Security (EMS)** (NCSC TEMPEST).
MI5 (*The Security Service*) & **NPSA**	Security Service Act 1989 (1989 c. 5); Investigatory Powers Act 2016	Home Secretary; Director General **Sir Ken McCallum** (MI5 Director General)	**Thames House** (Millbank, London); hosts **JTAC** and **NPSA** (NPSA CSE); operates **NEXUS BSS Desktop** (NSArchive GCHQ MyNOC)	Domestic counter-intelligence/CT consumer and originator of STRAP intelligence; conducts mandatory MI5 vetting checks for **CTC/SC/DV/eDV**; **NPSA** is the **NTA for Physical & Personnel Security** (**CSE** containers/acoustic barriers) (UKSV Clearance Levels).
SIS / MI6 (*Secret Intelligence Service*)	Intelligence Services Act 1994 (ss. 1–2, 5–7); Investigatory Powers Act 2016	Foreign Secretary **Rt Hon David Lammy MP**; Chief ("**C**") **Blaise Metreweli CMG** (18th Chief, appointed 15 June 2025, succeeding **Sir Richard Moore KCMG**) (SIS Chief Appointment)	**Vauxhall Cross** (85 Albert Embankment, London) & global stations; operates **CADDIS SIS Desktop** (NSArchive GCHQ MyNOC)	Foreign **HUMINT** (**CX**) collection and joint covert technical operations with GCHQ; originates compartmented/codeword reports and consumes **STRAP** SIGINT (SIS About Us).
MOD, **Defence Intelligence (DI)**, & **Military Intelligence Services (MIS)**	Investigatory Powers Act 2016 (s. 104); JSP 440; JSP 441; Strategic Defence Review 2025	Defence Secretary **Rt Hon John Healey MP**; Perm Sec **Jeremy Pocklington CB**; **CSOC** Commander **Gen Sir Jim Hockenhull**; **CDI Lt-Gen Matthew Jones OBE** (succeeding **Adrian Bird CB**) (MOD CDI Appointment)	MOD Whitehall, **RAF Wyton** (**DIFC**), **RAF Molesworth**, **Northwood (PJHQ)**, **Chicksands**; networks **RLI**, **MODNET Secret**, **SII** (MOD DI Levant Analyst)	All-source military intelligence (**MIS** unified Dec 2025) and counter-intelligence (**DCIU**); enforces **JSP 440** and **JSP 441**; maintains **CIO-DSAS STRAPSO Listing**, **STRAP Matrix**, **STRAP Location**, and **STRAP Incident** databases (MOD CIO-DSAS STRAP Database).
Cabinet Office (JIC, JIO, NSS, GSG, UKSV)	GSCP 2023/2024; Royal Prerogative	Cabinet Secretary **Dame Antonia Romeo**; **NSA Jonathan Powell**; **JIC** Chair & **JIO** Head **Adrian Bird CB** (succeeding **Dame Madeleine Alessandri**); **GCSO Christina Duncan** (New Head of the JIO Appointed, Cabinet Office)	70 Whitehall / 10 Downing Street; operates cross-government **ROSA SECRET** system (Foreign Affairs Committee HC 2681)	Produces all-source **JIC** assessments fusing **STRAP** and **HUMINT** inputs; sets **GSCP** and **UK NSA** international policy; operates **UK Security Vetting (UKSV)** delivering ~17,900 **DV** and ~164,700 **CTC/SC** clearances annually (PAC Report HC 994).

Entity / System / Instrument	Statutory / Policy Authority	Key Leadership / Custodian (2025–2026)	Primary Physical Sites / Networks	Role in the UK STRAP & Compartmented Intelligence Architecture
UK NACE (*UK National Authority for Counter-Eavesdropping*)	GSCP Guidance 1.3	Executive Agency / Trading Fund of the FCDO (**FCDO Services**) (UK NACE Portal)	**Hanslope Park**, Milton Keynes (MK19 7BH) (UK NACE Portal)	**National Technical Authority (NTA)** for Technical Security; accredits **TOP SECRET** and **STRAP** rooms against audio leakage, conducts **TSCM** counter-eavesdropping sweeps, and delivers **TEMPEST** tester training (UK NACE Operations).
ISC (*Intelligence and Security Committee of Parliament*) & **IPT** (*Investigatory Powers Tribunal*)	Justice and Security Act 2013; Regulation of Investigatory Powers Act 2000; Investigatory Powers Act 2016	Statutory Committee of Parliament (**ISC**, subject to s. 1(1)(b) Official Secrets Act 1989) (ISC HC 1113); Judicial Tribunal (**IPT**) & **IPCO**	Westminster / Royal Courts of Justice	Statutory parliamentary (**ISC**) and judicial (**IPT** / **IPCO**) oversight of **GCHQ**, **MI5**, **SIS**, **DI**, **JIO**, and **NSS**, with full clearance to inspect **TOP SECRET STRAP** operations, bulk warrants, and Five Eyes intelligence-sharing arrangements (ISC Annual Report 2023–2025).

Section 7: Categorized Primary-Source Reference Compendium

7.1 Legal, Statutory & Judicial Records

1. **UK Parliament**, *Security Service Act 1989 (1989 c. 5)*, Legislation.gov.uk (1989): https://www.legislation.gov.uk/ukpga/1989/5 and Section 2 (The Director-General).
2. **UK Parliament**, *Intelligence Services Act 1994 (1994 c. 13)*, Legislation.gov.uk (1994): https://www.legislation.gov.uk/ukpga/1994/13, Section 5 (Warrants: general), and Authorisation of certain actions (ss. 5–7).
3. **UK Parliament**, *Investigatory Powers Act 2016 (2016 c. 25)*, Legislation.gov.uk (29 November 2016): https://www.legislation.gov.uk/ukpga/2016/25/contents, Full Act Enacted PDF, Schedule 10 (Official Secrets Act 1989 Amendments), and The Investigatory Powers Act 2016 (Commencement No. 7 and Transitional and Saving Provisions) Regulations 2018 (S.I. 2018/873).
4. **UK Parliament**, *National Security Act 2023 (2023 c. 32)*, Legislation.gov.uk (2023): https://www.legislation.gov.uk/ukpga/2023/32 and Contents.
5. **Investigatory Powers Tribunal (IPT)**, *Liberty & Others v. The Security Service, SIS, GCHQ (**[2014] UKIPTrib 13_77-H**)*, Approved Judgment (5 December 2014): http://www.ipt-uk.com/docs/IPT_13_168-173_H.pdf.
6. **Investigatory Powers Tribunal (IPT)**, *Caroline Lucas MP & Others v. The Security Service, SIS, GCHQ (**[2015] UKIPTrib 14_79-CH**)*, Approved Judgment (14 October 2015): http://www.ipt-uk.com/docs/Caroline_Lucas_JUDGMENT.pdf.
7. **Investigatory Powers Tribunal (IPT)**, *Privacy International v. Secretary of State for Foreign and Commonwealth Affairs, Secretary of State for the Home Department, GCHQ, Security Service, SIS (**[2016] UKIPTrib 15_110-CH**)*, Bulk Personal Datasets and Section 94 Judgment (19 August 2016 / 17 October 2016): http://www.ipt-uk.com/docs/Bulk_Data_Judgment.pdf.

8. **House of Commons Library**, *What does the Supreme Court's ruling on the Investigatory Powers Tribunal mean for GCHQ? (Research Briefing CBP-11649)*, UK Parliament (28 May 2019): https://commonslibrary.parliament.uk/research-briefings/cbp-11649/.

7.2 Official UK Government Directives, Inquiries, and Parliamentary Committee Reports

1. **Cabinet Office**, *Government Security Classifications Policy (GSCP)*, GOV.UK (Updated 30 June 2023 & 5 August 2024): Collection Portal, GSCP Policy HTML, GSCP Quick Read PDF (29 July 2024), Guidance 1.1 - Working at OFFICIAL, Guidance 1.2 - Working at SECRET (5 August 2024), Guidance 1.3 - Working at TOP SECRET (5 August 2024), and Guidance 1.5 - Considerations for Security Advisors (5 August 2024).
2. **Cabinet Office / United Kingdom Security Vetting (UKSV)**, *National Security Vetting: Clearance Levels*, GOV.UK (Updated 23 September 2024): https://www.gov.uk/government/publications/united-kingdom-security-vetting-clearance-levels/national-security-vetting-clearance-levels, UKSV Applicant Guidance, and DV Guidance Pack for Applicants.
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7.3 Agency Portals, Executive Appointments & National Technical Authorities

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2. **National Cyber Security Centre (NCSC)**, *TEMPEST, Electromagnetic Security, and Leadership*: What We Do at the NCSC, Dr Richard Horne CEO NCSC, Richard Horne NCSC Annual Review 2026 Speech (3 November 2026), TEMPEST and Electromagnetic Security, and Formal TEMPEST Certification Scheme (CFTCS).
3. **UK National Authority for Counter-Eavesdropping (UK NACE)**, *Official Technical Security NTA Portal*: https://www.fcdoservices.gov.uk/uk-nace/, GOV.UK UK NACE Profile, and UK NACE Operations.
4. **MI5 (The Security Service) & National Protective Security Authority (NPSA)**: MI5 Director General Sir Ken McCallum, GOV.UK Ken McCallum Profile, MI5 Director General Annual Threat Update (13 October 2026), NPSA Catalogue of Security Equipment (CSE), and NPSA Physical Security.
5. **Secret Intelligence Service (SIS / MI6)**: Appointment of the New Chief of the Secret Intelligence Service — Blaise Metreweli CMG (15 June 2025), SIS Home, SIS About Us, and GOV.UK Sir Richard Moore KCMG Profile.

6. **Ministry of Defence (MOD), Defence Intelligence (DI), & Military Intelligence Services (MIS)**: Defence's Next Top Intelligence Officer Appointed — Lt-Gen Matthew Jones OBE as CDI and Establishment of MIS & DCIU (5 February 2026), New Ministry of Defence Permanent Secretary Announced — Jeremy Pocklington CB (31 October 2025), The Strategic Defence Review 2025 (8 July 2025), MOD Defence Intelligence Portal, and DI Levant Analyst Role Specification (STRAP Cleared / SII).
7. **Cabinet Office, Joint Intelligence Committee (JIC), & Joint Intelligence Organisation (JIO)**: New Head of the Joint Intelligence Organisation Appointed — Adrian Bird CB as JIC Chair (22 May 2026), Joint Intelligence Organisation Portal, Dame Madeleine Alessandri Profile, Appointment of Jonathan Powell as National Security Adviser (8 November 2024), and Cabinet Office EO Security Support Officer (COSR STRAPSO Team).

7.4 Archival, Academic & Institutional Primary Repositories

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2. **National Security Archive (George Washington University)**, *Declassified & Disclosed GCHQ **TOP SECRET STRAP** Primary Documents*: - *GCHQ Cyber Defence Operations Legal and Policy (**TOP SECRET STRAP 1 COMINT**, GCWiki, Jan 2011)*: https://nsarchive.gwu.edu/document/21656-document-03-government-communications. - *GCHQ Assessment of Intelligence Opportunity - Juniper (**TOP SECRET STRAP 1**, Feb 2011)*: https://nsarchive.gwu.edu/document/21530-document-2-6. - *GCHQ OPC-MCR: A potential technique to deanonymise users of the TOR network (**TOP SECRET STRAP 1 COMINT**, June 2011)*: https://nsarchive.gwu.edu/document/22582-document-05-names-redacted-opc-mcr-government. - *Head of GCHQ NAC: Mobile Networks in My NOC World (**TOP SECRET STRAP 2**, 2011)*: https://nsarchive.gwu.edu/document/22240-document-06-name-redacted-head-gchq-nac and Full OCR Text. - *GCHQ Network Analysis Centre: Automated NOC Detection (**TOP SECRET STRAP 2**, 2011)*: https://nsarchive.gwu.edu/document/22182-document-05.
3. **European Parliament Historical Archives**, *The ECHELON Affair (STOA PE 166.499 & PE 168.184)*: https://historicalarchives.europarl.europa.eu/files/live/sites/historicalarchive/files/03_PUBLICATIONS/03_European-Parliament/01_Documents/the-echelon-affair-en.pdf.
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https://bootcampmilitaryfitnessinstitute.com/2023/05/12/an-overview-of-uk-security-vetting-for-members-of-the-british-armed-forces/. - *The British Classification Marking STRAP*: https://www.electrospaces.net/2013/12/the-british-classification-marking-strap.html.

Part II: Special Access Programs (SAP), Sensitive Compartmented Information (SCI), Special Operational Information (SOI), and Persons Permanently Bound to Secrecy (PPBS) in Canada

Canada's compartmentalized national security, signals intelligence (**SIGINT**), and special operations architecture operates through a dual statutory and administrative framework: the strict-liability secrecy regime codified in the *Foreign Interference and Security of Information Act* (FISOIA / formerly SOIA, R.S.C., 1985, c. O-5) and the executive screening and compartmentation standards promulgated by the Treasury Board of Canada Secretariat (**TBS**) under the *Policy on Government Security* and the *Directive on Security Screening*.

Section 8: Legal, Statutory, and Policy Framework (FISOIA / SOIA & TBS)

8.1 Evolution and Statutory Mechanics of the *Foreign Interference and Security of Information Act* (R.S.C., 1985, c. O-5)

Originally enacted as the *Official Secrets Act* in 1939 (derived from the UK *Official Secrets Act 1911* and consolidated as R.S.C. 1985, c. O-5), Canada's primary secrecy statute was fundamentally restructured and renamed the *Security of Information Act* (**SOIA**) by the *Anti-terrorism Act* (**S.C. 2001, c. 41, s. 25**, in force December 24, 2001). Pursuant to Bill C-70 (*Countering Foreign Interference Act*, S.C. 2024, c. 16, s. 50) (Royal Assent June 20, 2024; amendments in force August 19, 2024), the statute was retitled the *Foreign Interference and Security of Information Act* (**FISOIA**, R.S.C., 1985, c. O-5).

Statutory Definition of Special Operational Information (SOI) (Section 8(1))

Under **Section 8(1)** of the Act, **Special Operational Information (SOI)** is defined as information that the Government of Canada is taking measures to safeguard that reveals, or from which may be inferred:

- **s. 8(1)(a) — Confidential Sources:** The identity of a person, agency, group, body, or entity that was, is, or is intended to be, has been approached to be, or has offered or agreed to be, a confidential source of information, intelligence, or assistance to the Government of Canada.
- **s. 8(1)(b) — Armed Conflict Operational Plans:** The nature or content of plans of the Government of Canada for military operations in respect of a potential, imminent, or present armed conflict.
- **s. 8(1)(c) — Covert Collection & SIGINT/Traduction Means:** The means that the Government of Canada used, uses, or intends to use, or is capable of using, to covertly collect or obtain, or to decipher, assess, analyse, process, handle, report, communicate, or otherwise deal with information or intelligence, including any vulnerabilities or limitations of those means.
- **s. 8(1)(d) — Covert Targets:** Whether a place, person, agency, group, body, or entity was, is, or is intended to be the object of a covert investigation, or a covert collection of information or intelligence, by the Government of Canada.

- **`s. 8(1)(e)` — Covert Operatives:** The identity of any person who is, has been, or is intended to be covertly engaged in an information- or intelligence-collection activity or program of the Government of Canada that is covert in nature.
- **`s. 8(1)(e.1)` — Military Vulnerabilities & Advantages (Added by Bill C-70, S.C. 2024, c. 16, s. 51(2)):** The military vulnerabilities or advantages of the Canadian Forces, including the operational or technical vulnerabilities and advantages of any ally or adversary.
- **`s. 8(1)(f)` — Cryptographic & Exploitation Systems (Amended by S.C. 2024, c. 16, s. 51(2)):** The means that the Government of Canada used, uses, or intends to use, or is capable of using, to protect or exploit any information or intelligence referred to in paragraphs **`(a)`** to **`(e.1)`**, including encryption and cryptographic systems, and any vulnerabilities or limitations of those means.
- **`s. 8(1)(g)` — Allied/Five Eyes & Foreign Intelligence:** Information or intelligence similar in nature to paragraphs **`(a)`** to **`(f)`** that is in relation to, or received from, a foreign entity or terrorist group.

`Persons Permanently Bound to Secrecy (PPBS)` (Sections 8–15)

- **Automatic Statutory & Schedule-Based Indoctrination (`s. 8(1)(a)–(a.4)`):** An individual becomes a **`Person Permanently Bound to Secrecy (PPBS)`** for life by operation of law if they are:
- A current or former member or employee of a federal department, division, branch, or office listed in Schedule 1 of the Act (**`s. 8(1)(a)`**);
- A current or former member of the National Security and Intelligence Review Agency (**`NSIRA`**, **`s. 8(1)(a.1)`**);
- A current or former member of the National Security and Intelligence Committee of Parliamentarians (**`NSICOP`**, **`s. 8(1)(a.2)`**); or
- Pursuant to the August 2024 Bill C-70 amendments, a current or former officer or non-commissioned member (or person appointed, attached, assigned, or seconded) of a Canadian Forces unit, element, or position listed in Schedule 2 of the Act (**`s. 8(1)(a.3)–(a.4)`**). The Governor in Council may amend Schedule 1 (**`s. 9`**) and Schedule 2 (**`s. 9.1`**) by Order in Council.
- **Complete Roster of Entities Listed in Schedule 1:**
  - Canadian Security Intelligence Service (**`CSIS`**) and **`CSIS Legal Services Unit`** (Department of Justice);
  - Communications Security Establishment (**`CSE`**) and **`CSE Legal Services Unit`** (Department of Justice);
  - Communications Branch of the National Research Council (**`CBNRC`**, prior to April 1, 1975);
  - Royal Canadian Mounted Police (**`RCMP`**): **`Criminal Intelligence Program`**, **`National Security Program`**, **`Security Service`**, and **`Technical Operations Program`** (excluding the Air Services Branch);
  - Privy Council Office (**`PCO`**): **`Foreign and Defence Policy Secretariat`**, **`Intelligence Assessment Secretariat`**, **`International Assessment Staff`**, **`Security and Intelligence Secretariat`**, **`Office of the National Security Advisor to the Prime Minister`**, and **`Office of the Security and Intelligence Coordinator`**;
  - Department of Justice (**`DOJ`**): **`National Security Group`** and **`National Security Litigation and Advisory Group`**;
  - Review and Oversight Secretariats: **`NSIRA Secretariat`**, **`NSICOP Secretariat`**, **`Office of the Intelligence Commissioner`**, former **`Office of the CSE Commissioner`**, former **`Office of the Inspector General of CSIS`**, former **`Security Intelligence Review Committee (SIRC)`**, and the **`Civilian Review and Complaints Commission for the RCMP`** (transitioning to the **`Public Complaints and Review Commission`** under S.C. 2024, c. 25, ss. 125–126);

- Designated Commissions of Inquiry: **Commission of Inquiry into the Actions of Canadian Officials in Relation to Maher Arar**, **Commission of Inquiry into the Investigation of the Bombing of Air India Flight 182**, **Internal Inquiry into the Actions of Canadian Officials in Relation to Abdullah Almalki, Ahmad Abou-Elmaati and Muayyed Nureddin**, and the **Public Inquiry into Foreign Interference in Federal Electoral Processes and Democratic Institutions** (added via **SOR/2023-204**).

- **Individual Designation by Deputy Heads (s. 8(1)(b), s. 8(2), s. 10, s. 11):** Under Section 10(1), a Deputy Head may designate any non-scheduled person by written notice if the person had, has, or will have authorized access to **SOI** (**s. 10(1)(a)**) and designation is in the interest of national security (**s. 10(1)(b)**). Under **s. 8(2)**, the designating Deputy Head authority is vested in:
  - The departmental **Deputy Minister** for departmental employees, attachés, secondments, and ministerial exempt staff (**s. 8(2)(a), (c)**);
  - The **Chief of the Defence Staff (CDS)** for officers and non-commissioned members of the Canadian Armed Forces (**s. 8(2)(b)**);
  - The **Deputy Minister of Public Works and Government Services (PSPC)** or the relevant contracting department Deputy Head for government contractors and subcontractors (**s. 8(2)(d)**);
  - The **Clerk of the Privy Council** for any other person (**s. 8(2)(e)**).

- **Service Mechanics & Ministerial Certificates (s. 11–12):** Lifetime **PPBS** status takes legal effect the moment the individual is personally served with the written notice (**s. 11(1)**). Under **s. 10(3)**, the Governor General, provincial lieutenant governors, *Judges Act* judges, and *National Defence Act* military judges cannot be newly designated (though pre-existing **PPBS** status continues). Under **s. 12(1)**, a Ministerial Certificate stating that an individual was or was not a **PPBS** at a given time is admissible in evidence without proof of signature and constitutes proof of status in the absence of evidence to the contrary. Pursuant to DND/CAF CANFORGEN 077/25 and NDSOD Chapter 4, Standard 4E, all **DND/CAF** personnel and contractors authorized to access **Sensitive Compartmented Information (SCI)** are mandatorily designated as **PPBS**.

Strict Criminal Liability Offences & Narrow Statutory Defence (Sections 13–24)

- **Section 13(1)–(3) (Purported Communication of SOI):** Every **PPBS** who, intentionally and without authority, communicates or confirms information that, *if it were true*, would be **SOI** commits an indictable offence punishable by up to **5 years less a day imprisonment** (**s. 13(3)**), regardless of whether the information is actually true (**s. 13(2)**).
- **Section 14(1)–(2) (Unauthorized Communication of Actual SOI):** Every **PPBS** who, intentionally and without authority, communicates or confirms actual **SOI** commits an indictable offence punishable by up to **14 years imprisonment** (**s. 14(2)**). Sections 13 and 14 eliminate the traditional espionage requirement for the Crown to prove "purpose prejudicial to the safety or interests of the State" or proof of actual injury —establishing near-strict criminal liability once intentional unauthorized communication and **PPBS** status are proven.
- **Section 15 (Statutory Public Interest Defence Procedure):** A narrow affirmative defence available only if the accused establishes on a balance of probabilities that they disclosed an offence under an Act of Parliament committed by another person in the purported performance of duties for the Government of Canada (**s. 15(2)(a)**), that the public interest in disclosure outweighed the public interest in non-disclosure (**s. 15(2)(b)**), and—unless necessary to avoid grievous bodily harm or death (**s. 15(6)**)—that they first brought the concern to their **Deputy Head** (or the **Deputy Attorney General of Canada**) without receiving a reasonable response, and subsequently brought the concern to **NSIRA** (and, if relating to **CSE**,

formerly the CSE Commissioner) without receiving a response within a reasonable time (**s. 15(5)(a)–(b)**).

- **Espionage & Foreign Interference Offences (ss. 16–24):** Communicating safeguarded **SOI** to a foreign entity or terrorist group (Section 16(1)–(2) and Section 17(1)–(2)) carries **imprisonment for life**. Preparatory acts towards committing an offence (Section 22(1)–(2)) carry up to **5 years imprisonment**, and Bill C-70 enacted new life-imprisonment offences for committing indictable offences (**s. 20.2**), surreptitious or deceptive conduct (**s. 20.3**), or political/democratic interference (**s. 20.4**) at the direction of a foreign entity, with mandatory consecutive sentencing. All prosecutions under the Act require the personal consent of the Attorney General of Canada (**s. 24**).

8.2 Treasury Board of Canada Secretariat (TBS) Security Policy & Screening Architecture

The administrative governance of classification levels, clearance tiers, and compartmented access across federal institutions is codified in the TBS *Policy on Government Security* (effective July 1, 2019), the TBS *Directive on Security Management*, and the TBS *Directive on Security Screening* (effective January 7, 2025, superseding the 2014 *Standard on Security Screening*).

Classification / Screening Tier	Statutory / Policy Injury Threshold & Scope	Vetting & Polygraph Requirements (TBS *Directive on Security Screening*, Appendix A)	Validity & Need-to-Know / Compartmentation
Protected A / B / C	Non-national interest injury (compromise could reasonably be expected to cause injury outside the national interest, up to extremely grave injury to an individual or organization for Protected C) (PSPC *Levels of Security*).	**Reliability Status** (Protected A/B): 5-year background verification, criminal record check, and credit check. **Enhanced Reliability**: adds law enforcement record inquiry and security questionnaire/interview (Protected C).	**10 years** (Reliability). Access restricted by administrative operational need.
Confidential (Level I)	Compromise could reasonably be expected to cause **injury to the national interest** (PSPC *Levels of Security*).	10-year background check, credit/financial check, law enforcement record check, and **CSIS** security assessment under *CSIS Act*, s. 13.	**10 years**. Requires formal security briefing and signed *Security Screening Certificate and Briefing Form* (**TBS/SCT 330-47**).
Secret (Level II)	Compromise could reasonably be expected to cause **serious injury to the national interest** (PSPC *Levels of Security*).	Same baseline as Confidential (10-year history, **CSIS** loyalty/reliability check). Required for access to Consolidated Secret Network Infrastructure (CSNI) and **NATO Secret** (with separate NATO briefing).	**10 years**. Strictly governed by **need-to-know** principle.
Top Secret (Level III)	Compromise could reasonably be expected to cause **exceptionally grave injury to the national interest** (PSPC *Levels of Security*).	10-year verifiable background, foreign travel/residency & immediate relatives vetting, financial check, law enforcement inquiry, and comprehensive **CSIS** field investigation.	**5 years**. Prerequisite for non-compartmented Top Secret material and **NATO Cosmic Top Secret** (PSPC *Contract Security Manual*).
Enhanced Top Secret (Enhanced Level III)	Required for continuous access to **Sensitive Compartmented Information (SCI)**, **Special Access Programs (SAP)**, **Top Secret/Special Intelligence**	All Top Secret (**Level III**) investigations **plus**: mandatory **polygraph examination**, structured psychological/security interview, and intrusive	**5 years**. Mandatory prerequisite for **SCI**/**SAP** indoctrination, CTSN access, and mandatory PPBS designation.

Classification / Screening Tier	Statutory / Policy Injury Threshold & Scope	Vetting & Polygraph Requirements (TBS *Directive on Security Screening*, Appendix A)	Validity & Need-to-Know / Compartmentation
	(TS/SI) (**SIGINT**/**COMINT**), and core intelligence agency duties (TBS *Directive on Security Screening*; CSE Security).	lifestyle/counter-intelligence vetting (CSE Hiring & Security; CSIS Jobs).	

Compartmented Access Rules (SCI, TS/SI, and SAP)

Under the TBS *Policy on Government Security* and *NDSOD* Chapter 4, holding a **Top Secret** or **Enhanced Top Secret** clearance confers **zero automatic entitlement** to compartmented holdings. Access requires four sequential gates:

1. Verified **Enhanced Top Secret** (or reciprocally recognized Allied **TS/SCI**) security clearance;
2. Affirmative determination of a strict operational **need-to-know** validated by the compartment's Sponsor / Office of Primary Interest (**OPI**);
3. Formal **indoctrination** (execution of a compartment-specific Non-Disclosure Agreement / Indoctrination Certificate and entry into the central compartment registry); and
4. Formal designation and service as a **Person Permanently Bound to Secrecy (PPBS)** under FISOIA s. 8/10 (DND PPBS Directive).
5. **TS/SI (Top Secret / Special Intelligence — SIGINT/COMINT):** Governed by **CSE** as national **SIGINT** authority under the CANUSA/UKUSA framework. Dissemination outside accredited **SCIFs** to non-indoctrinated senior officials is controlled exclusively via **CSE Client Relations Officers (CROs)** and **CSE**-accredited **SIGINT Dissemination Officers (SDOs)** embedded across 32 federal departments (DND/CSE 2025 Ministerial Transition Binder).

Section 9: Institutional Hierarchy, Agencies, and Command Structures in Canada

9.1 Communications Security Establishment (CSE / CST)

- **Headquarters Facilities:** Edward Drake Building (**EDB**) (1929 Ogilvie Road, Ottawa, ON — inaugurated February 26, 2015, named after Lt.-Col. Edward Drake) and the Canadian Centre for Cyber Security (**Cyber Centre**) facility at 1625 Vanier Parkway, Ottawa, ON.
- **Statutory Basis:** Originally established as the Communications Branch of the National Research Council (**CBNRC**) by Order in Council in 1946, transferred to the **DND** portfolio on April 1, 1975, codified in Part V.1 of the *National Defence Act* in 2001, and governed since August 1, 2019 by its standalone charter, the *Communications Security Establishment Act* (**S.C. 2019, c. 13, s. 76**) (enacted via Bill C-59, *National Security Act, 2017*).
- **Command & Executive Hierarchy:**
  - **Responsible Cabinet Minister:** Minister of National Defence (**Hon. David McGuinty**) (DND Organizational Structure; DM Farewell Message, Jan 2026), who issues Ministerial Authorizations (approved *ex ante* by the Intelligence Commissioner), Ministerial Orders, and Directives (CSE Ministerial Transition Binder).
  - **Chief of CSE (Deputy Head):** **Caroline Xavier** (appointed August 31, 2022).
  - **Subordinate Executive Structure (CSE Organizational Structure):**
    - Head of the Canadian Centre for Cyber Security (and Associate Head);

    - Deputy Chief, Signals Intelligence (manages Foreign **SIGINT**, **TS/SI** compartments, and Five Eyes intelligence sharing);
    - Deputy Chief, Authorities, Compliance and Transparency;
    - Deputy Chief, Innovative Business Strategy and Research Development – Strategic Policy, Planning and Partnerships;
    - Deputy Chief, Enterprise Technologies and Solutions (operates the Canadian Top Secret Network (**CTSN**));
    - Deputy Chief, Corporate Services; and
    - Government of Canada Senior Official for Cyber Security.

- **Five Statutory Aspects of the Mandate (*CSE Act*, ss. 15–20; CSE Transition Binder):**
- **Foreign Signals Intelligence (SIGINT, s. 16):** Acquiring information from or through the Global Information Infrastructure (**GII**) outside Canada. In FY 2024–25, **CSE** disseminated **2,878 classified intelligence reports** to **3,016 cleared clients** across **32 federal departments** (CSE Transition Binder).
- **Cybersecurity and Information Assurance (COMSEC / CRYPTO, s. 17):** Acting as Canada's national cryptographic authority—supplying Type-1 cryptographic hardware, software, keying material, and quantum-resistant algorithms, plus administering the Canadian Industrial TEMPEST Program (**CITP**).
- **Defensive Cyber Operations (DCO, s. 18):** Taking online action to disrupt foreign cyber threats targeting federal institutions and designated systems of importance (requiring consultation with the Minister of Foreign Affairs).
- **Active Cyber Operations (ACO, s. 19):** Taking proactive online action to degrade, disrupt, influence, or interfere with foreign threat capabilities (requiring mandatory consent of the Minister of Foreign Affairs). Joint military cyber operations are conducted with the **CAF** via **Canadian Armed Forces Cyber Command (CAFCYBERCOM)** (CSE Transition Binder).
- **Technical and Operational Assistance (s. 20):** Assisting **CSIS**, **RCMP**, **CBSA**, and **DND/CAF** under their respective legal and judicial warrant authorities (**51 assistance requests** executed in FY 2024–25).
- **Five Eyes (CANUSA / UKUSA) Custody: CSE** is Canada's national signatory authority under the 1949 CANUSA Agreement (negotiated May 27 – June 29, 1949 between Canadian Communications Research Committee Chairman G.G. "Bill" Crean and **USCIB** Chairman Maj.-Gen. Charles Cabell, integrating Canada into the 1946 **BRUSA** / 1952 **UKUSA** Five Eyes alliance alongside **NSA**, **GCHQ**, **ASD**, and **GCSB**).

9.2 Canadian Security Intelligence Service (CSIS / SCRS)

- **Headquarters:** 1941 Ogilvie Road, Ottawa, ON (immediately adjacent to **CSE**'s Edward Drake Building).
- **Statutory Basis:** *Canadian Security Intelligence Service Act* (**R.S.C., 1985, c. C-23**) (enacted June 21, 1984 following the McDonald Royal Commission, separating domestic security intelligence from the **RCMP Security Service**; modernized by Bill C-59 in 2019 and Bill C-70 in 2024).
- **Command Structure:**
    - **Responsible Cabinet Minister:** Minister of Public Safety (**Hon. Gary Anandasangaree**) (Question Period Note PS-2025-QP-00013; CSE Transition Binder).
    - **Director of CSIS (Deputy Head): Dan Rogers** (appointed under *CSIS Act*, s. 4).
- **Mandate & Compartmented Authorities:**
    - **Section 12 (Security Intelligence):** Collects, analyzes, and retains intelligence on activities suspected of constituting *threats to the security of Canada* (espionage, sabotage, foreign-influenced activities,

terrorism/politically motivated violence, and subversion as defined in **s. 2**). Under Bill C-70 (**S.C. 2024, c. 16**), **CSIS** gained expanded authority to collect foreign intelligence located outside Canada from within Canada (**s. 16(1.1)**), new single-attempt judicial production and preservation orders, and broader resilience disclosures to provincial, municipal, indigenous, and private-sector entities (**s. 19**).

- **Section 12.1 (Threat Reduction Measures):** Kinetic and non-kinetic disruption of threats to the security of Canada (subject to Federal Court warrant when limiting *Charter* rights or contravening Canadian law).
- **Sections 13–15 (Government Security Screening):** Conducts the field investigations, loyalty assessments, counter-intelligence interviews, and polygraph examinations required for **Top Secret** and **Enhanced Top Secret** clearances across the Government of Canada (CSIS Public Report; TBS *Directive on Security Screening*).
- **Section 16 (Foreign Intelligence within Canada):** Collects intelligence relating to the capabilities, intentions, or activities of foreign states or persons within Canada at the written request of the Minister of Foreign Affairs or Minister of National Defence.
- **Section 18.1 (Statutory Protection of Human Source Identities):** Establishes a statutory class privilege protecting the identity of **CSIS** human sources from disclosure in judicial proceedings (complementing FISOIA s. 8(1)(a) & s. 18).

9.3 Department of National Defence (DND) & Canadian Armed Forces (CAF)

- **Headquarters:** National Defence Headquarters (**NDHQ** Carling Campus, 60 Moodie Drive, Ottawa, ON, and 101 Colonel By Drive).
- **Statutory & Policy Basis:** *National Defence Act* (**R.S.C., 1985, c. N-5**), DAOD 2006-0 (*Defence Security*), and the **National Defence Security Orders and Directives (NDSOD)** (specifically *NDSOD* Chapter 4, Standard 4E on **PPBS** and **SCI**, and Chapter 6 on Security of Information).
- **Diarchic Command Hierarchy (DND/CAF Organizational Structure):**
  - **Minister of National Defence: Hon. David McGuinty** (DM Farewell Message).
  - **Deputy Minister of National Defence (DM): Christiane Fox** (succeeded Stefanie Beck in January 2026). Serves as Deputy Head under FISOIA s. 8(2)(a)/(d) for designating **DND** civilian employees and defence contractors as **PPBS** (DND PPBS Directive).
  - **Chief of the Defence Staff (CDS): General Jennie Carignan**. Serves as Deputy Head under FISOIA s. 8(2)(b) for designating **CAF** officers and non-commissioned members as **PPBS**.
  - **Vice Chief of the Defence Staff (VCDS) & Director General Defence Security (DGDS):** Departmental Security Officer (**DSO**) authority managing the **NDSOD** suite, **SCIF** accreditation policy, and the **DND/CAF PPBS** Registry (DND PPBS Directive).
- **Canadian Forces Intelligence Command (CFINTCOM / COMRENSFC):**
  - **Command Relationship:** Level 1 (**L1**) command established in 2013; its Commander serves dual-hatted as **Commander CFINTCOM and Chief of Defence Intelligence (CDI)**, reporting jointly to the **CDS** and **DM** (DND Organizational Structure; Commander CFINTCOM Binder).
  - **Compartmented Role:** Functional authority for all **DND/CAF** Defence Intelligence, **GEOINT** (Mapping and Charting Establishment), **HUMINT** (Joint Task Force X / **JTF X**), **OSINT**, counter-intelligence (Canadian Forces National Counter-Intelligence Unit — **CFNCIU**), and **DND SCIF** accreditation—including authoring the September 2025 *National Policy to Support Industry-Built Sensitive Compartmented Information Facility (SCIF) to Enable Government of Canada Capability Development* (**Ref: CFINTCOM2025-0007425**).

- **Industrial SAPs & PSPC Contract Security Program (CSP):**
  - Managed by Public Services and Procurement Canada (**PSPC**) under the *Contract Security Manual* and International Contract Security Requirements.
  - Administers Facility Security Clearances (**FSC**), Document Safeguarding Capability (**DSC**), and bilateral industrial security instruments (such as the Canada–US Industrial Security Agreement and NATO Security Agreements for **NATO Cosmic Top Secret** / **Atomal**). For US/Allied Special Access Programs (such as **F-35 Lightning II** Joint Strike Fighter **SAR**/**SAP** compartments and **NORAD** modernization), **DND** Project Management Offices and **PSPC** execute Program Security Instructions (**PSI**), co-utilization **SCIF** agreements, and contractor **PPBS** designations under FISOIA s. 8(2)(d).

9.4 Canadian Special Operations Forces Command (CANSOFCOM) & Joint Task Force 2 (JTF 2)

- **Command Hierarchy:** Established in 2006 as a Level 1 (**L1**) operational command reporting directly to the Chief of the Defence Staff (DND Organizational Structure; CANSOFCOM Defence 101).
- **Subordinate Units (DND Info Source):**
- **Command Headquarters (CANSOFCOM HQ)** (Ottawa, ON);
- **Joint Task Force 2 (JTF 2)** — Tier 1 counter-terrorism, hostage rescue, and special reconnaissance unit established in 1993 (succeeding the **RCMP** Special Emergency Response Team) and headquartered at the **Dwyer Hill Training Centre (DHTC)** in rural southwest Ottawa, ON, currently undergoing a 1.4 billion, 10-year infrastructure modernization project (construction commenced May 2023 to build 23 new net-zero-ready facilities replacing 89 temporary structures) (2024–25 Main Estimates);
- **Canadian Special Operations Regiment (CSOR)** (Garrison Petawawa, ON);
- **Canadian Joint Incident Response Unit (CJIRU)** — Chemical, Biological, Radiological, and Nuclear (**CBRN**) SOF unit (CFB Trenton, ON);
- **427 Special Operations Aviation Squadron (427 SOAS)** (Garrison Petawawa, ON).
- **SOI Protection & Five Eyes SOF/SIGINT Integration:**
  - Operational plans, tactics, techniques, and procedures (**TTPs**), military vulnerabilities and advantages, and the identities of **JTF 2** and covert **CANSOFCOM** assaulters and operators are legally protected as **Special Operational Information (SOI)** under FISOIA s. 8(1)(b), (c), (e), and (e.1).
  - Following the August 2024 enactment of FISOIA s. 8(1)(a.3)–(a.4) and s. 9.1 (Schedule 2) and CANFORGEN 077/25, **CANSOFCOM** and **JTF 2** personnel with access to **SOI** and **SCI** are bound for life as **PPBS**.
  - **CANSOFCOM** and **JTF 2** maintain dedicated **SCIFs** and direct tactical/strategic **SIGINT** cells (supported by **CSE** and **CFINTCOM**) to integrate directly with Five Eyes Special Operations Forces counterparts (US **JSOC**, UK **UKSF**, Australian **SOCOMD**, New Zealand **NZSOC**).

9.5 Statutory Review and Oversight Bodies (NSICOP, NSIRA, and IC)

- **National Security and Intelligence Committee of Parliamentarians (NSICOP):**
  - **Statutory Basis:** *National Security and Intelligence Committee of Parliamentarians Act* (**S.C. 2017, c. 15; R.S.C. c. N-16.6**).
  - **Composition & PPBS Status:** Multi-party committee of Senators and Members of Parliament appointed by the Governor in Council. Under FISOIA s. 8(1)(a.2) and Schedule 1, all current and former **NSICOP** members and Secretariat staff are **Persons Permanently Bound to Secrecy (PPBS)**, must hold **Top Secret** clearances, and statutorily waive parliamentary privilege regarding unauthorized disclosures of **SOI** (**NSICOP Act, s. 12**).

- **Access to SAPs/SOI (ss. 13–16):** Entitled to any information under the control of any department (**s. 13**), including solicitor-client privileged information (**s. 13(2.1)**), subject to narrow statutory exclusions (**s. 14**, such as ongoing defence intelligence or law enforcement operations where the responsible Minister determines disclosure would be injurious to national security under **s. 16**, which must be reported to **NSIRA** under **s. 16(3)**).

- **National Security and Intelligence Review Agency (NSIRA):**
  - **Statutory Basis:** *National Security and Intelligence Review Agency Act* (**S.C. 2019, c. 13, s. 2; R.S.C. c. N-16.62**), enacted via Bill C-59 (replacing **SIRC** and the **Office of the CSE Commissioner**). Coordinated with **NSICOP** via the NSICOP–NSIRA Memorandum of Understanding.
  - **PPBS & Unfettered Access Authority (ss. 8–12):** Under FISOIA s. 8(1)(a.1) and Schedule 1, all **NSIRA** members and Secretariat staff are **PPBS**. Unlike **NSICOP**, **NSIRA** holds an absolute statutory right of access (**NSIRA Act, ss. 9–10**)—notwithstanding any other Act of Parliament or privilege (excluding only Cabinet confidences)—to examine **all** Special Access Programs, **SCI** compartments, human source files, and **SIGINT**/cyber operations across **CSE**, **CSIS**, **DND/CAF**, **RCMP**, **CBSA**, and **GAC**, and serves as the statutory tribunal and whistleblower recipient under FISOIA s. 15(5)(b).

- **Office of the Intelligence Commissioner (IC):**
  - **Statutory Basis:** *Intelligence Commissioner Act* (**S.C. 2019, c. 13, s. 50**). Listed in FISOIA Schedule 1. A retired superior court judge who conducts quasi-judicial *ex ante* review and must approve Ministerial Authorizations for **CSE** foreign **SIGINT**/cyber operations and **CSIS** dataset/foreign intelligence activities before they can legally take effect (CSE Transition Binder).

Section 10: Technical and Security Engineering Architecture in Canada

Technical Layer	Governing Authority & Standard	Engineering & Security Specifications
Sensitive Compartmented Information Facilities (SCIFs) & Shielded Enclosures	**CSE** / **Cyber Centre**, **CFINTCOM** (**CDI**), and **RCMP** Lead Security Agency standards; Five Eyes Intelligence Community Standard (**ICS 705-1** / **705-2** alignment); CFINTCOM Industry-Built SCIF Policy (**CFINTCOM2025-0007425**).	**High-Security Zones:** Acoustic attenuation (Sound Transmission Class **STC 45–50+** perimeter walls, baffled HVAC ducts, acoustic door seals), slab-to-slab construction, no exterior windows (or dielectric/acoustic/RF-attenuating glazing), GSA/RCMP-approved **X-10**/**CDX-10** high-security combination/biometric access control, Intrusion Detection Systems (**IDS**) with volumetric motion and duress sensors, complete ban on Personal Electronic Devices (**PEDs**), and two-person integrity / Continuous Access Roster verification.
TEMPEST & Emission Security (EMSEC)	**CSE** / **Cyber Centre** **Canadian Industrial TEMPEST Program (CITP)** (1625 Vanier Parkway, Ottawa) and **NATO SDIP-27** (**Level A / B / C**) / **NSTISSAM TEMPEST/1-92** standards.	Mitigates compromising emanations (conducted and radiated electromagnetic and acoustic signals) from ICT processing **Top Secret** / **SCI** data. Certifies **TEMPEST Level I** (**NATO Level A** — immediate proximity intercept contour), **Level II** (**NATO Level B**), and **Level III** (**NATO Level C**) hardware, RF-shielded enclosures (Faraday cages achieving **>80–100 dB** attenuation across **10 kHz–40 GHz**), power/signal line filters, and Protected Distribution Systems (**PDS**) / non-conductive fiber-optic runs.
Secret-Level Networks (GC-CSI, CSNI & StoneGhost)	Shared Services Canada (SSC Networks Program) and **DND/CAF** (CSNI).	**Government of Canada Secret Infrastructure (GC-CSI)** (managed by **SSC** across civil departments) and **Consolidated Secret Network Infrastructure (CSNI)**

Technical Layer	Governing Authority & Standard	Engineering & Security Specifications
		(managed by **DND**/**CAF** for military/defence Secret C2). Protected by **CSE**-approved Type-1 High Assurance Internet Protocol Encryptors (**HAIPE**). **CJI (Canadian Joint Infrastructure) / StoneGhost:** Five Eyes Defence Intelligence Secret/Top-Secret-bridged intelligence-sharing network linking **CFINTCOM**, US **DIA**, UK **DI**, Australian **DIO**, and New Zealand **NZDF**.
Top Secret & SCI Networks (CTSN)	Communications Security Establishment (**Canadian Top Secret Network — CTSN**, PIB **CSE PPE 001** / Class of Records **CSE MIS 001**).	Canada's primary **Top Secret** / **SCI** collaboration and dissemination fabric linking **32 federal Departments, Agencies, and Organizations (DAOs)** and bridging into the Five Eyes **Top Secret/SCI** fabric (e.g., US **JWICS** / **BICES**). Modernized by **CSE** (CTSN Modernization Report) to deliver high-assurance PKI, voice, VTC, email, web, storage, and role-based identity/access management (**CSE PPE 001**) inside accredited **SCIFs**.
Enhanced Top Secret Polygraph & PPBS Registers	TBS *Directive on Security Screening* (Appendix A); CSE, CSIS, and DND/CAF NDSOD Chapter 4 Standard 4E.	Psychophysiological detection of deception (polygraph) testing counter-intelligence (espionage, sabotage, unauthorized disclosure, unreported foreign contacts) and suitability/lifestyle honesty (minimum 1-year pre-application illicit drug abstinence at **CSE**). Indoctrinated personnel are recorded in departmental **PPBS** registers (**DND DGDS**, **CSE**, **CSIS**, **RCMP**) backed by signed **Section 10(1)** Notices and Ministerial Certificates (FISOIA ss. 10–12).

Section 11: Deterministic Causal Chains, Historical Evolution, and Key Espionage Cases in Canada

1. **Cold War *Official Secrets Act* to the 2001 *Security of Information Act* (SOIA):**
2. **Historical Trigger:** Following the September 1945 Igor Gouzenko defection in Ottawa (investigated by the Kellock-Taschereau Royal Commission), the 1969 MacKenzie Royal Commission on Security, the 1981 McDonald Royal Commission Concerning Certain Activities of the RCMP, and the failed *Official Secrets Act* prosecution in *R. v. Toronto Sun Publishing Ltd.* (1979), Canadian officials determined that the 1939 *Official Secrets Act* was practically unenforceable against modern insider disclosures and lacked a statutory mechanism to permanently bind intelligence officers or protect compartmented Five Eyes equities without proving intent to aid an enemy.
3. **Statutory Result:** Driven by the September 11, 2001 attacks and Five Eyes intelligence-sharing harmonization imperatives, Parliament enacted the *Anti-terrorism Act* (S.C. 2001, c. 41, ss. 24–30), replacing the *Official Secrets Act* with the *Security of Information Act* (**SOIA**) and creating the **SOI** (**s. 8(1)**) and **PPBS** (**ss. 8–15**) strict-secrecy regime.
4. **The Sub-Lieutenant Jeffrey Paul Delisle Espionage Case (HMCS Trinity / StoneGhost, 2007–2013):**
5. **Operational & Technical Breach:** Between July 2007 and his arrest in January 2012, Sub-Lieutenant Jeffrey Paul Delisle, a **CAF** naval intelligence officer posted to the all-source intelligence centre **HMCS Trinity** at CFB Halifax (as well as earlier postings at **NDHQ**/**CDI** in Ottawa), exploited unmonitored floppy-disk and USB thumb-drive ports on classified terminals inside an accredited facility. Delisle queried and copied Five Eyes intelligence from the **StoneGhost (CJI)** and Secret/Top Secret networks onto removable media and sold the files to the Russian Main Intelligence Directorate (**GRU**).

6. **Legal & Military Outcome:** Delisle pleaded guilty in October 2012 to communicating safeguarded information to a foreign entity (SOIA s. 16(1)) and breach of trust (*Criminal Code*, **s. 122**)—becoming the **first person convicted under the 2001 SOIA**. On February 8, 2013, he was sentenced to **20 years imprisonment** (less remand credit) and fined **111,817**. Following sentencing, Governor General David Johnston, on the recommendation of **CDS** General Tom Lawson and Defence Minister Peter MacKay, revoked Delisle's commission, released him from the CAF for service misconduct, forfeited his Canadian Forces Decoration and severance pay, and ordered recovery of his military pay (Minister MacKay Statement, Feb 2013; CDS Statement, Feb 2013).
7. **Direct Causal Engineering & Policy Reforms:**
   - **DND Security Transformation & CFINTCOM Creation:** In October 2012, **DND** established the **Director General Security Transformation** organization to rewrite the entire National Defence Security Orders and Directives (**NDSOD**) and centralized military intelligence under **CFINTCOM** (created in 2013).
   - **Hardware & Network Lockdown:** Across all **DND/CAF SCIFs**, **HMCS Trinity**, **CSNI**, **StoneGhost**, and **CTSN** terminals, USB and removable media ports were physically disabled/epoxied and logically blocked at the kernel and group-policy level; mandatory two-person media transfer controls and automated Data Loss Prevention (**DLP**) audit logging were deployed (CTSN Architecture).
   - **Creation of the TBS Enhanced Top Secret Tier:** Directly driven by the Delisle post-mortem (where Delisle had passed a standard **Top Secret** renewal despite severe financial distress and marital breakdown), **TBS** promulgated the October 20, 2014 *Standard on Security Screening* (now the 2025 *Directive on Security Screening*), creating the formal government-wide **Enhanced Top Secret** clearance tier mandating **polygraph examinations** and deep financial and counter-intelligence vetting for all personnel accessing **SIGINT/SCI** compartments.
8. **The Cameron Jay Ortis Case (RCMP ORI / NICC & Judicial Rulings on SOIA/PPBS, 2015–2026):**
9. **Operational Breach:** Cameron Jay Ortis, a civilian PhD hired into the **RCMP** who rose to become Director General of the **RCMP**'s **National Intelligence Coordination Centre (NICC)** and previously led **Operations Research and Information (ORI)**—a highly compartmented unit acting as the interface between **CSE**/Five Eyes **TS/SI SIGINT** and **RCMP** national security and transnational organized crime investigations—was designated a **Person Permanently Bound to Secrecy (PPBS)**. In 2015, Ortis printed and exfiltrated classified **SOI** and Five Eyes intelligence packages and anonymously contacted transnational money-laundering and criminal targets (including Phantom Secure CEO Vincent Ramos) offering to sell them the intelligence. After US/Five Eyes authorities seized Phantom Secure servers and uncovered the leaked Canadian documents, the **RCMP** arrested Ortis on September 12, 2019 (Public Safety QP Note PS-2019-QP-00004).
10. **Judicial Rulings (*Canada Evidence Act* s. 38 & SOIA Constitutionality):**
    - **Bifurcated Jurisdiction & Section 38 Litigation:** Because Ortis was charged with communicating **SOI** under SOIA s. 14(1), his prosecution required intensive litigation under **Section 38** of the *Canada Evidence Act* (**CEA**) before the Federal Court and Federal Court of Appeal to balance the protection of Five Eyes **TS/SI** secrets against Ortis's right to make full answer and defence under **s. 7** and **s. 11(d)** of the *Canadian Charter of Rights and Freedoms*, allowing Ortis to testify in a heavily guarded *in camera* / redacted framework (PPSC 2024 Transition Book).
    - **First Contested Jury Conviction under SOIA s. 14(1):** On November 22, 2023, an Ontario Superior Court of Justice jury in Ottawa convicted Ortis on all six counts: four counts of intentionally and without authority communicating **SOI** while being a **PPBS** contrary to **SOIA s. 14(1)**, one count of breach of trust (*Criminal Code*, **s. 122**), and one count of fraudulently obtaining a computer service (*Criminal Code*, **s. 342.1(1)**). On February 7, 2024, Ortis was sentenced to **14 years imprisonment**,

and the Public Prosecution Service of Canada (**PPSC**) filed an appeal to the Court of Appeal for Ontario seeking a higher sentence (20+ years).

1. **Causal Legislative & Institutional Reforms (2019–2026):**
    - **RCMP & Insider Threat Overhaul:** Following Ortis's arrest, the **RCMP** appointed a dedicated Chief Security Officer under the 2019 **TBS** *Policy on Government Security* (Public Safety QP Note PS-2019-QP-00004), subjected senior civilian intelligence executives and **ORI**/**NICC** personnel to **CSE**/**CSIS**-grade **Enhanced Top Secret** polygraph vetting, and **NSICOP** and **NSIRA** conducted reviews of internal security and intelligence handling inside the **RCMP**.
    - **Bill C-70 Statutory Reforms (2024–2025):** Delays caused by pre-trial **Section 38 CEA** appeals in national security prosecutions like *Ortis* directly prompted Parliament in Bill C-70 (**S.C. 2024, c. 16**) to amend the *Canada Evidence Act* so that an accused in a criminal proceeding may appeal a **Section 38** disclosure decision **only after conviction**, absent exceptional circumstances. Simultaneously, Bill C-70 expanded FISOIA s. 8(1)(e.1) to protect military vulnerabilities and advantages, added **Schedule 2** (**s. 8(1)(a.3)–(a.4)**, **s. 9.1**) for unit-based **CAF PPBS** indoctrination, and catalyzed **DND**'s July 2025 CANFORGEN 077/25 mandate designating every **SCI**-cleared military member, civilian employee, and contractor in Canada as a **Person Permanently Bound to Secrecy**.

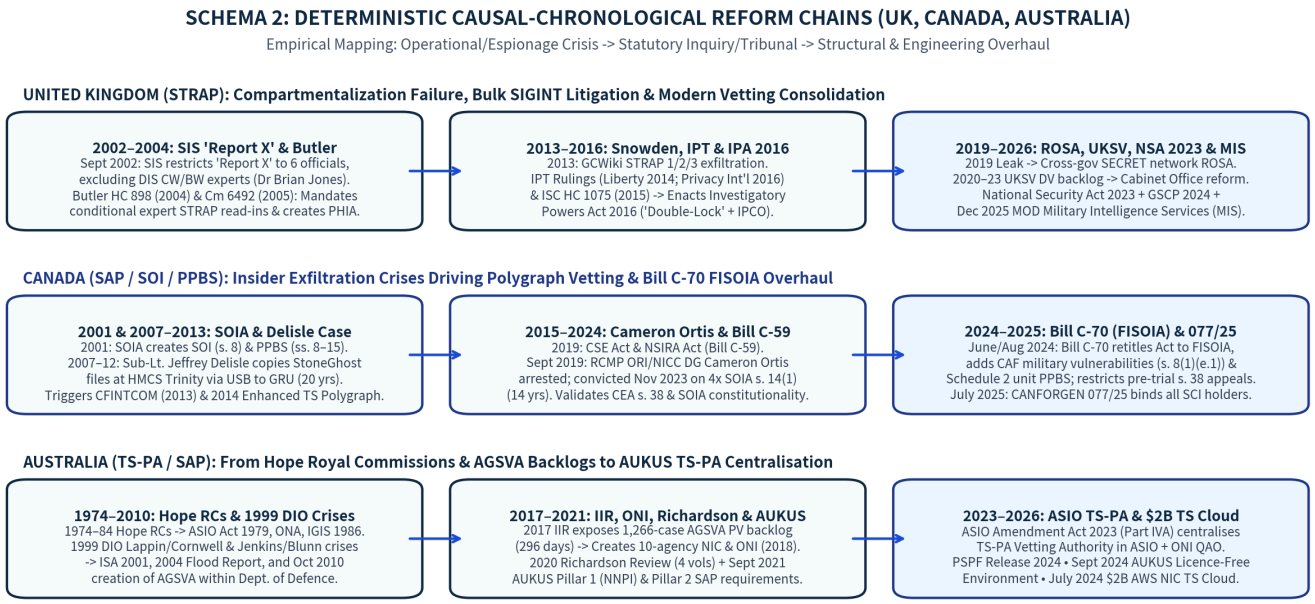


Figure 2: Deterministic Causal-Chronological Chains across the United Kingdom (Butler/Chilcot, IPT/Snowden, ROSA/NSA 2023), Canada (SOIA 2001, Delisle/StoneGhost, Ortis/Bill C-70 FISOIA), and Australia (Hope Royal Commissions, 1999 DIO Crises, 2017 IIR, AUKUS & ASIO Amendment Act 2023).

Part III: Special Access Programs (SAP), Top Secret-Privileged Access (TS-PA), and the National Intelligence Community (NIC) in Australia

Australia's highest-tier protective security and compartmentalized intelligence architecture underwent its most significant structural transformation since the 1977 Hope Royal Commission between 2023 and 2026. Driven by state-sponsored espionage, foreign interference, and the trilateral technology-sharing imperatives

of the AUKUS Nuclear-Powered Submarine Pathway, the Commonwealth enacted the Australian Security Intelligence Organisation Amendment Act 2023 (No. 33, 2023) and amended the Protective Security Policy Framework (PSPF) to replace decentralized **Top Secret (Positive Vetting)** (**PV**) arrangements with a single, standardized, and transferable **Top Secret-Privileged Access** (**TS-PA**) clearance administered centrally by the Australian Security Intelligence Organisation (ASIO) and independently audited by a statutory **Quality Assurance Office (QAO)** within the Office of National Intelligence (ONI).

Section 12: Protective Security Policy Framework (PSPF) & Clearance Engineering (PV to TS-PA)

12.1 Statutory & Policy Transition from Positive Vetting (PV) to Top Secret-Privileged Access (TS-PA)

Historically, Australia's highest security clearance level—**Top Secret (Positive Vetting)** (**TS (PV)**)—was governed under the Sensitive Material Security Management Protocol (SMSMP) – Personnel Security – Positive Vetting Guidelines and administered across five separate Top Secret Authorised Vetting Agencies (**AVAs**), alongside lower-tier vetting agencies (ASIO Annual Report 2023–24: Top Secret-Privileged Access Vetting; PSPF Glossary):

- **Decentralized Pre-2023 Vetting Fragmentation:** Under the legacy model, the Australian Government Security Vetting Agency (**AGSVA**) (established on 1 October 2010 within the Department of Defence), **ASIO**, the Australian Secret Intelligence Service (**ASIS**), the Office of National Intelligence (**ONI**), the Department of Foreign Affairs and Trade (**DFAT**), the Australian Federal Police (**AFP**), and the Australian Securities and Investments Commission (**ASIC**) operated as Authorised Vetting Agencies. As documented by AGSVA Assistant Secretary Vetting Mel Flett on 20 November 2025, differing agency implementations of **Positive Vetting** prevented seamless inter-agency recognition and required personnel moving between National Intelligence Community (**NIC**) and Defence compartments to undergo duplicative re-vetting.
- **Statutory Centralisation (ASIO Amendment Act 2023):** Prior to 2023, **ASIO**'s statutory functions under subsection **17(1)** of the Australian Security Intelligence Organisation Act 1979 were restricted to furnishing advisory security assessments under **Part IV** and did not empower **ASIO** to issue final security clearance decisions for non-ASIO personnel (Australian Security Intelligence Organisation Amendment Bill 2023 Explanatory Memorandum). Enacted as Act No. 33 of 2023 and commencing on 1 July 2023 (IGIS Act Series Register C2023C00168), the **Australian Security Intelligence Organisation Amendment Act 2023** inserted a new **Part IVA** (**Security vetting and security clearance related activities**, encompassing sections **82A** through **82N**) into the **ASIO Act 1979**, establishing the **TOP SECRET-Privileged Access (TS-PA) Vetting Authority** within **ASIO**.
- **PSPF Policy Amendments & PSPF Release 2024:** To operationalize the statutory reform across non-corporate Commonwealth entities under the **Public Governance, Performance and Accountability Act 2013** (**PGPA Act**), the Department of Home Affairs amended PSPF Policy 12: Eligibility and suitability of personnel and PSPF Policy 13: Ongoing assessment of personnel to give effect to the **TOP SECRET-Privileged Access Standard** and designate the **TS-PA Authority** as a vetting authority. Concurrently, mandatory direction powers were operationalized under **PSPF Policy 1: Role of accountable authority** and **PSPF Policy 3: Security planning and risk management**—beginning with PSPF Direction 001-2023 (issued 4 April 2023 restricting high-risk foreign applications on government devices) and **PSPF Direction 001-2024** (mandating Foreign Ownership, Control or Influence [**FOCI**] risk identification across technology assets) (Protective Security Directions under the PSPF). On 1 November 2024, Home Affairs launched PSPF Release 2024 (followed by revised PSPF Release 2024 Guidelines on 18 December 2024), transitioning the Commonwealth from a self-assessed maturity model to a strict

compliance reporting model (`Fully implemented`, `Risk managed`, `Not yet implemented`), expanding protective security domains to incorporate dedicated **Risk** and **Technology** domains alongside Governance, Information, Personnel, and Physical security, and removing the legacy Commonwealth statutory declaration requirement under PSPF Policy 12 to align lower-tier clearances (`Baseline`, `NV1`, `NV2` on `AGSVA`'s `myClearance` portal) with the digital `TS-PA` workflow.

Statutory Provision / Instrument	Administering Entity	Legal & Operational Mechanism	Primary Source Verification
`ASIO Act 1979 s 82C(1) & s 82D` (inserted by Act No. 33, 2023)	ASIO TS-PA Vetting Authority	Confers statutory functions and powers on **`ASIO`** (exercisable by the Director-General of Security under **`s 82D`**) to undertake security vetting, make **`security clearance decisions`** (**`SCDs`**), and issue **`security clearance suitability assessments`** (**`SCSAs`**) for both **`ASIO`** and non-**`ASIO`** Commonwealth/industry personnel, and to assume responsibility for existing **`PV`** clearances granted by other vetting agencies.	ASIO Amendment Bill 2023 Explanatory Memorandum
`ASIO Act 1979 s 82E & s 82F` (inserted by Act No. 33, 2023)	ASIO TS-PA Vetting Authority	Authorizes **`ASIO`** to request (**`s 82F`**) and communicate (**`s 82E`**) personal and security-vetting information continuously across Commonwealth, state/territory, and sponsoring entities without prior subject notification, enabling real-time continuous assessment and insider-threat detection.	ASIO Amendment Bill 2023 Explanatory Memorandum
`ASIO Act 1979 Part IVA Div 3` (Internal, Independent & AAT Merits Review)	**`ASIO`** / Attorney-General / **`AAT`** (**`ART`**) / Minister for Home Affairs	Establishes a three-tier review architecture for **`Prejudicial Security Clearance Decisions`** and **`Prejudicial Security Clearance Suitability Assessments`** (**`PSCSAs`**): (1) statutory internal merits review within **`ASIO`**; (2) independent review by an Attorney-General appointee for non-Commonwealth employees/non-clearance holders; and (3) external merits review in the Security Division of the Tribunal, subject to the Minister for Home Affairs' power to issue a conclusive certificate in exceptional national security circumstances.	ASIO Amendment Bill 2023 Explanatory Memorandum
`ONI Act 2018 Amendments (Quality Assurance Office - QAO)`	Office of National Intelligence (ONI)	Establishes the statutory **`Quality Assurance Office (QAO)`** within **`ONI`** to independently audit and assure the quality, consistency, and inter-agency transferability of **`TS-PA`** clearances and drive insider-threat capability uplift across all sponsoring agencies.	ASIO Amendment Bill 2023 Explanatory Memorandum
`PSPF Policy 12 & Policy 13 / PSPF Release 2024`	Department of Home Affairs	Enforces the **`TOP SECRET-Privileged Access Standard`**, mandates agency-level Insider Threat Programs (aligned with **`PGPA Rule 2014 s 10`** fraud/corruption controls), and governs the phased transfer of **`AGSVA`** and agency **`PV`** cohorts to **`ASIO`**'s **`TS-PA Vetting Authority`** at revalidation.	PSPF Personnel Security Amendment; ATO Fraud & Corruption Plan 2026

12.2 Technical & Procedural Vetting Engineering: Baseline to `TS-PA` / `Positive Vetting`

Under the PSPF and AGSVA Security Clearance Definitions, personnel security vetting evaluates whether a candidate possesses and demonstrates an appropriate standard of **integrity**—defined in Australian protective security doctrine across six core character traits: **honesty, trustworthiness, maturity, tolerance, resilience, and loyalty**—and ensures the subject is free from undue vulnerability to foreign influence, blackmail, or coercion (ANAO Audit Report: myClearance Implementation; ONI Security Vetting).

Clearance Tier	Vetting Authority	Maximum Classification & Compartment Access	Checkable Background Period	Referee & Interview Requirements	Psychological & Financial Vetting	Revalidation Cycle & AGSVA Fee (2026–27 ex GST)
Baseline	AGSVA	Ongoing access up to **PROTECTED** (**BIL 3 - High**)	Preceding **5 years** (AGSVA Application Information)	1 professional referee (supervising ≥ 3 months); no routine interview (AGSVA Nominating Referees)	Financial documents and psychological assessment only if triggered by risk indicators	**15 years** (AGSVA Revalidation); **810.91** (AGSVA Fees)
Negative Vetting Level 1 (NV1)	AGSVA	Ongoing access up to **SECRET** (**BIL 4 - Extreme**) and temporary access to **TOP SECRET** under strict conditions	Preceding **10 years** (AGSVA Application Information)	1 professional referee (≥ 3 months) + 1 personal referee covering past 10 years (AGSVA Nominating Referees)	**ASIO** security assessment (**Part IV**), **AFP** criminal history check, credit/financial probity check	**10 years** (AGSVA Revalidation); **1,724.55** (AGSVA Fees)
Negative Vetting Level 2 (NV2)	AGSVA	Ongoing access up to **TOP SECRET** (**BIL 5 - Catastrophic**), excluding **TS-PA** / **PV** caveated and codeword compartments	Preceding **10 years** (AGSVA Application Information)	1 professional referee (≥ 3 months) + 1 personal referee covering past 10 years; **mandatory security interview** (Why Does AGSVA Get Personal)	Mandatory vetting interview, **ASIO** check, digital footprint & financial/credit verification	**5 to 7 years** (AGSVA Revalidation); **3,445.45** (AGSVA Fees)
Positive Vetting (PV) / Top Secret-Privileged Access (TS-PA)	Phased transition from AGSVA / legacy AVAs to **ASIO TS-PA Vetting Authority** (ASIO Annual Report 2024–25)	Ongoing access up to **TOP SECRET**, including **Privileged Access (TS//Special Access)**, **Codeword**, **Sensitive Compartmented Information (SCI)**, and **Special Access Programs (SAP)** (AGSVA Guide Book)	**Whole-of-life** / since **age 16 or past 10 years** (whichever is greater), plus **10 years** documented residential & employment proofs and **all** current/expired passports (AGSVA Application Information; INSLM Eligibility)	**5 referees** minimum (1 professional supervisor ≥ 3 months + 4 personal referees collectively covering life since age 16/past 10 years); **in-depth face-to-face subject & referee security interviews** (AGSVA	**Mandatory two-part Psychological Assessment** (written psychometric testing + clinical interview with registered psychologist), exhaustive forensic financial disclosures, digital footprint/social media audit, and **Annual**	**5 to 7 years** full revalidation + **Annual Security Appraisal** (AGSVA Revalidation); **16,003.64** (**PV** fee; benchmark processing 180 business days / 12–24 months for NIC graduates) (AGSVA

Clearance Tier	Vetting Authority	Maximum Classification & Compartment Access	Checkable Background Period	Referee & Interview Requirements	Psychological & Financial Vetting	Revalidation Cycle & AGSVA Fee (2026–27 ex GST)
				Nominating Referees)	**Security Appraisal** (AGSVA Application Information; AGSVA Revalidation)	Fees; NIC Graduate Stream)

12.3 Granular Technical Requirements of `TS-PA` and `Positive Vetting`

1. **Sponsorship & Designated Security Assessed Positions (`DSAP`):** A security clearance must be sponsored by an Australian Government **Sponsoring Entity** (PSPF Glossary) against a validated billet on the entity's **Designated Security Assessed Positions (`DSAP`) Register** (DISP FAQs). Under the Defence Security Principles Framework (**`DSPF`**), private-sector **Defence Industry Security Program (`DISP`) members are prohibited from directly sponsoring `Positive Vetting` / `TS-PA` clearances**; where a contractor requires **`PV`** / **`TS-PA`** access for a classified Defence contract or Special Access Program, the clearance must be sponsored directly by the Commonwealth Project Security Officer (**`SO`**) or sponsoring government agency. Within **`ASIO`**, employees maintaining a **`TS-PA`** clearance receive a dedicated **7.5% salary allowance** recognising the intrusive ongoing lifestyle and reporting obligations (ASIO Job Details – TS-PA Allowance).
2. **Two-Stage Clinical Psychological & Psychometric Assessment:** Mandatory for all **`PV`** and **`TS-PA`** applicants (AGSVA Application Information; ONI Security Vetting; DSTG Employment Eligibility):
3. **Stage 1 (Psychometric Testing):** Standardized written clinical and personality psychometric test batteries administered under controlled conditions to screen for personality pathology, emotional instability, impulsivity, narcissism, and susceptibility to manipulation or coercion.
4. **Stage 2 (Registered Psychologist Clinical Interview & Report):** An in-person, face-to-face clinical interview conducted by a registered psychologist in compliance with Australian psychology registration board standards, probing personal relationships, living circumstances, personal values, financial stressors, physical and mental health history, alcohol and illicit substance use, and civil/military behavioral records, culminating in a formal psychological suitability report to the vetting delegate (DSTG Information Pack). (Where a **`PV`** clearance is cancelled at the psychological assessment stage, **`AGSVA`**'s 2026–27 schedule levies a **5,445.45** stage-cancellation fee, rising to **13,560.00** at the vetting assessment stage and **16,003.64** at delegation: AGSVA Fees).
5. **Forensic Financial & Asset Vetting:** **`PV`** / **`TS-PA`** mandates a comprehensive **Financial Questionnaire** and primary-source verification (AGSVA Application Information):
6. Gross and net annual salary, average fortnightly expenditure, and all secondary income streams (investment properties, trusts, pensions, government payments, child support).
7. Full itemization of current and previous real estate holdings (purchase dates, prices, valuations, mortgage providers, and loan account numbers), motor vehicles (make, model, registration, value, loan details), and high-value physical assets (boats, caravans, jewellery, artwork).
8. Complete disclosure of cash accounts, investment/brokerage portfolios, superannuation balances, business ownership or board directorships, personal loans, credit card limits and balances, and **Buy-Now-Pay-Later (`BNPL`)** accounts (e.g., Afterpay, Zip Pay, Humm).

9. Mandatory submission of the applicant's **latest Australian Taxation Office (`ATO`) tax return and Notice of Assessment** (not older than 18 months) plus **3 consecutive months of complete transaction statements** covering the exact same timeframe across *all* financial accounts held by the applicant and joint account holders (AGSVA Application Information).
10. **Whole-of-Life Background, Kinship, Foreign Contact & Digital Footprint Vetting:**
11. **Residential, Educational & Travel Provenance:** Full residential address history, employment history, educational history, and international travel history **since the age of 16 or the past 10 years (whichever is greater)**, backed by documentary evidence for the last 10 years of addresses and employment, plus physical inspection of **all current and expired Australian and foreign passports** including every stamped page (AGSVA Application Information).
12. **Extended Kinship & Former Partner Mapping: `PV` / `TS-PA`** mandates full biographical, citizenship, residential, employment, overseas travel, and foreign contact disclosures for **all siblings, children, and cohabitants (regardless of age)** as well as **former partners** (AGSVA Application Information), alongside regular contacts outside Australia and any contact with foreign government officials.
13. **Digital Footprint & Open-Source Intelligence (`OSINT`) Vetting:** Applicants must declare all current and historical online accounts, social media handles, professional networking profiles, and blogs (AGSVA Application Information), which are subjected to dedicated digital footprint checks (ONI Security Vetting) and cross-referenced against **`ASIO`** intelligence holdings, **`AFP`** law enforcement databases, and external credit bureaus (Why Does AGSVA Get Personal).
14. **Forensic Security Interviews, Continuous Assessment & Annual Security Appraisals:**
15. Australian personnel security vetting under the PSPF and AGSVA Applicant Guide Book relies on **in-depth, structured forensic security interviews** of both the candidate and their 5+ nominated referees (who are interviewed face-to-face, by video, or by telephone by vetting analysts), combined with clinical psychological evaluation and **`ASIO`** counterintelligence holdings checks under **`Part IVA`** of the **`ASIO Act 1979`** (AGSVA Nominating Referees; ONI Security Vetting).
16. **Annual Maintenance & Continuous Assessment:** Holding a **`PV`** / **`TS-PA`** clearance requires completing a mandatory **Annual Security Appraisal**, which audits compliance with change-of-circumstances reporting, security incident history, mandatory security awareness training, and a formal **annual supervisory referee report** from the holder's current manager (AGSVA Revalidation and Review Activities), alongside continuous real-time assessment under **`ss. 82E–82F`** of the **`ASIO Act 1979`** and full revalidation every **5 to 7 years**.

12.4 Need-to-Know, Security Caveats, `SCI`, `TS//Special Access`, and `SAP`

Under PSPF Policy 8: Sensitive and classified information, **`PSPF Policy 9: Access to information`**, and the **Australian Government Security Caveat Standard** (updated by the Government Security Committee on 18 November 2024 and published 18 December 2024: PSPF Guidelines and Release 2024 Update), access to compartmented holdings requires the conjunction of three independent gates:

1. **Security Clearance Eligibility:** A current, active **`TS-PA`** or **`PV`** clearance commensurate with **TOP SECRET** (**`Business Impact Level [BIL] 5 – Catastrophic`**, defined as material whose compromise would cause *exceptionally grave damage to the national interest*) (INFOSEC-2: Protecting official information).
2. **Need-to-Know Determination:** Affirmative operational verification by the **Originator** or **Controlling Authority** (PSPF Glossary) that the individual requires access to perform their official duties.
3. **Compartment Indoctrination / Security Caveat Briefing (`TS//Special Access`, `SCI`, `SAP`):** Formal briefing and registration into the specific **Security Caveat** administered by a designated **Controlling Authority**:

4. **Codeword, `TS//Special Access` & Sensitive Compartmented Information (`SCI`) Compartments:** Intelligence-derived compartments protecting sensitive collection sources, methods, and analytical products (such as **`UKUSA`** / Five Eyes signals, geospatial, and human intelligence compartments) that require **`TS-PA`** / **`PV`** clearance, formal indoctrination ("reading in"), and storage/processing exclusively inside an accredited **Zone 5 Sensitive Compartmented Information Facility (`SCIF`)** (AFP LEX271; AGSVA Applicant Guide Book).
5. **Special Access Programs (`SAP`):** Compartmented capability acquisition, research, technology development, and tactical/operational programs established under the **`Defence Security Principles Framework (`**DSPF**`)`** and managed within the Department of Defence—including a dedicated **Special Access Programs** branch inside the **Planning and Assurance** Directorate of **Defence Science and Technology Group Headquarters (`DSTG HQ`)** (DSTG Headquarters Structure) and **`ADF`** capability development streams (ADF Network-Driven System Design)—to impose heightened access lists, billet caps, and physical/ICT isolation beyond standard classification controls.
6. **Releasability Caveats (`AGAO`, `AUSTEO`, `REL`):** Under an amendment to PSPF Policy 8 approved by the National Intelligence Subcommittee, the **Australian Government Access Only (`AGAO`)** caveat was updated to allow all 10 **`NIC`** agencies and the Department of Defence to share **`AGAO`**-marked material with embedded Five Eyes (**`FVEY`**) exchange or attached officers, while **Australian Eyes Only (`AUSTEO`)** remains strictly restricted to Australian citizens. Additionally, the Government Security Committee amended electronic handling rules for the **Releasable To (`REL`)** caveat (e.g., **`REL AUS, GBR, USA`** or **`REL FVEY`**) so that even when applied to **`PROTECTED`** information, electronic **`REL`**-caveated material must be processed exclusively on **SECRET** or **TOP SECRET** ICT networks capable of enforcing technical releasability controls (PSPF Security Caveat Changes).

Section 13: National Intelligence Community (`NIC`) Hierarchy, Statutory Authorities, and Functions

Australia's **National Intelligence Community (`NIC`)** comprises **10 agencies** working under common Australian Intelligence Missions (National Intelligence Community Agencies): the six core intelligence agencies—**`ONI`**, **`ASD`**, **`ASIO`**, **`ASIS`**, **`AGO`**, and **`DIO`**—plus the Australian Criminal Intelligence Commission (**`ACIC`**), the Australian Transaction Reports and Analysis Centre (**`AUSTRAC`**), the intelligence functions of the Australian Federal Police (**`AFP`**), and the intelligence functions of the Department of Home Affairs.

NIC Entity / Oversight Body	Governing Statute	Portfolio & Ministerial Accountability	Agency Head / Statutory Officeholder	Core Statutory Mandate & Compartmented Functions
Office of National Intelligence (`ONI`)	Office of National Intelligence Act 2018 (No. 155, 2018)	Prime Minister and Cabinet (**`PM&C`**) — Prime Minister of Australia	**Director-General of National Intelligence (`DGNI`)**: Mr Andrew Shearer (ONI DGNI)	Enterprise leadership and integration of the 10-agency **`NIC`**; principal intelligence advisor to the Prime Minister; all-source national assessments; Open Source intelligence; program sponsor of the **Top Secret Cloud**; houses the statutory **Quality Assurance Office (`QAO`)** auditing **`TS-PA`** clearances (ONI Overview; Top Secret Cloud Announcement).
Australian Signals	Intelligence Services Act 2001 (No. 152,	Defence Portfolio — Deputy Prime Minister & Minister	**Director-General ASD**: Ms Abigail Bradshaw CSC	Foreign Signals Intelligence (**`SIGINT`**), offensive cyber operations, Australian Cyber

NIC Entity / Oversight Body	Governing Statute	Portfolio & Ministerial Accountability	Agency Head / Statutory Officeholder	Core Statutory Mandate & Compartmented Functions
Directorate (ASD)	2001) (as amended in 2018)	for Defence (The Hon Richard Marles MP)	(appointed September 2024, succeeding Ms Rachel Noble PSM) (ASD Who We Are)	Security Centre (**ACSC**), **REDSPICE** (9.9B capability expansion), **UKUSA**/Five Eyes **SIGINT** operations, Australian lead at **Joint Defence Facility Pine Gap (JDFPG)**, and national **TOP SECRET ICT System Authorisation to Operate** & cryptographic **Australian Key Management Infrastructure (AKMI)** (ASD Annual Report 2025–26).
Australian Security Intelligence Organisation (ASIO)	Australian Security Intelligence Organisation Act 1979 (No. 113, 1979) & ASIO Amendment Act 2023 (No. 33, 2023)	Home Affairs / Attorney-General's Portfolio (warrants issued by Attorney-General)	**Director-General of Security**: Mr Mike Burgess AM (ASIO Annual Threat Assessment 2026)	Domestic security intelligence (counter-espionage, foreign interference, sabotage, counter-terrorism); operates the **TS-PA Vetting Authority** (**Part IVA ASIO Act**) centralising all Commonwealth Top Secret clearances; operates **ASIO T4 Protective Security** and manages the **Security Construction and Equipment Committee (SCEC)** certifying Zone 5/**SCIF** facilities (PSPF Glossary; ASIO Corporate Plan 2026–30).
Australian Secret Intelligence Service (ASIS)	Intelligence Services Act 2001 (No. 152, 2001)	Foreign Affairs and Trade (**DFAT**) — Minister for Foreign Affairs (Senator the Hon Penny Wong)	**Director-General ASIS**: Ms Kerri Hartland (ASIS Leadership / ASIS Official)	Overseas human intelligence (**HUMINT**) collection, counter-intelligence outside Australia, and covert special operations/other activities directed by the Minister for Foreign Affairs under **s 6(1)(e)** of the **ISA 2001** (ASIS Who We Are).
Australian Geospatial-Intelligence Organisation (AGO)	Intelligence Services Act 2001 (**s 6B**)	Department of Defence (Defence Intelligence Group - **DIG**)	**Chief of Defence Intelligence (CDI)**: Mr Tom Hamilton (from 6 July 2024); **Director AGO**: Ms Kathryn McMullan (CDI Profile; AGO 25th Anniversary)	Lead agency for geospatial data, overhead imagery intelligence (**IMINT**), and geospatial intelligence (**GEOINT**) under the AGO Strategy 2025–2027, supporting **ADF** targeting, Five Eyes **GEOINT** sharing, and whole-of-government security.
Defence Intelligence Organisation (DIO)	Executive agency governed by the DIO Mandate & **ISA 2001** (**s 15** privacy rules)	Department of Defence (Defence Intelligence Group - **DIG**)	**Chief of Defence Intelligence (CDI)**: Mr Tom Hamilton (CDI Profile)	Strategic, all-source military, scientific, and technical intelligence assessment agency evaluating foreign weapons systems, WMD proliferation, and order-of-battle capabilities for the Minister for Defence and **ADF** operations (DIO Overview).

NIC Entity / Oversight Body	Governing Statute	Portfolio & Ministerial Accountability	Agency Head / Statutory Officeholder	Core Statutory Mandate & Compartmented Functions
Inspector-General of Intelligence and Security (IGIS)	Inspector-General of Intelligence and Security Act 1986 (No. 101, 1986)	Independent statutory officeholder appointed by the Governor-General	**Inspector-General**: Mr Michael Barnes (commenced 7 April 2026) (IGIS Profile)	Independent Royal Commission-equivalent oversight of the legality, propriety, ministerial direction compliance, and human rights impact of **NIC** agencies, including mandatory annual inspections of the **ASIO TS-PA Vetting Authority** (ASIO Annual Report 2024–25).
Parliamentary Joint Committee on Intelligence and Security (PJCIS)	Intelligence Services Act 2001 (**Part 4, s 28–s 31**)	Parliament of Australia (Joint Senate & House of Representatives Committee)	**Chair (48th Parliament)**: Senator Raff Ciccone; **Deputy Chair**: Mr Andrew Wallace MP (Senate Hansard PJCIS)	Parliamentary oversight of the administration and expenditure (including financial statements) of the intelligence agencies and scrutiny of national security bills, while statutorily barred from reviewing specific intelligence operations or sources (Defence Intelligence Controls).

13.1 Detailed Agency Statutory Authorities & Compartmented Operations

- **Office of National Intelligence (ONI):** Established on 20 December 2018 under the Office of National Intelligence Act 2018 (No. 155, 2018) (superseding the **Office of National Assessments Act 1977**), **ONI** is led by **Director-General of National Intelligence (DGNI) Andrew Shearer** (ONI Agency Profile). Under **s 7** and **s 9** of the **ONI Act 2018**, the **DGNI** is the Prime Minister's principal advisor on national intelligence priorities and leads the integration of the 10-agency **NIC** (National Intelligence Community). **ONI** hosts the multi-agency Cyber and Critical Technology Intelligence Centre (**CCTIC**) and the National Intelligence Academy, chairs the National Intelligence Subcommittee (**NISC**) setting whole-of-government intelligence security caveats (**PSPF Policy 8** Sensitive and Classified Information Amendment), acts as program sponsor for the **2 billion NIC Top Secret Cloud** (Top Secret Cloud Announcement), and houses the statutory **Quality Assurance Office (QAO)** auditing **TS-PA** clearances (ASIO Amendment Bill 2023 Explanatory Memorandum).
- **Australian Signals Directorate (ASD), REDSPICE, and Joint Defence Facility Pine Gap (JDFPG):** Elevated to an independent statutory agency within the Defence portfolio on 1 July 2018 under the Intelligence Services Act 2001 and led by **Director-General Abigail Bradshaw CSC** (supported by Head of the **ACSC** Ms Stephanie Crowe, Deputy Director-General Signals Intelligence and Effects Mr Cameron Ashe AIM, Deputy Director-General Capability and Transformation Ms Sarah Collins, and COO Mr Dale Furse: ASD Organisation Chart):
  - **Foreign SIGINT & Offensive Cyber (s 7 ISA 2001):** Collects foreign **COMINT**/**ELINT** and conducts offshore offensive cyber operations (ASD Offensive Cyber).
  - **Project REDSPICE (9.9 Billion):** Announced in the 2022–23 Budget over a decade to triple **ASD**'s offensive cyber capacity, double persistent cyber-hunting, deliver 1,900 new **TS-PA**/**PV**-cleared specialists, and expand AI/cloud infrastructure (ASD REDSPICE Blueprint; ASD REDSPICE Overview).
  - **ACSC, AKMI & Top Secret ATO Authority:** Authors the **Information Security Manual (ISM)**, administers the **Infosec Registered Assessors Program (IRAP)** (410 endorsed assessors in 2025–26), operates the **Australian Key Management Infrastructure (AKMI)**, and serves as the sole national authority

issuing **Authorisations to Operate (ATO) for TOP SECRET ICT Systems and Networks** (ASD Annual Report 2025–26).

- **Five Eyes (UKUSA) & Joint Defence Facility Pine Gap (JDFPG): ASD** is Australia's signatory authority to the 1946/1948 **UKUSA** Agreement and serves as the primary Australian intelligence partner at **Joint Defence Facility Pine Gap (JDFPG)** near Alice Springs, NT (established by treaty in December 1966, commissioned 1967/1970, redesignated **JDFPG** in 1988: Ministerial Statement: Full Knowledge and Concurrence; NT Space History), operated jointly with the US **NRO**, **NSA**, and **CIA** under the bipartisan Australian sovereign doctrine of **"Full Knowledge and Concurrence"** (Securing Australia's Sovereignty Statement; DFAT Full Knowledge and Concurrence Exchange).

- **Australian Security Intelligence Organisation (ASIO):** Governed by the ASIO Act 1979 and led by **Director-General of Security Mike Burgess AM** (ASIO Annual Threat Assessment 2026), **ASIO** executes domestic counter-espionage and foreign interference operations (**Part III** Special Powers Warrants), operates the central **TS-PA Vetting Authority** (**Part IVA**), and runs **ASIO T4 Protective Security (ASIO-T4)** and the **Security Construction and Equipment Committee (SCEC)** certifying Zone 5/**SCIF** facilities (PSPF Glossary; ASIO Corporate Plan 2026–30).
- **Australian Secret Intelligence Service (ASIS):** Governed by the Intelligence Services Act 2001 and led by **Director-General Kerri Hartland** (with Deputy Director-General Operations Mr Ewan Macmillan—who previously served in **ASIO** from 2022 to April 2024 as Deputy Director-General Vetting Service Delivery standing up the **TS-PA** reform—and Deputy Director-General Capability Ms Catherine Burn: ASIS Leadership), conducting overseas **HUMINT** and ministerial-directed special operations (**s 6(1)(e)**).
- **AGO, DIO, and the Defence Intelligence Group (DIG):** Consolidated on 1 July 2020 into the **Defence Intelligence Group (DIG)** under the **Chief of Defence Intelligence (CDI)**—led since 6 July 2024 by **Mr Tom Hamilton** (Strengthening Defence Intelligence Release; CDI Profile)—comprising **AGO** (led by **Director Kathryn McMullan**: AGO 25th Anniversary; AGO Strategy 2025–2027), **DIO** (DIO Overview), Intelligence Capability Division (**ICD**), and Intelligence Policy and Priorities (**IPP**) (Defence Intelligence Group Structure), governed under the **Defence Security Principles Framework (DSPF)** (ANAO Audit: Defence Contractors & DSPF).

Section 14: AUKUS, Five Eyes Compartmentalization, DISP, and Technical Infrastructure

14.1 Causal Link Between AUKUS Pillar 1 (NNPI) & Pillar 2 and the 2023–2024 TS-PA / SAP Overhaul

The announcement of the trilateral **AUKUS** partnership on 15 September 2021 (Portfolio Budget Statements 2022–23) and the **2023 Defence Strategic Review (DSR)** (Defence Strategic Review PDF) directly drove the overhaul of Australia's **TS-PA** and **SAP** regime:

1. **AUKUS Pillar 1 (Naval Nuclear Propulsion Information - NNPI):** Under the Agreement among the Governments of Australia, the United Kingdom, and the United States for Cooperation Related to Naval Nuclear Propulsion (**Annex A, Section I: Information Security**) and the Australian Naval Nuclear Power Safety Act 2024 (No. 91, 2024) (administered alongside the **Australian Submarine Agency [ASA]** under Director-General Vice Admiral Jonathan Mead and the **Australian Naval Nuclear Power Safety Regulator [ANNPSR]**: ASA First US Submarine Assigned to SRF-West; ANNPSR Privacy Policy), the US and UK transfer **Naval Nuclear Propulsion Information (NNPI)** and **Special Nuclear Material** for Submarine Rotational Force-West (**SRF-West** at HMAS Stirling from 2027), Virginia-class SSNs, and **SSN-AUKUS** construction. The treaty (**Annex A, Section I**) mandates that *"NNPI that constitutes Classified Information requires both a security clearance and a separate determination related to access to NNPI"*

(AUKUS Naval Nuclear Propulsion Treaty Text), with **ASA** enforcing security controls equivalent to US Naval Reactors (**NOFORN/NNPI**) and UK **Developed Vetting (DV)** standards (ASA FOI 008/2425 Release).

2. **AUKUS Pillar 2 (Advanced Capabilities & US DoD SAP / UK STRAP Interoperability):** Pillar 2 accelerates co-development in quantum technologies, AI/autonomy, hypersonics, undersea warfare, cyber, and electronic warfare (Securing Australia's Sovereignty Statement). Because these capabilities reside inside US DoD **Special Access Programs (SAPs)** (**DoD Directive 5205.07**: ADF Kill-Web & SAP Analysis) and UK **STRAP / Special Access** compartments, and because prior to 2023 five Australian agencies conducted **Positive Vetting** to disparate standards while **AGSVA** outsourced 92% of assessments (ANAO Delivery of Security Vetting Services Follow-up), the ASIO Annual Report 2024–25 confirms that transferring Defence **PV** cohorts into **ASIO**'s **TS-PA Vetting Authority** directly implemented the **2023 Defence Strategic Review recommendation**, paired with the **AUKUS Licence-Free Environment** (effective **1 September 2024** via the **My Australian Defence Exports [MADE] Portal**: ASA Industry Security Requirements).

14.2 Defence Industry Security Program (DISP), Zone 5 / SCIF Accreditation (ASIO T4), and Top Secret Cloud

- **Defence Industry Security Program (DISP):** Governed by **DSPF Principle 16, Control 16.1** (DISP Eligibility and Suitability), **DISP** subjects industry members to mandatory **Foreign Ownership, Control or Influence (FOCI)** checks, Australian Standard **AS 4811:2022 (Workforce Screening)**, and **ASD Essential Eight (E8) Maturity Level 2 (ML2)** (DISP Cyber and Assurance; DISP Member Portal). Across its four tiers (**Entry Level** for **OFFICIAL: Sensitive**, **Level 1** for **PROTECTED/Baseline**, **Level 2** for **SECRET/NV1**, and **Level 3** for **TOP SECRET/NV2**), **Level 3** requires a **Zone 5 / SCIF** certified by **ASIO T4** and **TOP SECRET ICT** authorised (**ATO**) by **ASD**.
- **Physical Security Zones (Zones 1 to 5) & ASIO T4 SCIF Certification:** Under the PSPF Physical Security Framework and **ASIO Technical Note 5/12: Physical security of Zone 5 (TOP SECRET) areas**, **Zone 5 (TOP SECRET Areas / SCIFs)** is mandatory for **TOP SECRET**, **SCI**, **Codeword**, and **SAP** material (**BIL 5 – Catastrophic**) (PHYSEC-1 Summary; AFP LEX271), requiring exclusive certification and **TSCM** sweeps by **ASIO T4 Protective Security (ASIO-T4)**, **SCEC-approved Class A/B containers, vaults, locks, and Type 1 Security Alarm Systems (SAS)** (PSPF Australian Government Resources), dual-factor authentication on Electronic Access Control Systems (**EACS**), and a strict ban on personal electronic devices.
- **Australian Top Secret Network (ATSN) & the 2 Billion AWS Top Secret (TS) Cloud:** All Top Secret networks receive their **Authorisation to Operate (ATO)** and cryptographic keys (**AKMI**) from **ASD** under the **Information Security Manual (ISM)** (ASD Annual Report 2025–26; ASD Corporate Plan 2026–27). On **4 July 2024**, the Australian Government announced a **2 billion+ over 10 years** partnership between **ASD**, **ONI**, and Amazon Web Services (**AWS**) (under **REDSPICE**) to build a sovereign, air-gapped **Top Secret (TS) Cloud** in Australia interconnecting all 10 **NIC** agencies and Defence at the **TOP SECRET** and **SCI/SAP** levels (NIC Top Secret Cloud Announcement; Minister for Defence Media Release; ONI STEM Speech).

Section 15: Historical Causal Chains & Royal Commissions in Australia (1974–2026)

Period / Milestone	Lead Commissioner / Reviewer	Triggering Event / Mandate	Statutory & Institutional Outcomes
1. First Hope Royal Commission (RCIS, 1974–1977)	Justice Robert Marsden Hope (**Royal Commission on Intelligence and**	Cold War domestic surveillance controversies, lack of statutory charters for **ASIS/DSD**, and unstandardized departmental	Delivered **8 Reports** (**A8908, 1–8**), including the **Second Report on Security Vetting and the Establishment of a Security Appeals Tribunal** (**A8908, 2** &

Period / Milestone	Lead Commissioner / Reviewer	Triggering Event / Mandate	Statutory & Institutional Outcomes
	Security, appointed 21 August 1974 by PM Gough Whitlam)	security vetting (NAA RCIS 1974–77).	Annex **2B**). Led directly to the **Office of National Assessments Act 1977** (**ONA**), the **ASIO Act 1979 (No. 113, 1979)**, the **Security Appeals Tribunal** (later merged into the **AAT** Security Division), and standardized protective security manuals (NAA Fact Sheet 262; Senate Report 1994 PP121).
2. Second Hope Royal Commission (RCASIA, 1983–1984)	Justice Robert Marsden Hope (**Royal Commission on Australia's Security and Intelligence Agencies**, appointed May 1983 by PM Bob Hawke)	The Combe–Ivanov Affair (1983 expulsion of Soviet KGB officer Valeriy Ivanov) and the November 1983 **ASIS** Sheraton Hotel training incident in Melbourne (Senate Hansard 1983).	Refined post-1977 vetting architecture (Senate Report 1994 PP121), brought **ASIO** within the **Archives Act 1983** (Parliament Report PP68), established the **Parliamentary Joint Committee on ASIO** (**ASIO Amendment Act 1986**), and created the **Inspector-General of Intelligence and Security (IGIS)** via the IGIS Act 1986 (No. 101, 1986).
3. The 1999 DIO Espionage & Caveat Crises (Lappin / Cornwell & Jenkins / Blunn) & Cornall Reviews (1999–2011)	**Blunn Inquiry** (Anthony Blunn AO, 2000), **Flood Inquiry** (Philip Flood AO, 2004), and **Cornall / Black IRIC** (Robert Cornall AO & Dr Rufus Black, 2011)	(a) In April 1999, **DIO** clerk **Jean-Philippe Lappin** (and co-accused **Sheridan Cornwell**) was arrested for stealing Top Secret compartmented satellite/SIGINT documents from **DIO** (**Crimes Act 1914 s 78**). (b) In June 1999, **DIO** officer **Mervyn Jenkins** died in Washington DC while under investigation for passing **AUSTEO**-caveated reports to US officials (DFAT Blunn Report on the Jenkins Case, Dec 2000).	Overhauled **Positive Vetting** psychological screening and caveat handling (**AUSTEO** vs. **REL USA**), followed by the Intelligence Services Act 2001, the 2004 Flood Report, the establishment of **AGSVA on 1 October 2010** (ANAO Central Administration of Security Vetting), and the **2011 Independent Review of the Intelligence Community (IRIC)** by **Robert Cornall AO** and Dr Rufus Black (2011 IRIC Report; ASIO Annual Report 2021–22 Appendix C).
4. 2017 Independent Intelligence Review (IIR)	**Mr Michael L'Estrange AO & Mr Stephen Merchant PSM** (report released 18 July 2017)	Fragmented coordination across the 6-agency **AIC** and a severe **1,266-case backlog in AGSVA Top Secret (Positive Vetting) clearances** averaging **296 business days (18+ months)** (2017 IIR Report PDF; ANAO ASIO Workforce Planning Audit).	(1) Created the **10-agency National Intelligence Community (NIC)** led by **ONI** under the ONI Act 2018; (2) Converted **ASD** into a statutory agency (**ISA Amendment Act 2018**); (3) **Recommendation 12** mandated urgent remediation of the 1,266 **TS (PV)** backlog and directed structural alternatives if **PV** times exceeded 6 months (ANAO ASIO Workforce Planning Audit; ANAO Mitigating Insider Threats).
5. 2020 Comprehensive Review of the Legal Framework of the	**Mr Dennis Richardson AC** (4-volume unclassified report and	Recommendation 21 of the 2017 **IIR** calling for the first root-and-branch statutory audit of all **NIC** legislation since the	Produced a 4-volume report (203 recommendations) auditing **ASIO**, **ISA**, **ONI**, **TIA**, and **IGIS** legislation, leading to the Foreign

Period / Milestone	Lead Commissioner / Reviewer	Triggering Event / Mandate	Statutory & Institutional Outcomes
NIC (`Richardson Review`)	Government Response released 4 December 2020)	1974–77 Hope Royal Commission (Richardson Review Report & 4 Volumes).	Intelligence Legislation Amendment Bill 2021 and **`National Security Legislation Amendment Acts`** (Government Response to the Richardson Review).
6. 2023 ASIO Amendment Act & 2024 Independent Intelligence Review (`IIR`)	Parliament of Australia (Act No. 33, 2023); **Dr Heather Smith PSM & Mr Richard Maude** (2024 IIR Report, released March 2025)	Escalating foreign espionage against high-clearance personnel, **`AUKUS`** Pillar 1 (**`NNPI`**) and Pillar 2 (**`SAP`**) interoperability requirements, and the 2023 **`Defence Strategic Review`** mandate to unify Top Secret vetting.	Enacted **`Part IVA`** of the **`ASIO Act 1979`**, replacing **`PV`** across 5 vetting agencies with a single **`TS-PA Vetting Authority`** in **`ASIO`** and the **Quality Assurance Office (`QAO`)** in **`ONI`**, reinforced by PSPF Release 2024 and the 67 recommendations of the 2024 Independent Intelligence Review.



Figure 3: Hierarchical Command Schema of China's National Science and Technology Major Projects (国家科技重大专项), Central Science and Technology Commission (中央科技委员会), Three-Ministry Comprehensive Balancing (三部门综合平衡), Qian Xuesen Two-Chief System (两总系统), and Military-Civil Fusion (军民融合) Dual-Use Pipeline.

Part IV: China's National Science and Technology Major Projects (国家科技重大专项 / 国家重大科技专项) and Military-Civil Fusion (军民融合 / 军民结合 / 寓军于民)

China's **National Science and Technology Major Projects (国家科技重大专项 / 国家重大科技专项, *Guójiā Kējì Zhòngdà Zhuānxiàng*)** constitute the core state-directed engineering and industrial-technological

mobilization instrument established under the State Council's (国务院) 2006–2020 Medium and Long-Term S&T Development Plan and managed through the Ministry of Science and Technology (科学技术部, MOST) Major Projects Portal.

Section 16: Foundational State Directive & Strategic Origin (`MLP 2006–2020` to `CSTC 2023–2026`)

- **Promulgating the 2006–2020 Master Directive (国发〔2005〕44号 & 国发〔2006〕6号):** Promulgated by the **`State Council`** (**国务院**, *Guówùyuàn*) on February 9, 2006 (Document **国发〔2005〕44号**, with accompanying implementation policies issued via State Council Notice **国发〔2006〕6号**), the **`National Medium and Long-term Science and Technology Development Plan Outline (2006–2020)`** **(国家中长期科学和技术发展规划纲要(2006–2020年), *Guójiā Zhōngchángqī Kēxué hé Jìshù Fāzhǎn Guīhuà Gāngyào*, MLP)** established the 16-character strategic guideline: **自主创新，重点跨越，支撑发展，引领未来** (*"indigenous innovation, leapfrogging in priority fields, enabling development, and leading the future"*) (CNSA Official Release).
- **Establishing the Five Statutory Selection Principles:** Grounded in the historical precedent of **`"Two Bombs, One Satellite"`** (**两弹一星**), manned spaceflight, and hybrid rice, the MOST Official Program Charter defines five statutory selection criteria for the 16 Mega-Projects:
- Aligning closely with major socio-economic demands to cultivate strategic industries with core autonomous intellectual property (**核心自主知识产权**).
- Highlighting key generic technologies (**关键共性技术**) with strong spillover effects on overall industrial competitiveness.
- Resolving major bottleneck constraints (**重大瓶颈问题**) restricting socio-economic growth in energy, resources, environment, and public health.
- Embodying military-civilian integration and locating military potential in civilian capabilities (**体现军民结合、寓军于民**) to safeguard national security and comprehensive national power.
- Matching China's national conditions and fiscal bearing capacity (**切合我国国情，国力能够承受**), executing dynamic adjustments under the principle of **`"launching one as soon as it matures"`** (**成熟一项，启动一项**) (MOST Management Regulations **国科发专〔2017〕145号**).
- **Reforming Central Fiscal S&T Funding Channels (国发〔2014〕64号):** In December 2014, the State Council issued the *Opinions on Deepening the Management Reform of Central Fiscal S&T Plans (Special Projects, Funds, etc.)* (**关于深化中央财政科技计划（专项、基金等）管理改革的方案**), consolidating fragmented central R&D channels into five unified categories (**五大类科技计划**):
- **`National Natural Science Foundation`** (**国家自然科学基金**);
- **`National Science and Technology Major Projects`** (**国家科技重大专项**);
- **`National Key R&D Programs`** (**国家重点研发计划**, which absorbed the former **`863 Program`** and **`973 Program`**);
- **`Technology Innovation Guidance Special Funds`** (**技术创新引导专项（基金）**); and
- **`Bases and Talents Special Projects`** (**基地和人才专项**) (CAC / State Council Reform Audit).
- **Transitioning Through the 13th and 14th Five-Year Plans & `Sci-Tech Innovation 2030` (科技创新2030–重大项目):** In May 2016, the CCP Central Committee and State Council released the Outline of the National Innovation-Driven Development Strategy (**国家创新驱动发展战略纲要**), followed on August 8, 2016 by the **`13th Five-Year National S&T Innovation Plan`** (**“十三五”国家科技创新规划, 国发〔2016〕43号**) (CSET English Translation). These directives bridged the 2006–2020 MLP horizon to 2030 by launching **`Sci-Tech Innovation 2030 – Major Projects`** **(科技创新2030–重大项目, *Kējì Chuàngxīn 2030 — Zhòngdà Xiàngmù*)** in a **`"15 + 1"`** architecture (including **`Aero-Engine and Gas Turbine`** **航空发动机及燃气轮机,**

Quantum Communication and Quantum Computer 量子通信与量子计算机, **Brain Science and Brain-Inspired Research 脑科学与类脑研究**, **National Cyberspace Security 国家网络空间安全**, **Deep Space Exploration 深空探测**, **Deep Sea Space Station 深海空间站**, **Integrated Space-Ground Information Network 天地一体化信息网络**, **Big Data 大数据**, **Intelligent Manufacturing and Robotics 智能制造和机器人**, **Key New Materials 重点新材料**, **Seed Industry Autonomous Innovation 种业自主创新**, **Clean and Efficient Utilization of Coal 煤炭清洁高效利用**, **Smart Grid 智能电网**, and the added 16th project **New Generation Artificial Intelligence (新一代人工智能 / AI 2.0)**) (MOST AI 2030 Notice). During the **14th Five-Year Plan (“十四五”规划**, 2021–2025) and continuing into the 2026–2030 window, unfinished core tasks from the original 16 Mega-Projects were integrated with **Sci-Tech Innovation 2030** and regional/municipal major projects under the **"New Whole-of-Nation System" (新型举国体制**, *Xīnxíng Jǔguó Tǐzhì*) (Shanghai STCSM 02 Project Report).

- **Restructuring Party/State Governance in 2023 (中央科技委员会 & MOST):** Under the **Plan on Reforming Party and State Institutions (党和国家机构改革方案)** adopted at the Second Plenary Session of the 20th CCP Central Committee and promulgated in March 2023 (MOJ Official Explanation), the CCP Central Committee established the **Central Science and Technology Commission (中央科技委员会, *Zhōngyāng Kējì Wěiyuánhuì*, CSTC)** to exercise unified Party leadership over national S&T strategy, military-civilian tech coordination, and major project approval. The general office functions (**办事机构职责**) of the **CSTC** are wholly assumed by the reorganized **Ministry of Science and Technology (科学技术部, MOST)** (MOST 2026 Departmental Budget). While industry-specific project management funds and specialized agencies were transferred to mission ministries (**MIIT**, **MARA**, **NHC**, and **MEE**), the reorganized **MOST** explicitly retained statutory responsibility for macro-planning, resource balancing, and supervision of the **National Science and Technology Major Projects (国家科技重大专项)** (MOE Institutional Reform Summary).

Section 17: Governance, Administrative Hierarchy, and the "Two-Chief System" (两总系统)

The statutory governance of the civilian and dual-use Major Projects (**民口重大专项**) is codified in the *Interim Provisions on the Management of National S&T Major Projects* (**国家科技重大专项管理暂行规定, 国科发计〔2008〕453号**), the *Working Rules for the Organization and Implementation of National S&T Major Projects* (**国家科技重大专项组织实施工作规则, 国办发〔2016〕105号**), the **Management Provisions for National S&T Major Projects (Civilian Sector) (国家科技重大专项（民口）管理规定, 国科发专〔2017〕145号)**, and the **Acceptance Management Measures for National S&T Major Projects (Civilian Sector) (国家科技重大专项（民口）验收管理办法, 国科发专〔2018〕37号)**.

Governance Tier	Institutional Actors (Chinese / Pinyin / English)	Statutory Mandate & Causal Mechanics
Tier 1: Supreme Decision & Strategic Command	**中共中央** (CCP Central Committee), **国务院** (State Council), **国家科技教育领导小组** (National Leading Group for S&T and Education) & **国家科技体制改革和创新体系建设领导小组** (National Leading Group for S&T System Reform) → consolidated in 2023 into **中央科技委员会** (Central Science and Technology Commission, **CSTC**)	Approves the establishment of each Mega-Project, ratifies Master Implementation Plans (**实施方案**), authorizes major target or budget adjustments, and reviews final overall acceptance conclusions (**整体验收结论**) (Art. 3–4, 18, 42 of **国科发专〔2017〕145号**).
Tier 2: Inter-Ministerial Joint Conference & "Three-Ministry	**国家科技计划（专项、基金等）管理部际联席会议** (Inter-Ministerial Joint Conference), advised by **战略咨询与综合评审委员会** (Strategic Consultation &	Conducts **"Comprehensive Balancing" (综合平衡)** of 5-year stage plans (**阶段实施计划**) and annual plans (**年度计划**). **MOST** coordinates links with national S&T plans and manages the unified

Governance Tier	Institutional Actors (Chinese / Pinyin / English)	Statutory Mandate & Causal Mechanics
Comprehensive Balancing" (三部门综合平衡)	Comprehensive Review Committee, **咨评委**), and executed by the **"Three Ministries" (三部门)**: **科学技术部 (MOST)**, **国家发展和改革委员会 (NDRC)**, and **财政部 (MOF)**	information platform (**国家科技管理信息系统**); **NDRC** formulates supporting industrial policies and links with National Major Engineering Projects (**国家重大工程**); **MOF** audits and approves total, five-year, and annual fiscal budgets (**总概算、阶段概算、年度预算**) (Art. 8–10, 20, 29 of **国科发专〔2017〕145号**).
Tier 3: Sectoral & Defense-Industrial Authorities	**工业和信息化部 (MIIT)**, **国务院国有资产监督管理委员会 (SASAC)**, **国家国防科技工业局 (SASTIND)**, and **中央军委装备发展部 (CMC Equipment Development Department, EDD**, successor to the PLA General Armaments Department **总装备部**)	Serve as Leading Organizing Departments (**牵头组织部门 / 牵头组织单位**) for industrial, dual-use, and military-sector (**军口**) projects. **SASAC** supervises central SOEs (**中央企业**) executing project tasks, while **SASTIND** and **CMC EDD** co-manage defense-industrial, aerospace, nuclear, and classified projects.
Tier 4: Project Execution & "Two-Chief System" (两总系统)	**牵头组织单位** (Leading Organizing Department), **实施管理办公室** (Implementation Management Office, **实施办**), **项目管理专业机构** (Specialized Project Management Agency), **总体专家组** (Overall Expert Group) led by **技术总师** (Chief Technical Expert / Chief Designer), and **牵头承担单位** (Lead Undertaking Enterprise/Institute)	Executes day-to-day governance, annual bidding guidelines (**年度指南**, requiring ≥ 50 days open notice for public bidding), contract signing (**任务合同书**), and technical integration (Art. 11–14, 26, 31 of **国科发专〔2017〕145号**).

17.1 Causal Mechanics of the "Two-Chief System" (两总系统: 总指挥 + 总设计师)

Originated by **Qian Xuesen (钱学森)** in the Fifth Academy of the Ministry of National Defense (**国防部五院**) and Seventh Ministry of Machine Building (**七机部**) during the **"Two Bombs, One Satellite"** era, China's complex engineering mega-projects institutionalize dual parallel command lines (**两条指挥线**) (CAS Bulletin Historical Audit):

- **Administrative Command Line (行政指挥线) led by the Chief Commander (总指挥, *Zǒngzhǐhuī*) / Implementation Management Office Director (实施管理办公室主任):** Typically a ministerial or vice-ministerial official (e.g., **SASTIND** Director **Luan Enjie 栾恩杰**, **Chen Qiufa 陈求发**, or **Xu Dazhe 许达哲** for Lunar Exploration; **GAD** Director **Chang Wanquan 常万全** for Manned Spaceflight; or **MIIT** departmental directors **Diao Shijing 刁石京** and **Wen Ku 闻库** for civilian ICT projects). The administrative line commands inter-agency resource allocation, fiscal disbursement, schedule enforcement, supply-chain mobilization, and security/secrecy compliance.
- **Technical Command Line (技术指挥线) led by the Chief Designer / Chief Technical Expert (总设计师 / 技术总师, *Zǒngshèjìshī / Jìshù Zǒngshī*):** Typically an Academician of the Chinese Academy of Sciences (**中国科学院院士**) or Chinese Academy of Engineering (**中国工程院院士**) who heads the **Overall Expert Group (总体专家组)** (CMSA Chief Designer System). Under Article 12 of **国科发专〔2017〕145号**, the **技术总师** exercises technical authority over architecture design, technical roadmap selection (**技术路线**), interface standardization across sub-systems (**分系统**), and milestone performance verification. To prevent conflicts of interest in civilian grant-based projects, members of the Overall Expert Group are statutorily barred in principle from personally acting as PIs on sub-projects (**总体专家组成员原则上不得承担重大专项项目（课题）**).

Section 18: Complete Technical and Engineering Breakdown of All 16 Mega-Projects (`Projects 01 to 16`)

18.1 Master Summary Matrix of the 16 National S&T Major Projects

Code	Official Chinese Title & Short Name	Official English Title	Lead Organizing Body (牵头组织单位)	Key Chief Designers / Technical Chiefs (技术总师 / 总设计师)	Primary Source Portal
01专项	**核心电子器件、高端通用芯片及基础软件产品 (核高基)**	Core Electronic Devices, High-end General Purpose Chips and Basic Software	**工业和信息化部 (MIIT)**	**魏少军** (Wei Shaojun, Tsinghua Univ.)	MOST 01 Portal
02专项	**极大规模集成电路制造装备及成套工艺 (集成电路装备)**	Large-scale IC Manufacturing Equipment and Complete Process	**北京市政府** & **上海市政府** (Beijing & Shanghai Municipal Govts)	**叶甜春** (Ye Tianchun, CAS IME)	MOST 02 Portal
03专项	**新一代宽带无线移动通信网 (宽带移动通信)**	New Generation Broadband Wireless Mobile Communication Network	**工业和信息化部 (MIIT)**	**邬贺铨** (Wu Hequan, CAE Academician)	MOST 03 Portal
04专项	**高档数控机床与基础制造装备 (数控机床)**	High-end CNC Machine Tools and Basic Manufacturing Equipment	**工业和信息化部 (MIIT)**	**卢秉恒** (Lu Bingheng, CAE Academician)	MOST 04 Portal
05专项	**大型油气田及煤层气开发 (油气开发)**	Large Oil and Gas Fields and Coalbed Methane Development	**中国石油 (CNPC)**, **中国石化 (Sinopec)**, **中国海油 (CNOOC)**	**贾承造** (Jia Chengzao, CAS Academician)	MOST 05 Portal
06专项	**大型先进压水堆及高温气冷堆核电站 (大型核电站)**	Large Advanced PWR and High-Temperature Gas-Cooled Reactor Nuclear Power Plant	**国家能源局** (National Energy Administration, **NEA**)	**郑明光** (Zheng Mingguang, **CAP1400**); **张作义** (Zhang Zuoyi, **HTR-PM**)	MOST 06 Portal
07专项	**水体污染控制与治理 (水专项)**	Water Pollution Control and Treatment	**生态环境部 (MEE)** & **住房和城乡建设部 (MOHURD)**	**孟伟** (Meng Wei, former) / **吴丰昌** (Wu Fengchang, CAE Academician)	MOST 07 Portal
08专项	**转基因生物新品种培育 (转基因专项)**	Genetically Modified New Variety Breeding	**农业农村部 (MARA)**	**万建民** (Wan Jianmin, CAE) & **吴孔明** (Wu Kongming, CAE)	MOST 08 Portal
09专项	**重大新药创制 (新药专项)**	Major New Drug Creation	**国家卫生健康委员会 (NHC)**	**桑国卫** (Sang Guowei, CAE) & **陈凯先** (Chen Kaixian, CAS, Deputy Chief)	MOST 09 Portal

Code	Official Chinese Title & Short Name	Official English Title	Lead Organizing Body (牵头组织单位)	Key Chief Designers / Technical Chiefs (技术总师 / 总设计师)	Primary Source Portal
10专项	**艾滋病和病毒性肝炎等重大传染病防治 (传染病专项)**	AIDS, Viral Hepatitis and Other Major Infectious Diseases Prevention and Control	**国家卫生健康委员会 (NHC)**	**侯云德** (Hou Yunde, CAE) & **李兰娟** (Li Lanjuan, CAE, Deputy Chief)	MOST 10 Portal
11专项	**大型飞机 (大型飞机重大专项)**	Large Aircraft	**工业和信息化部 (MIIT) / 国防科工局 (SASTIND)**	**吴光辉** (Wu Guanghui, **C919**); **唐长红** (Tang Changhong, **Y-20**)	MOST 11 Portal
12专项	**高分辨率对地观测系统 (高分专项 / CHEOS)**	High-Resolution Earth Observation System	**国家国防科技工业局 (SASTIND) / 国家航天局 (CNSA)**	**童旭东** (Tong Xudong) / **赵坚** (Zhao Jian)	MOST 12 Portal
13专项	**载人航天与探月工程 (载人航天 / 探月工程)**	Manned Spaceflight and Lunar Exploration	**中央军委装备发展部 (CMC EDD / CMSA) & 国防科工局 / 国家航天局 (SASTIND / CNSA)**	**周建平** (Zhou Jianping), **杨利伟** (Yang Liwei); **孙家栋** (Sun Jiadong), **欧阳自远** (Ouyang Ziyuan), **叶培建** (Ye Peijian), **吴伟仁** (Wu Weiren)	MOST 13 Portal
14–16专项 (+两机专项)	Classified / Defense-Strategic Projects (**军口重大专项**): **北斗卫星导航系统** (Beidou), **惯性约束聚变 / 神光 (ICF** / ShenGuang), Strategic Defense Systems, plus **航空发动机及燃气轮机 (两机专项)**	Beidou Navigation Satellite System, Laser Inertial Confinement Fusion, Strategic Defense, and Aero-Engine & Gas Turbine Special Project	**中央军委装备发展部 (CMC EDD)**, **国防科工局 (SASTIND)**, **中国工程物理研究院 (CAEP)**, **工业和信息化部 (MIIT)**	**孙家栋** (Sun Jiadong) & **杨长风** (Yang Changfeng, Beidou); **范滇元** (Fan Dianyuan) & **张杰** (Zhang Jie, ShenGuang **ICF**); **尹泽勇** (Yin Zeyong, Aero-Engine)	MOST Beidou Report & CAS ShenGuang Portal

18.2 Project-by-Project Engineering & Institutional Audit

Project 01 (01专项 / 核高基): Core Electronic Devices, High-end General Purpose Chips and Basic Software (核心电子器件、高端通用芯片及基础软件产品)

- **Governance & Leadership:** Led by **MIIT** (**工业和信息化部**) (MOST 01 Portal), with the Implementation Management Office (**实施管理办公室**) housed in the **MIIT** Electronic Information Department (Executive Deputy Director **Diao Shijing 刁石京**) and Chief Technical Expert (**技术总师**) Professor **Wei Shaojun (魏少军)**, Director of the Institute of Microelectronics at Tsinghua University (**清华大学微电子学研究所**).
- **Funding Magnitude:** During the 11th Five-Year Plan phase alone (through December 2010), approved total project funding reached **19.6+ billion RMB**, comprising **8.4+ billion RMB** from central fiscal appropriations (**中央财政资金**) and **11.1+ billion RMB** in local government and enterprise self-raised matching funds (NEA / Official 01 Audit), scaling to over 30 billion RMB across the 2008–2020 execution cycle.
- **Core Entities & Engineering Milestones:**
  - **High-End General-Purpose CPUs (高端通用芯片):**

- **Loongson (龙芯):** Developed by the **CAS Institute of Computing Technology** (**中国科学院计算技术研究所**) and **Loongson Technology** (**龙芯中科**) under Chief Architect **Hu Weiwu (胡伟武)**, completing the 64-bit multi-core **Loongson-3A/3B** (32nm, 1.5 GHz octa-core) under the **01 Project** and subsequently transitioning to the fully autonomous **LoongArch** (**龙架构**) instruction set (**3A5000/3A6000**).
- **Sunway (申威):** Developed by the **Shanghai High-Performance IC Design Center** (**国家高性能集成电路（上海）设计中心** / Wuxi **江南计算技术研究所**) under **01 Project** funding, producing the 64-bit autonomous RISC/heterogeneous **Sunway SW26010 (申威26010)** processor (260 cores on a single die at 1.5 GHz, achieving 3.168 TFLOPS double-precision peak performance), which powered the **Sunway TaihuLight (神威·太湖之光)** supercomputer (93 PFLOPS sustained, ranked #1 on the global TOP500 list in June 2016).
- **Phytium (飞腾):** Developed by the **National University of Defense Technology** (**国防科技大学**, **NUDT**) and **Phytium Technology** (**天津飞腾**, under **CEC 中国电子**), beginning with the 65nm, 1 GHz, 350-million-transistor multi-core **FT-1000 (飞腾1000)** used in the TOP500 #1 **Tianhe-1A (天河一号)** supercomputer (2010), and evolving into the ARMv8-compatible **FT-1500A**, **FT-2000+**, and **Tengyun S2500/S5000** server CPUs.
- **Hygon (海光):** Developed by **Hygon Information Technology** (**海光信息**) in partnership with **Sugon** (**中科曙光**) and **CAS** institutes, creating the x86-compatible **C86** architecture (**Hygon 7000** series up to the 96-core **Hygon 7448**) and the general-purpose AI/HPC **Deep Computing Unit (深算DCU)** GPGPU accelerator.

- **Core Electronic Devices (核心电子器件 — DSP / FPGA):**
  - **Huarui-1 (华睿1号) & Hunxin (魂芯) DSPs: CETC 14th Research Institute** (**中国电科14所**) developed the **Huarui-1 (华睿1号)** high-performance radar/defense DSP (achieving 3× the processing performance at 1/3 the power consumption of foreign equivalents), while **CETC 38th Research Institute** (**中国电科38所**) developed the **Hunxin-1** (**魂芯一号**) and **Hunxin-2A** (**魂芯二号A**) floating-point DSPs; domestic FPGAs were advanced by **Unigroup Guoxin** (**紫光国微 / 紫光同创**), **CETC 58th Institute** (**中国电科58所**), and **Anlogic** (**安路科技**).

- **Basic Software (基础软件产品 — OS & Database):**
  - **Kylin OS (麒麟操作系统):** Developed by **NUDT** and **CEC** (**中国电子**) as **Galaxy Kylin** (**银河麒麟**) and **NeoKylin** (**中标麒麟**, merged into **KylinSoft 麒麟软件**), deployed across **Tianhe-1**, government cloud infrastructure, and defense command systems (NEA 01 Report).
  - **Dameng Database (达梦数据库):** Developed by **Wuhan Dameng Database Co., Ltd.** (**武汉达梦数据库股份有限公司**, spun out of Huazhong University of Science and Technology), along with **Kingbase** (**人大金仓**) and **GBase** (**南大通用**), achieving shared-storage cluster architectures (**DM7/DM8**) to replace Oracle/IBM DB2 in state grid, finance, and e-government systems (Hubei 13th FYP Industrial Plan).

Project 02 (02专项): Large-scale Integrated Circuit Manufacturing Equipment and Complete Process (极大规模集成电路制造装备及成套工艺)

- **Governance & Leadership:** Uniquely among all 16 Mega-Projects, **Project 02** was jointly led by local governments—the **Beijing Municipal People's Government (北京市政府)** and **Shanghai Municipal People's Government (上海市政府)** (MOST 02 Portal; Shanghai STCSM Report). The Chief Technical Expert (**技术总师**) is **Ye Tianchun (叶甜春)**, Director of the **CAS Institute of Microelectronics** (**中国科学院微电子研究所**).
- **Core Entities & Engineering Milestones:**
  - **Lithography Systems (光刻机) — SMEE (上海微电子装备(集团)股份有限公司, 上微):** Developed 90nm front-end projection lithography machines (**SSA600/20**) and 280nm/110nm/90nm KrF/ArF step-and-scan

systems, alongside high-market-share advanced packaging lithography tools (**SSB500** series) (Shanghai STCSM 02 Report).

- **Plasma Etching (等离子体刻蚀机) & Thin-Film Deposition (薄膜沉积):**
  - **AMEC (中微半导体设备(上海)股份有限公司, 中微公司):** Founded by **Yin Zhiyao (尹志尧)**, developed 65nm–14nm–7nm/5nm Capacitively Coupled Plasma (**CCP**) dielectric etchers (**Primo D-RIE** / **Primo AD-RIE**) and Inductively Coupled Plasma (**ICP**) etchers, entering TSMC and SMIC high-volume production lines (MOST 02 Press Conference).
  - **NAURA Technology Group (北方华创科技集团股份有限公司):** Formed via the strategic merger of Sevenstar Electronics (**七星华创**) and Beijing NMC (**北方微电子**), developing 14nm/28nm **ICP** silicon/metal plasma etchers, Hard-Mask PVD (**硬掩膜物理气相沉积设备**), LPCVD/oxidation furnaces, and single-wafer annealing systems (**单片退火炉**) designated as baseline tools (**基准机台**) in domestic fabs (SASAC NAURA Audit).
  - **Piotech (拓荆科技股份有限公司) & ACM Research Shanghai (盛美上海): Piotech** (co-founded in Shenyang with **CAS** Shenyang Institute of Scientific Instrument) commercialized 12-inch Plasma-Enhanced Chemical Vapor Deposition (**PECVD**), **ALD**, and **SACVD** thin-film equipment (CSRC Piotech Filing), while **ACM Research** (**盛美半导体**) commercialized wet cleaning (**系列湿法设备**) and electroplating tools (Shanghai STCSM 02 Report).

- **Complete Wafer Fabrication Process (成套工艺) & Advanced Packaging:** Led by **SMIC (中芯国际)**, **HLMC** (**上海华力微电子**), and **XMC** (**武汉新芯** / **YMTC 长江存储**) in collaboration with **CAS IME**, Tsinghua, and Fudan, achieving mass production of 65nm, 45/40nm, 28nm PolySiON and High-K Metal Gate (**HKMG**), and **14nm FinFET** logic processes (where the **CAS IME** team ranked #1 and #5 globally in 20–14nm **HKMG**/**FinFET** patent filings) (CAS 2014 Outstanding S&T Achievement Award), alongside 3D NAND flash and wafer-level TSV packaging (**JCET 长电科技**, **Tongfu 通富微电**, **HuaTian 华天科技**) (Wuhan Science & Tech Bureau Report).

Project 03 (03专项): New Generation Broadband Wireless Mobile Communication Network (新一代宽带无线移动通信网)

- **Governance & Leadership:** Led by **MIIT** (**工业和信息化部**) (MOST 03 Portal), with the Implementation Management Office led by **Wen Ku (闻库)**, Director of the **MIIT** Department of Information and Communication Development (**工信部信息通信发展司**), and Chief Technical Expert (**技术总师**) CAE Academician **Wu Hequan (邬贺铨)**.
- **Core Entities & Engineering Milestones:**
  - **From "2G Following, 3G Breakthrough" to "4G Synchronization (TD-LTE)":** Coordinated **China Mobile (中国移动)**, **CAICT (中国信息通信研究院)**, **Datang Telecom / CICT (大唐电信 / 中国信科)**, **Huawei (华为)**, **ZTE (中兴通讯)**, and **Spreadtrum (展讯 / 紫光展锐)** to commercialize **TD-SCDMA** (3G) and establish **TD-LTE-Advanced** as one of the two official ITU 4G international standards (**IMT-Advanced**), building a complete domestic chain across baseband chips, RF transceivers, base stations, and test instrumentation (**StarPoint 星河亮点**, **CETC 41st Institute 中国电科41所**) (MOST 03 Press Conference; MOST 2016 Progress Exchange).
  - **5G IMT-2020 Global Standard Leadership:** Established the **IMT-2020 (5G) Promotion Group (IMT-2020(5G)推进组)** in 2013 under **CAICT**, driving breakthroughs in Massive MIMO (**大规模天线**), Polar Codes (**极化码**), novel multiple access, and 5G Standalone (**SA**) service-based core network architecture within 3GPP Release 15/16 (MOST 03 Portal).

Project 04 (04专项): High-end CNC Machine Tools and Basic Manufacturing Equipment (高档数控机床与基础制造装备)

- **Governance & Leadership:** Led by **MIIT** (**工业和信息化部**) (MOST 04 Portal), chaired at the ministerial Leadership Group level by **MIIT** Minister **Miao Wei** (**苗圩**), with Implementation Management Office Deputy Director **Li Dong** (**李东**, **MIIT** Equipment Industry Department), Expert Advisory Committee Chair **Lu Yansun** (**陆燕荪**), and Chief Technical Expert (**技术总师**) CAE Academician **Lu Bingheng** (**卢秉恒**).
- **Core Entities & Engineering Milestones:**
  - **5-Axis CNC Controllers (五轴联动高档数控系统) — Huazhong CNC (华中数控): Wuhan Huazhong Numerical Control Co., Ltd.** (**华中数控**, backed by Huazhong University of Science and Technology), alongside **Dalian Guangyang** (**大连光洋** / **科德数控**) and **Shenyang Gaojing** (**沈阳高精**), broke foreign export-control monopolies by developing the **HNC-8 (华中8型)** all-digital bus-based high-end 5-axis linkage CNC controller, deployed in aerospace structural machining (MOST 04 Press Conference).
  - **ISO "S-Shaped Test Piece" International Standard (S形试件国际标准):** Under **04 Project** funding, **AVIC Chengdu Aircraft Industrial (Group) (成飞)** and domestic machine tool enterprises jointly formulated and won ISO approval for the **"S-shaped test piece" (S形试件)** accuracy testing standard for 5-axis linkage machining centers, establishing China's first international ISO testing standard for complex curved-surface aviation machining (MOST 04 Press Conference).
  - **Machine Tool Conglomerates — Genertec (通用技术集团), DMTG (大连机床), SMTCL (沈阳机床), Jier (济南二机床):** Restructured under central SOE **China General Technology Group (中国通用技术集团, Genertec)** (including **Shenyang Machine Tool 沈阳机床**, **Dalian Machine Tool 大连机床**, **Qier Machine Tool 齐二机床**, and **Harbin Measuring & Cutting Tool 哈量**) (SASAC Genertec Report), developing heavy-duty gantry 5-axis machining centers, dual-5-axis mirror milling equipment for rocket tank domes and aircraft skins (Shanghai S&T Award Gazette), and automotive high-speed stamping lines.
  - **80,000-Ton Heavy Die Forging Press (8万吨大型模锻压力机):** Built by **China National Erzhong Group (中国二重 / 国机重装 under Sinomach 国机集团)** in Deyang, Sichuan (NEA Sinomach Audit), representing the world's largest single-cylinder die forging press (800 MN forging force) with ± 1 mm precision, which forged over 130 critical aerospace and nuclear components including the main landing gear (**主起落架锻件**) and bulkhead forgings for the **COMAC C919** and **Y-20** (Official Labor Model & Erzhong Report).

Project 05 (05专项): Large Oil and Gas Fields and Coalbed Methane Development (大型油气田及煤层气开发)

- **Governance & Leadership:** Jointly led by China's three central petroleum SOEs—**China National Petroleum Corporation (中国石油, CNPC)**, **China Petrochemical Corporation (中国石化, Sinopec)**, and **China National Offshore Oil Corporation (中国海油, CNOOC)** (MOST 05 Portal)—with CAS Academician **Jia Chengzao** (**贾承造**) serving as Chief Technical Expert (**技术总师**).
- **Core Entities & Engineering Milestones:**
  - **Marine Carbonate & Ultra-Deep Drilling (海相和深层天然气勘探开发):** Developed 7,000–8,000m+ ultra-deep drilling, rotary steerable systems (**旋转导向钻井系统**), and high-temperature/high-pressure logging tools, unlocking the Tarim Basin (**塔里木盆地** Kuqa/Shunbei) and Sichuan Basin (**四川盆地** Anyue/Puguang) ultra-deep gas fields (MOST 05 Press Conference; MOST 2016 Exchange).
  - **Shale Gas & Shale Oil (页岩气与页岩油):** Breakthroughs in 3,500m+ horizontal well drilling and multi-stage hydraulic fracturing (**长水平井分段压裂技术**) enabled **Sinopec** to commercialize the **Fuling Shale Gas Field (涪陵页岩气田)** (the world's first large commercial shale gas field outside North America) and **CNPC** to develop the Southern Sichuan (**川南**) shale gas and **Changqing Oilfield (长庆油田)** Ordos shale oil fracturing demonstration projects (MOST 05 Shale Gas Report; MOST Index).

- **Coalbed Methane (煤层气) & Deepwater Offshore (深水油气):** Established surface and underground Coalbed Methane (**CBM**) extraction demonstration bases in the **Qinshui Basin (沁水盆地)** and Eastern Margin of the **Ordos Basin (鄂尔多斯盆地东缘)** (NDRC 13th FYP Natural Gas Plan); **CNOOC** deployed the 3,000m deepwater semi-submersible drilling rig **Haiyang Shiyou 981 (海洋石油981)**, the **Deep Sea No. 1 (深海一号 / Lingshui 17-2)** 1,500m deepwater gas field, and China's first domesticated deepwater subsea Christmas tree (**首套国产化深水水下采油树**) (MOST Index).

Project 06 (06专项): Large Advanced Pressurized Water Reactor and High Temperature Gas-cooled Reactor Nuclear Power Plant (大型先进压水堆及高温气冷堆核电站)

- **Governance & Leadership:** Led by the **National Energy Administration (国家能源局, NEA)** (MOST 06 Portal; NEA 06 Management Measures), structured into two major sub-projects (**压水堆分项** and **高温气冷堆分项**).
- **Sub-Project 1 — Large Advanced PWR (CAP1400 / Guohe One 国和一号):**
  - **Lead Enterprise & Chief Designer:** Undertaken by **State Power Investment Corporation (国家电力投资集团, SPIC / formerly SNPTC 国家核电技术公司)** and **Shanghai Nuclear Engineering Research and Design Institute** (**上海核工程研究设计院**, **SNERDI**), with CAE Academician **Zheng Mingguang (郑明光)** serving as Chief Technical Expert (**核电专项技术总师 / CAP1400总设计师**).
  - **Engineering Milestones:** Absorbing and re-innovating Westinghouse **AP1000** Generation-III passive nuclear safety technology (**非能动安全技术**), China developed the **1,400 MWe (1,500 MWe gross) CAP1400 (Guohe One / 国和一号)** with independent intellectual property (≥ 85% equipment domestication, including heavy reactor pressure vessels, steam generators, and canned motor main pumps) and constructed the Shidaowan **CAP1400** demonstration units in Rongcheng, Shandong (MOST 06 Portal; Tsinghua Interview with Zheng Mingguang & Zhang Zuoyi).
- **Sub-Project 2 — High-Temperature Gas-Cooled Reactor (HTR-PM / 石岛湾高温气冷堆示范工程):**
  - **Lead Institute/Enterprises & Chief Designer:** Designed by the **Institute of Nuclear and New Energy Technology of Tsinghua University (清华大学核能与新能源技术研究院, INET)** in consortium with **China Huaneng Group (中国华能)** and **China National Nuclear Corporation (中核集团, CNNC)**, with **INET** Director **Zhang Zuoyi (张作义)** serving as Chief Designer (**高温气冷堆分项总师**).
  - **Engineering Milestones:** Scaled up from Tsinghua's 10 MWth experimental reactor (**HTR-10**), the **200 MWe (20万千瓦级) pebble-bed modular high-temperature gas-cooled reactor (HTR-PM)** at **Shidaowan (石岛湾)**, Weihai, Shandong features Generation-IV inherent safety (**固有安全性** — core meltdown is physically impossible even under complete loss of coolant), spherical TRISO coated fuel elements (**球形燃料元件批量制备**), and helium turbine/steam generator systems. Unit 1 achieved **first criticality on September 12, 2021** (MOST Official Milestone Notice), **first grid connection on December 20, 2021** (MOST Index), and entered commercial operation in December 2023 as the world's first commercial Generation-IV nuclear power plant.

Project 07 (07专项): Water Pollution Control and Treatment (水体污染控制与治理 / 水专项)

- **Governance & Leadership:** Jointly led by the **Ministry of Ecology and Environment (生态环境部, MEE)** and the **Ministry of Housing and Urban-Rural Development (住房和城乡建设部, MOHURD)** (MOST 07 Portal; 2019 Management Measures). The Overall Expert Group (**总体专家组**) was initially led by Chief Technical Expert **Meng Wei** (**孟伟**, Chinese Research Academy of Environmental Sciences **中国环境科学研究院**, later removed in 2017) and succeeded by CAE Academician **Wu Fengchang** (**吴丰昌**) alongside CAE Academicians **Qu Jiuhui** (**曲久辉**) and **Peng Yongzhen** (**彭永臻**).
- **Core Basins & Engineering Milestones:** Focused on the **"Three Rivers, Three Lakes, One Reservoir, One Source" (三河三湖一库一源)** and key urban clusters (MOST 07 Portal):

- **Lake Eutrophication & Cyanobacteria Control:** Comprehensive governance demonstrations in **Lake Taihu (太湖)**, **Lake Dianchi (滇池)**, and **Lake Chaohu (巢湖)**.
- **Major River Basins:** Industrial point-source, agricultural non-point-source, and ecological restoration engineering across the **Haihe River (海河 / Beijing-Tianjin-Hebei 京津冀)**, **Huaihe River (淮河)**, **Liaohe River (辽河)**, **Songhua River (松花江)**, **Three Gorges Reservoir (三峡库区)**, and **Yellow River (黄河流域)** drinking water safety corridors (MOST Index; MEE Environmental Bulletin).

Project 08 (08专项): Genetically Modified New Variety Breeding (转基因生物新品种培育)

- **Governance & Leadership:** Led by the **Ministry of Agriculture and Rural Affairs (农业农村部, MARA)** (MOST 08 Portal), with CAE Academician **Wan Jianmin (万建民)** (**Chinese Academy of Agricultural Sciences**, **CAAS 中国农业科学院**) serving as Chief Technical Expert (**技术总师**) alongside CAE Academician **Wu Kongming (吴孔明)** (**CAAS** Vice President).
- **Core Entities & Engineering Milestones:** Executed by **CAAS** (**中国农科院**), **Huazhong Agricultural University** (**华中农业大学**, CAS Academician **Zhang Qifa 张启发**), **China Agricultural University** (**中国农业大学**), and commercial seed leaders (**DBN 大北农**, **Origin Agritech 奥瑞金**, **Longping High-Tech 隆平高科**, **Syngenta China 先正达中国**) (MARA Official Interview with Wu Kongming & Wan Jianmin):
  - **Pest-Resistant Cotton (转基因抗虫棉):** Achieved >95% domestic market share for autonomous Bt cotton varieties, reversing early foreign seed dominance.
  - **Bt Pest-Resistant Rice (转基因抗虫水稻) & Phytase Corn (转植酸酶基因玉米):** Issued China's landmark 2009 biosafety certificates (**农业转基因生物安全证书**) for **Huahui-1** (**华恢1号**) and **Bt Shanyou 63** (**Bt汕优63**) rice (developed by Huazhong Agricultural University) and **BVLA430101** phytase transgenic corn (**转植酸酶基因玉米**, developed by **CAAS** Institute of Biotechnology).
  - **Insect-Resistant & Herbicide-Tolerant Corn and Soybean (抗虫耐除草剂玉米与大豆):** Approved biosafety certificates for 4 insect-resistant/herbicide-tolerant GM corn events (achieving >95% pest control, 7%–17% yield gain, and 60% pesticide reduction) and 3 herbicide-tolerant GM soybean varieties (**DBN9004**, **SHZD32-01**, **ZH10-6**), including commercial planting approval in Argentina (MARA Official Report).

Project 09 (09专项): Major New Drug Creation (重大新药创制)

- **Governance & Leadership:** Led by the **National Health Commission (国家卫生健康委员会, NHC)** (MOST 09 Portal), with CAE Academician **Sang Guowei (桑国卫)** (former Vice Chairman of the NPC Standing Committee) serving as Chief Technical Expert (**技术总师**), CAS Academician **Chen Kaixian (陈凯先)** (former Director of the **CAS Shanghai Institute of Materia Medica 中国科学院上海药物研究所**) and CAE Academician **Zhang Boli** (**张伯礼**) serving as Deputy Chief Technical Experts (**技术副总师**), and **NHC** S&T and Education Department official **Liu Dengfeng (刘登峰)** serving as Deputy Director of the Implementation Management Office.
- **Core Entities & Engineering Milestones:**
  - **National Drug Innovation System (网格化新药创新体系):** Centered on the **CAS Shanghai Institute of Materia Medica (中科院上海药物所)** and the **National Center for Drug Screening** (**国家新药筛选中心**) in Shanghai's **"Zhangjiang Medicine Valley" (张江药谷)**, alongside the **Chinese Academy of Medical Sciences** (**中国医学科学院**) and **Academy of Military Medical Sciences** (**军事医学科学院**) (Shanghai S&T Party Committee Profile of Chen Kaixian).
  - **Category-1 Innovative Drugs (一类新药):** Shifted China's pharmaceutical industry from >90% generics imitation (**仿制为主**) to autonomous innovation (**创新为主**), approving dozens of Category-1 new drugs including **Icotinib** (**埃克替尼**, Betta Pharma), **Apatinib** (**阿帕替尼**, Hengrui), **Chidamide** (**西达本胺**, Chipscreen), **Fruquintinib** (**呋喹替尼**, Hutchmed), domestic PD-1 monoclonal antibodies (**Toripalimab**

特瑞普利单抗, **Sintilimab 信迪利单抗**), **Albuvirtide** (**艾博卫泰**, Frontier Biotech anti-HIV), **Tegoprazan** (**替戈拉生片**, P-CAB), and the recombinant Ebola vaccine (MOST 09 Press Conference; MOST Index). By 2019, 29 project-supported drugs won US/EU marketing approval, 23 formulations and 4 vaccines achieved WHO prequalification, and domestic pharma enterprises exceeding 10 billion RMB in revenue grew from 2 to 17 (MOST 09 Press Conference).

Project 10 (10专项): AIDS and Viral Hepatitis and other Major Infectious Diseases Prevention (艾滋病和病毒性肝炎等重大传染病防治)

- **Governance & Leadership:** Led by the **National Health Commission (国家卫生健康委员会, NHC)** (MOST 10 Portal), with CAE Academician and 2017 State Preeminent S&T Award laureate **Hou Yunde (侯云德)** (**China CDC** National Institute for Viral Disease Control and Prevention) appointed by the State Council in 2008 as full-time Chief Technical Expert (**专职技术总师**) (MOST Profile of Hou Yunde), and CAE Academician **Li Lanjuan (李兰娟)** serving as Deputy Chief Technical Expert (**技术副总师**).
- **Core Entities & Engineering Milestones:**
  - **National BSL-3 / BSL-4 Laboratory & Pathogen Surveillance Network:** Coordinated **China CDC (中国疾病预防控制中心)**, **CAS**, and **CAMS** to build a nationwide networked pathogen detection platform capable of identifying 300+ known and unknown acute respiratory/infectious pathogens within 72 hours (MOST 10 Portal; MOST Hou Yunde Report), anchored at the apex by the **Wuhan National Biosafety Laboratory (中国科学院武汉国家生物安全实验室 / 武汉P4实验室)** at the **CAS Wuhan Institute of Virology** (**中科院武汉病毒研究所**, directed by **Yuan Zhiming 袁志明**), which completed construction in January 2015 and entered formal BSL-4 operation in January 2018 (NHC P4 Official Notice; CAS Major S&T Infrastructure Portal).
  - **"Three Major Diseases" (三病: HIV/AIDS, Hepatitis B, Tuberculosis) & Acute Epidemics:** Reduced mother-to-child and transfusion HIV transmission, lowered under-5 Hepatitis B surface antigen (**HBsAg**) prevalence below 0.5% through universal vaccination and antiviral protocols, improved multidrug-resistant TB cure rates, and countered the 2009 H1N1 and 2013 H7N9 avian influenza outbreaks (MOST 10 Portal).

Project 11 (11专项): Large Aircraft (大型飞机重大专项)

- **Governance & Leadership:** Approved by the State Council Executive Meeting in February 2007 under a dual civil-military architecture led by **MIIT** (**工业和信息化部**) and **SASTIND** (**国家国防科技工业局**) (MOST 11 Portal), with CAE Academician **Zhang Yanzhong** (**张彦仲**) chairing the Expert Advisory Committee.
- **Civil Trunk Airliner — COMAC C919 (大型客机 C919):**
  - **Lead Enterprise & Chief Designer:** **Commercial Aircraft Corporation of China, Ltd. (中国商用飞机有限责任公司, 中国商飞, COMAC)** was inaugurated in Shanghai in May 2008 specifically to execute the civil trunkliner program, with CAE Academician **Wu Guanghui (吴光辉)** serving as Chief Designer (**C919总设计师**).
  - **Engineering Milestones:** The 158–192 seat narrow-body trunkliner **C919** achieved roll-out in November 2015, **maiden flight on May 5, 2017**, CAAC Type Certificate (**型号合格证**) in September 2022, and **first commercial flight (China Eastern Airlines MU9191, Shanghai to Beijing) on May 28, 2023**, expanding to multi-airline fleet operations and the follow-on wide-body **C929** program (Wuhan Municipal Gov / Wu Guanghui Profile; Hunan MIIT Nanchang C919/C929 Report).
- **Military Strategic Transport — AVIC Xi'an Y-20 (大型军用运输机 运-20 / 鲲鹏):**
  - **Lead Enterprise & Chief Designer:** Developed by **AVIC First Aircraft Institute (航空工业一飞院)** and **AVIC Xi'an Aircraft Industrial Corporation (中航工业西飞)** under Chief Designer and CAE Academician **Tang Changhong (唐长红)**.

    - **Engineering Milestones:** The 200-ton-class strategic airlifter **Y-20 (运-20)** featured supercritical wing aerodynamics, composite structures, and full-authority digital fly-by-wire, completing its **maiden flight on January 26, 2013** (in under 2,000 days from formal launch) (CAE Tang Changhong Decode; CAE Flight Test Report) and **formally entering PLA Air Force service on July 6, 2016** (SASTIND Official Y-20 Commissioning Report).

- **Aero-Propulsion Systems — AECC (中国航空发动机集团有限公司, 中国航发):**
    - To resolve the aero-engine bottleneck, the State Council detached aero-engine assets from **AVIC** (**110 billion RMB** in assets plus **35 billion RMB** in fresh capital) and formally established **Aero Engine Corporation of China (中国航发, AECC)** on August 28, 2016 (Belt and Road Portal / SASAC Announcement).
    - **AECC Commercial Aircraft Engine Co. (中国航发商用航空发动机有限责任公司, 中国航发商发, ACAE)** in Shanghai leads development of the high-bypass turbofan **CJ-1000A (长江-1000A)** for the **C919** (and **CJ-2000** for the **C929**) (Shanghai Economy and IT Commission ACAE Report), while **AECC** Shenyang/Xi'an developed the high-bypass **WS-20 (涡扇-20)** turbofan engine for the **Y-20B**.

Project 12 (12专项): High-resolution Earth Observation System (高分辨率对地观测系统 / 高分专项, CHEOS)

- **Governance & Leadership:** Approved by the State Council in May 2010 and led by **SASTIND (国家国防科技工业局)** / **China National Space Administration (国家航天局, CNSA)**, with the **CNSA Earth Observation and Data Center (国家航天局对地观测与数据中心 / 国防科工局重大专项工程中心)** serving as the Implementation Management Office (MOST 12 Portal; CNSA CHEOS Portal). The Engineering Chief Designer (**高分专项工程总设计师**) has been served by **Tong Xudong (童旭东)** and **Zhao Jian** (**赵坚**).
- **Core Entities & Satellite Constellation (Gaofen 高分 Series):** Built primarily by **China Aerospace Science and Technology Corporation (中国航天科技集团有限公司, CASC)** Fifth Academy (**CAST 航天五院**) and Eighth Academy (**SAST 航天八院**), with payload optics and **SAR** sensors from **CAS** (**CIOMP 长春光机所**, **AIRCAS 空天信息创新研究院**) and **CETC** (CNSA CHEOS Milestone Report):
    - **Gaofen-1 (高分一号, launched April 26, 2013):** First **CHEOS** satellite, combining 2m panchromatic / 8m multispectral resolution with an 800 km wide-swath camera (CNSA GF-1 Launch Release).
    - **Gaofen-2 (高分二号, launched August 19, 2014):** Marked China's civil remote sensing entry into the **sub-meter era (亚米级, 0.8m panchromatic)** (CNSA CHEOS Overview).
    - **Gaofen-3 (高分三号, launched August 10, 2016):** China's first **1m-resolution C-band multi-polarization Synthetic Aperture Radar (SAR 合成孔径雷达)** satellite with 12 imaging modes and an 8-year design life for all-weather maritime and disaster monitoring (CNSA GF-3/GF-4 Technical Spec).
    - **Gaofen-4 (高分四号, launched December 29, 2015):** The world's first **geostationary orbit (地球同步轨道, 36,000 km) high-resolution optical staring imaging satellite** (50m visible / 400m mid-wave infrared) (CNSA GF-3/GF-4 Technical Spec).
    - **Gaofen-5 (高分五号, launched May 2018):** Hyperspectral observation satellite (**高光谱观测卫星**, 330 spectral channels) for atmospheric greenhouse gas and environmental pollution monitoring (CNSA CHEOS News).
    - **Gaofen-6 (高分六号, launched June 2, 2018):** "Red-edge" multispectral agricultural and forestry satellite paired with **GF-1** (SASAC GF-6 Launch Notice).
    - **Gaofen-7 (高分七号, launched November 2019) through Gaofen-14 (高分十四号, launched December 2020) & Gaofen Multi-Mode (高分多模, 2020, China's first PPP civil satellite):** Sub-meter stereo mapping (**1:10,000** scale optical/laser altimetry on **GF-7** and **GF-14**) and classified military-reconnaissance series (**GF-8**, **GF-9**, **GF-10**, **GF-11**, **GF-12**, **GF-13**) completing the all-weather, all-day space-based "Sky Eye" (**天眼工程**) (CNSA CHEOS News).

Project 13 (13专项): Manned Spaceflight and Lunar Exploration (载人航天与探月工程)

- **Sub-Program A — China Manned Space Program (中国载人航天工程 / 921工程):**
  - **Governance & Leadership:** Approved on September 21, 1992 (**921工程**) under a "Three-Step Strategy" (**三步走战略**) (Hunan Gov 921 Historical Audit) and managed by the **China Manned Space Agency (中国载人航天工程办公室, CMSA)** under the **CMC Equipment Development Department (中央军委装备发展部)** (CMSA Engineering Chief Designer System). CAE Academician **Zhou Jianping (周建平)** succeeded **Wang Yongzhi (王永志)** in 2006 as Chief Designer (**中国载人航天工程总设计师**), with China's first astronaut **Yang Liwei (杨利伟)**, **Chen Shanguang (陈善广)**, and **Zhang Bainan (张柏楠)** serving as Deputy Chief Designers (**副总设计师**) and CAS Academician **Gu Yidong (顾逸东)** as Space Science Chief Expert (CMSA Leadership Roster).
  - **Core Entities & Engineering Milestones:** Executed by **CASC (中国航天科技集团**, 1st Academy **CALT** for rockets, 5th Academy **CAST** and 8th Academy **SAST** for spacecraft), **CAS (CSU 中科院空间应用工程与技术中心)**, and **CETC**:
    - **Phase 2 Space Laboratory (空间实验室阶段):** Achieved EVA spacewalk (**Shenzhou-7**, 2008), automated and manual orbital rendezvous and docking (**Shenzhou-8/9/10** with **Tiangong-1**, 2011–2013) (CMSA Shenzhou-9 Report), and propellant orbital refueling (**Tianzhou-1** with **Tiangong-2**, 2017) (CMSA Tiangong-2 Review).
    - **Phase 3 Tiangong Space Station (天宫空间站) & Launch Vehicles:** Developed the heavy-lift **Long March 5B (长征五号B)** (25-ton LEO capacity) and **Long March 7 (长征七号)** cargo launch vehicles at the Wenchang Spacecraft Launch Site, assembling the three-module T-shaped **Tiangong Space Station (天和 Core Module + 问天 & 梦天 Lab Modules)** in 2021–2022 for permanent crewed rotation (**Shenzhou-12** through **Shenzhou-19+** and **Tianzhou** cargo ships) (MOST Index).

- **Sub-Program B — Chinese Lunar Exploration Program (中国探月工程 / 嫦娥工程, CLEP):**
  - **Governance & Leadership:** Formally approved in January 2004 under the three-phase unmanned roadmap **"Orbiting, Landing, Returning" (绕、落、回)** and managed by the **CNSA Lunar Exploration and Space Engineering Center (国家航天局探月与航天工程中心)** under **SASTIND** / **CNSA** (SASAC / Guangming Daily CLEP Audit). The founding "Iron Triangle" comprised Chief Commander **Luan Enjie (栾恩杰)**, First Chief Designer **Sun Jiadong (孙家栋)**, and Chief Scientist CAS Academician **Ouyang Ziyuan (欧阳自远)**; CAS Academician **Ye Peijian (叶培建)** served as **Chang'e-1** Chief Designer & Chief Commander, CAE Academician **Wu Weiren (吴伟仁)** succeeded as Chief Designer of the Chinese Lunar Exploration Program (**中国探月工程总设计师**), and **Hu Hao (胡浩)** served as Phase-3 Chief Designer (**探月工程三期总设计师**).
  - **Engineering Milestones (Chang'e-1 to Chang'e-6):**
    - **Chang'e-1 (嫦娥一号, Oct 2007) & Chang'e-2 (嫦娥二号, Oct 2010):** Completed Phase 1 global 3D lunar mapping and L2 Lagrangian point / Asteroid **4179 Toutatis** flyby (SASAC CLEP Audit).
    - **Chang'e-3 (嫦娥三号, Dec 2013) & Chang'e-4 (嫦娥四号, Jan 2019):** Achieved China's first extraterrestrial soft landing (**Yutu** rover) and—via the **Queqiao (鹊桥)** halo-orbit relay satellite—humanity's **first ever soft landing and roving exploration on the lunar far side (月球背面 Von Kármán crater)** (SASAC CLEP Audit; SASTIND 2016 Space White Paper).
    - **Chang'e-5 (嫦娥五号, Nov–Dec 2020) & Chang'e-6 (嫦娥六号, May–June 2024):** Launched by **Long March 5**, **Chang'e-5** achieved five "Chinese Firsts" (lunar surface sampling/packaging, lunar liftoff, unmanned lunar-orbit rendezvous/docking, near-second-cosmic-velocity skip re-entry, and sample curation, returning 1,731 g of nearside regolith) (SASAC CLEP Audit), followed in June 2024 by **Chang'e-6** returning 1,935.3 g in humanity's first sample return from the lunar far side, leading into

Phase 4 (**Chang'e-7** in 2026 and **Chang'e-8** in 2028 for the International Lunar Research Station **国际月球科研站**) (CNSA Lunar Program Chronicle).

Projects 14, 15, 16 (Classified Defense/Strategic Projects 军口重大专项) & the Added Aero-Engine Mega-Project (两机专项)

When the State Council promulgated the 2006–2020 MLP, it announced 16 National S&T Major Projects in total (**16个重大专项**), publicly naming the 13 civilian/dual-use projects (**01–13**) while withholding the titles of the remaining 3 defense/strategic projects (**14–16**) under national security classification (**涉密/军口专项**) (MOST Major Projects Overview). Official Chinese government white papers, **MOST** award gazettes, and **CAS** national laboratory archives explicitly identify the strategic programs executed within this framework:

1. **Beidou Navigation Satellite System (北斗卫星导航系统 / 北斗二号卫星工程):**
2. **Official Confirmation:** Both the State Council white paper *China's Beidou Navigation Satellite System (《中国北斗卫星导航系统》白皮书*, June 2016) and MOST's official Major Projects portal (**北斗二号卫星工程是国家科技重大专项**) (Beidou Official Portal) explicitly state that China listed the Beidou system (**BDS-2** / **BDS-3**) as a **National Science and Technology Major Project (国家科技重大专项)**, appearing directly on the MOST National S&T Major Projects homepage news feed.
3. **Leadership & Milestones:** Managed by the **China Satellite Navigation Office** (**中国卫星导航系统管理办公室**) under **CMC EDD**, with CAS Academician **Sun Jiadong** (**孙家栋**) as **BDS-1**/**BDS-2** Chief Designer and CAE Academician **Yang Changfeng** (**杨长风**) as **BDS-3** Chief Designer. **BDS-2** achieved Asia-Pacific regional service in December 2012 (winning the 2016 State S&T Progress Grand Prize **特等奖**), and **BDS-3** completed its 35-satellite global constellation (**GEO + IGSO + MEO** with rubidium/hydrogen atomic clocks and inter-satellite links) in July 2020 (MOST Beidou Feature).
4. **Laser Inertial Confinement Fusion (激光惯性约束聚变 / ICF / 神光 ShenGuang Facility):**
5. **Official Confirmation:** The **Chinese Academy of Sciences Major S&T Infrastructure Sharing Platform (中国科学院重大科技基础设施共享服务平台)** explicitly states in its official report on the **ShenGuang-II (神光II) High-Power Laser Facility** that *"the three landmark achievements obtained on the ShenGuang-II facility directly drove the establishment of the National Major Special Project (**直接推动了国家重大专项的确立**)"* and that **CAS**'s Double-Cone Ignition (**双锥对撞点火方案**) and **ShenGuang-II Upgrade** (**神光II升级装置**) serve China's **National Ignition Special Project (我国点火专项 / 国家重大专项)** (CAS Official ShenGuang/ICF Report).
6. **Leadership & Entities:** Jointly executed by the **China Academy of Engineering Physics (中国工程物理研究院, CAEP / Laser Fusion Research Center 激光聚变研究中心 in Mianyang, operating ShenGuang-III 神光III and the ShenGuang-IV / 主机装置)** and the **CAS Shanghai Institute of Optics and Fine Mechanics (中科院上海光机所, SIOM, High Power Laser Physics Joint Laboratory 高功率激光物理联合实验室)**, led by Academicians **Fan Dianyuan** (**范滇元**), **He Xiantu** (**贺贤土**), **Zhu Jianqiang** (**朱健强**), and **Zhang Jie** (**张杰**).
7. **Strategic Defense & Hypersonic/Aerospace Systems (国防尖端武器装备专项) & the Added Aero-Engine and Gas Turbine Major Project (航空发动机及燃气轮机重大专项 / 两机专项):**
8. While the remaining defense slot in the original 2006 MLP covers classified strategic strike/hypersonic/antisatellite systems under **CMC EDD** and **SASTIND**, in **2015–2016** the CCP Central Committee and State Council elevated **Aero-Engine and Gas Turbine (航空发动机及燃气轮机重大专项, 两机专项)**—backed by the creation of **AECC (中国航发)** and led technically by CAE Academician **Yin Zeyong** (**尹泽勇**)—as an independent national major project (**第17个国家科技重大专项**) and simultaneously placed it as **Project #1** under **Sci-Tech Innovation 2030 – Major Projects (科技创新2030–重大项目)** (State Council 13th FYP S&T Plan **国发〔2016〕43号**; MOST 15+1 Announcement).

Section 19: Military-Civil Fusion (军民融合 / 军民结合 / 寓军于民)

19.1 Doctrinal & Institutional Evolution: From the 16-Character Policy to the 2017 Central Commission

- **Deng Xiaoping's 16-Character Policy (十六字方针, 1982):** Following the post-1978 defense-industrial transition (**军转民**), **`Deng Xiaoping`** (**邓小平**) formulated the foundational 16-character directive for China's defense science, technology, and industry (SASTIND Official MCF Strategy Review):

  > ***军民结合，平战结合，军品优先，以民养军*** *(Jūnmín jiéhé, píngzhàn jiéhé, jūnpǐn yōuxiān, yǐ mín yǎng jūn — **"Combining the military and civil, combining peacetime and wartime, giving priority to military products, and letting the civil support the military."**)*

- **`Locating Military Potential in Civilian Capabilities` (寓军于民, *Yù jūn yú mín*, 2002–2006):** Codified at the 16th CCP National Congress (2002) and embedded as the **Fourth Basic Principle** of the 2006–2020 MLP (**四是体现军民结合、寓军于民，对保障国家安全和增强综合国力具有重大战略意义**) (MOST Major Projects Charter), **寓军于民** shifted defense industrialization away from isolated state arsenals toward embedding defense R&D and mobilization capacity directly inside China's commercial high-tech industrial base (State Council General Office Opinion **`国办发〔2017〕91号`**).

- **Elevation to a National Strategy (上升为国家战略, March 2015):** At the plenary meeting of the PLA delegation during the Third Session of the 12th National People's Congress in March 2015, General Secretary **`Xi Jinping`** (**习近平**) formally elevated **`Military-Civil Fusion Development`** (**军民融合发展战略, *Jūnmín Rónghé Fāzhǎn Zhànlüè*, `MCF`**) to a **National Strategy (国家战略)** aimed at building an *"integrated national strategic system and capabilities"* (**构建一体化国家战略体系和能力**) and achieving an *"all-element, multi-domain, high-efficiency pattern of deep military-civil fusion"* (**全要素、多领域、高效益的军民深度融合发展格局**) (SASTIND MCF Doctrine; MOD Official Commentary; CSET MCF Analysis). In July 2016, the CCP Central Committee, State Council, and CMC jointly issued the *Opinions on the Integrated Development of Economic Construction and National Defense Construction* (**关于经济建设和国防建设融合发展的意见**).

- **Establishment of the `Central Commission for Integrated Military and Civilian Development` (中央军民融合发展委员会, January 2017):** On January 22, 2017, the CCP Politburo established the **`Central Commission for Integrated Military and Civilian Development`** (**中央军民融合发展委员会, *Zhōngyāng Jūnmín Rónghé Fāzhǎn Wěiyuánhuì*, `CCIMCD`**) as the supreme Party-State-Military decision-making and deliberative body (**决策议事协调机构**), chaired by **`Xi Jinping`** (**习近平**) with Premier **`Li Keqiang`** (**李克强**), **`Liu Yunshan`** (**刘云山**) / **`Wang Huning`** (**王沪宁**), and **`Zhang Gaoli`** (**张高丽**) / **`Han Zheng`** (**韩正**) as Deputy Directors, and a permanent General Office (**中央军民融合发展委员会办公室**) (CAC Report on 1st Plenary Session, June 20, 2017; NRA Report on 19th CCIMCD 1st Plenary Session, March 2018). In December 2017, the State Council General Office issued the landmark 31-article directive **`Opinions on Promoting the Deep Development of Military-Civil Fusion in National Defense S&T Industry`** (**国务院办公厅关于推动国防科技工业军民融合深度发展的意见, 国办发〔2017〕91号**).

19.2 Dual-Use Technology Transfer Pipelines, Defense Key Laboratories & Licensing Reform (四证 → 三证 → 两证)

- **Bidirectional Technology Transfer (双向转移转化 — 军转民 & 民参军):**
  - **Institutionalized under the `13th Five-Year Special Plan for S&T Military-Civil Fusion Development` (“十三五”科技军民融合发展专项规划, August 2017) and 国办发〔2017〕91号:** China established a *"small core, grand collaboration, specialized, and open"* (**小核心、大协作、专业化、开放型**) weapons R&D and production system.

- **Declassification & "Three Rights" IP Reform (国防科技成果“三权”改革): SASTIND** and **CMC EDD** reformed national defense intellectual property (**国防知识产权**), delegating the **right of use, right of disposal, and right of income (使用权、处置权、收益权 — "Three Rights")** of defense S&T achievements to undertaking institutes and universities, while periodically declassifying defense patents (**国防专利解密**) and publishing open bidding guides on the **All-Army Weapons and Equipment Procurement Information Network (全军武器装备采购信息网)** (SASTIND Reform Report; CSET Innovation Strategy Translation).
- **National Defense S&T Industry Military-Civil Fusion Industrial Investment Fund (国家国防科技工业军民融合产业投资基金):** Established under **Article 31 of 国办发〔2017〕91号**, alongside the restructuring of defense research institutes into corporate entities (**军工科研院所转制**, pioneered by China South Industries Group's 58th Research Institute **中国兵器装备集团自动化研究所/58所**) (Wuhan Gov MCF Report).

- **National Defense S&T Key Laboratories (国防科技重点实验室) & Resource Sharing:**
  - Under **Article 8 of 国办发〔2017〕91号** and the **13th FYP S&T MCF Plan**, **SASTIND** and the **CMC Science and Technology Commission** (**中央军委科学技术委员会**) integrated **National Defense S&T Key Laboratories (国防科技重点实验室)**, **National Defense Key Discipline Laboratories (国防重点学科实验室)**, and **National Defense S&T Industry Innovation Centers (国防科技工业创新中心)** with civilian **State Key Laboratories** (**国家重点实验室 / 全国重点实验室**) (MOE Higher Education MCF Conference; CSET Defense Workforce Study). By 2018, **SASTIND** had opened over **3,000 major defense testing facilities and laboratory instruments** in three batches to civilian enterprises and universities (Wuhan Gov MCF Report).

- **Defense Industrial Licensing Reform (“四证”简化为“三证”及“两证”):**
  - **Historical "Four Certificates" (军工四证):** Historically, any civilian enterprise (**民口企业**) seeking to participate in PLA weapons R&D and manufacturing (**民参军**) was required to obtain four separate, sequential qualifications (SASTIND Press Conference on **国办发〔2017〕91号**; Hunan SASTIND Licensing Guide):
  - **Weapons and Equipment Quality Management System Certificate** (**武器装备质量管理体系认证**, **GJB9001**);
  - **Secrecy Qualification for Weapons and Equipment Research and Production Units** (**武器装备科研生产单位保密资格认定** — classified into Level 1 **一级**, Level 2 **二级**, and Level 3 **三级**, administered under the *Measures for the Determination of Secrecy Qualifications* **武器装备科研生产单位保密资格认定办法**);
  - **Weapons and Equipment Research and Production License** (**武器装备科研生产许可证** — classified into Category I **一类许可** and Category II **二类许可**, managed by **SASTIND** with **CMC EDD** coordination under the *Regulations on the Administration of Weapons and Equipment Research and Production Licensing* **武器装备科研生产许可管理条例**); and
  - **Equipment Manufacturing Unit Qualification** (**装备承制单位资格认证** — managed by the **CMC Equipment Development Department 中央军委装备发展部**).
  - **Step 1 — "Two Certificates in One" (两证合一: 军工四证 → 军工三证, October 1, 2017):** Approved by the **CMC Equipment Development Department** (**中央军委装备发展部**) effective **October 1, 2017**, the **Weapons and Equipment Quality Management System Certificate** (**武器装备质量管理体系认证**) was merged directly into the **Equipment Manufacturing Unit Qualification** (**装备承制单位资格认证**), reducing the four certificates to **"Three Certificates" (军工三证)** (Hunan SASTIND Official Licensing Guide).
  - **Step 2 — Catalog Slashing & Joint Licensing (军工三证 → 两证, 2018–present):** Through consecutive revisions of the *Weapons and Equipment Research and Production Licensing Catalog* (**武器装备科研生产许可目录**, slashed by over 60%–70% across multiple batches so that only core complete systems and critical subsystems require pre-licensing, while general components transition to post-event filing **武器装备科研生产备案管理**) and the joint review mechanism between **SASTIND** (**武器装备科研生产许可**) and **CMC**

EDD (装备承制单位资格 联合审查/多证合一), most civilian high-tech suppliers now require only **two core threshold credentials (两证)**: the **Secrecy Qualification (保密资格认定)** and the unified **Equipment Manufacturing / Research & Production Qualification (装备承制单位资格 / 科研生产许可或备案)** (SASTIND Press Conference; Hunan SASTIND Licensing Guide).

Part V: Consolidated Cross-Jurisdictional Governance Matrix & Categorized Primary-Source Compendium

Section 20: Master Cross-Jurisdictional Synthesis Matrix (UK, Canada, Australia, China)

Analytical Dimension	United Kingdom (STRAP System)	Canada (SAP / SOI / PPBS)	Australia (TS-PA / SAP / NIC)	China (国家科技重大专项 & 军民融合)
Primary Statutory & Policy Anchor	ISA 1994; SSA 1989; IPA 2016; NSA 2023; MOD JSP 440 & JSP 441; Cabinet Office GSCP (2024).	FISOIA (R.S.C. 1985, c. O-5; Bill C-70 2024); CSE Act 2019; CSIS Act; TBS Directive on Security Screening (2025); NDSOD Ch. 4.	ASIO Act 1979 Part IVA (Act No. 33, 2023); ISA 2001; ONI Act 2018; PSPF Release 2024; Naval Nuclear Power Safety Act 2024.	2006–2020 MLP (**国发〔2005〕44号**); Management Provisions **国科发专〔2017〕145号**; MCF Opinion **国办发〔2017〕91号**; 2023 Institutional Reform Plan (**中央科技委员会**).
Supreme Coordinating & Vetting Authority	**Cabinet Office (JIC, JIO, GSG, UKSV)** & **STRAP Management Board** (**GCHQ** technical custodian; departmental **STRAPSOs**).	**Privy Council Office (PCO)**, **TBS**, **CSIS** (vetting/polygraph), **CSE** (**TS/SI** & **CTSN**), and **Deputy Heads** (**FISOIA s. 8(2)**).	**Office of National Intelligence (ONI + QAO)** and **ASIO (TS-PA Vetting Authority + ASIO-T4)** alongside **ASD** (**ATO** & **AKMI**).	**Central Science & Technology Commission (CSTC / MOST)**, **Central MCF Commission (CCIMCD)**, **Three-Ministry Balancing (MOST, NDRC, MOF)**, **SASTIND**, and **CMC EDD**.
Highest Personnel Vetting & Secrecy Bond	**Developed Vetting (DV)** & **Enhanced DV (eDV)** via **UKSV** + **MI5** records check + Role-Specific **STRAPSO** Briefing & signed **STRAP Indoctrination Register**.	**Enhanced Top Secret (Enhanced Level III)** with mandatory **CSIS Polygraph** + lifetime **Person Permanently Bound to Secrecy (PPBS)** under **FISOIA ss. 8–15** (**14 yrs** strict liability).	**Top Secret-Privileged Access (TS-PA)** (replacing **Positive Vetting PV**): whole-of-life since age 16, 2-part clinical psych assessment, ATO/3-month bank statements, 5+ referees, Annual Appraisal.	**Weapons & Equipment Research/Production Secrecy Qualification (保密资格认定 Level 1–3)** + **Two-Chief System (两总系统: 总指挥 + 总设计师)** responsibility contracts (**任务合同书**).
Compartmented Tiers & Programs	**STRAP 1** (**FVEY SI/COMINT**), **STRAP 2** (**SI-GAMMA** / active **CNE** / **QI**), **STRAP 3** (**ECI** / fragile sources; zero automatic right of access for	**Special Operational Information (SOI)** (**s. 8(1)(a)–(g)**), **Top Secret/Special Intelligence (TS/SI)**, Allied **SAR/SAP** (**PSPC CSP**),	**TS//Special Access**, **SCI**, **Codeword Caveats**, **DSTG/ADF Special Access Programs (SAP)**, **AUKUS Pillar 1 (NNPI)** & **Pillar 2**	**16 National S&T Major Projects (01专项 核高基 to 13专项 + Classified 14–16专项 Beidou, ShenGuang ICF, Strategic Defense + 两机专项)** & **Sci-Tech**

Analytical Dimension	United Kingdom (STRAP System)	Canada (SAP / SOI / PPBS)	Australia (TS-PA / SAP / NIC)	China (国家科技重大专项 & 军民融合)
	successors under JSP 441).	and CANSOFCOM/JTF 2 compartments.	compartments, AUSTEO/AGAO/REL.	Innovation 2030 (15+1).
Physical (SCIF) & Accredited Network Substrate	**Tri-NTA**: NCSC (TEMPEST/CFTCS), UK NACE (TSCM/audio), NPSA (CSE equipment). Networks: SII (TOP SECRET STRAP), ROSA (SECRET), NEXUS, CADDIS, UKROS/OSE.	CSE CITP (TEMPEST) & CFINTCOM SCIF standards. Networks: CTSN (Top Secret/SCI across 32 DAOs), CSNI & GC-CSI (Secret), CJI / StoneGhost.	ASIO-T4 Zone 5 / SCIF & SCEC hardware; DISP Level 3 (Essential Eight ML2). Networks: ATSN & 2B AWS NIC Top Secret Cloud (ASD ATO & AKMI).	National Defense S&T Key Laboratories (国防科技重点实验室), Wuhan BSL-4 (P4), ShenGuang Laser Facility, Sunway TaihuLight / Tianhe HPC, Tiangong & Gaofen (CHEOS).

Section 21: Categorized Primary-Source Reference Compendium (Canada, Australia, China)

21.1 Canada: Statutory Legislation, Treasury Board Directives, DND/CSE Records & Judicial Precedents

- *Foreign Interference and Security of Information Act* (FISOIA / formerly SOIA, R.S.C., 1985, c. O-5 — Full Text, Schedules 1 & 2, and Sections 8–24) — Justice Laws Website, Government of Canada.
- Bill C-70: *An Act respecting countering foreign interference* (S.C. 2024, c. 16 — Royal Assent Text, June 20, 2024) — Parliament of Canada.
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